



Electrical Engineering Department
Computer Systems Engineering

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X-Fed NeuroScan

Federated Learning with Explainable AI
for Medical Image Classification

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**X-Fed NeuroScan:
Federated Learning with
Explainable AI for
Medical Image
Classification**

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Abstract

Medical image analysis plays a critical role in the early detection and diagnosis of various diseases. However, manual interpretation of radiographic images is time-consuming and subject to inter-observer variability. This project presents an end-to-end deep learning-based system for the automated detection of chest diseases and bone fractures from X-ray images. The proposed solution integrates multiple transfer learning models within a unified diagnostic framework, enabling efficient and scalable analysis of heterogeneous medical images.

The system architecture consists of a deep learning pipeline that includes image preprocessing, feature extraction using pre-trained convolutional neural networks, an intelligent medical image routing model, and task-specific diagnostic models. ResNet50 is employed as a core feature extractor due to its residual learning mechanism and suitability for medical image analysis, while YOLOv8 is utilized for bone fracture detection using a one-stage object detection approach. The chest disease detection task is addressed using a dedicated classification model optimized for thoracic imaging.

Preliminary experimental results demonstrate the feasibility of the proposed approach and indicate promising performance in detecting both bone fractures and chest-related conditions. The modular design of the system allows for extensibility and supports the integration of additional diagnostic tasks. This work highlights the potential of deep learning and transfer learning techniques to assist clinical decision-making and improve the efficiency of medical image diagnosis.

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CHAPTER 1

INTRODUCTION

Chapter 1 Introduction

Medical imaging plays a critical role in modern healthcare, supporting clinicians in diagnosing a wide range of conditions, from bone fractures to chest tumors and other life-threatening emergencies. As imaging technologies become more advanced and widely available, the volume of medical scans produced each day continues to grow. This rapid increase has intensified the need for efficient, reliable, and scalable diagnostic support systems that can assist healthcare professionals in interpreting complex visual data.

Recent advances in computer vision have made automated medical image analysis a promising tool for improving diagnostic accuracy and reducing clinical workload. Deep learning models, in particular, have demonstrated exceptional performance in tasks such as classification, localization, and segmentation across many imaging modalities. However, despite their capabilities, traditional centralized learning approaches often face critical limitations—especially in sensitive domains like healthcare, where data privacy, institutional policies, and regulatory constraints restrict the sharing of patient information.

To address these challenges, new paradigms in machine learning and model transparency have emerged. Federated learning enables collaborative training across multiple medical institutions without requiring direct access to patient data, offering a pathway toward more diverse and robust models while preserving privacy. Complementing this, explainable AI techniques provide insights into model decisions, helping clinicians understand why a prediction was made and increasing trust in AI-assisted diagnostics.

This project aims to integrate these three powerful components—computer vision, explainable AI, and federated learning—into a unified system designed to support medical image analysis across a variety of diseases and emergency conditions. By leveraging decentralized learning and transparent decision-making, the system seeks to improve accuracy, reliability, and interpretability while respecting the privacy and variability inherent in real-world clinical environments.

1.1 Problem Definition

Accurate diagnosis of bone fractures and chest diseases from X-ray images remains a critical challenge in modern healthcare. Many clinically important abnormalities—such as hairline fractures, early-stage pneumonia, subtle tuberculosis patterns, or small tumor-like chest masses—are visually faint, easily overlooked, or masked by overlapping anatomical structures. These challenges are intensified in high-volume clinical environments and can lead to delayed or incorrect diagnosis, negatively impacting patient outcomes.

Furthermore, deep learning models have achieved state-of-the-art performance in medical image analysis, but their clinical adoption is limited by two major obstacles: **(1) the inability to train diverse and robust models without aggregating sensitive**

patient data, and (2) the black-box nature of deep models, which makes their predictions difficult for clinicians to trust or verify. Hospitals typically hold isolated datasets that differ in imaging quality, patient demographics, disease prevalence, and device configurations, resulting in highly non-IID data distributions that degrade model performance when trained centrally or institutionally.

Given these challenges, the central problem addressed by this project is the following:

How can we build a privacy-preserving, explainable, and clinically reliable deep learning system for detecting four major chest disease classes—pneumonia, tuberculosis (TB), COVID-19, and tumor-like chest masses—using X-ray and CT images, despite highly heterogeneous data and strict privacy constraints across medical institutions?

To contextualize this problem, the diseases and abnormalities targeted by the system are defined as follows:

- **Pneumonia:** An inflammatory lung infection characterized by alveolar consolidation, visible as opacities or infiltrates on chest X-rays.
- **Tuberculosis (TB):** A bacterial infection caused by *Mycobacterium tuberculosis*, typically presenting as lung cavities, nodular lesions, or patchy infiltrates.
- **COVID-19:** A viral infection (SARS-CoV-2) associated with ground-glass opacities, bilateral consolidations, and diffuse lung involvement on X-rays.
- **Chest masses/tumor-like abnormalities:** Suspicious opacities or lesions that may indicate malignant or benign tumors, requiring early detection for timely treatment.
- **Normal (healthy) cases:** X-rays exhibiting clear lung fields and healthy bone structures, used as the baseline class.

Addressing these conditions within a single diagnostic pipeline introduces several technical challenges:

- **Subtlety and variability of abnormalities:** Many chest pathologies present with faint or overlapping features that are difficult to detect even for experienced radiologists.
- **Data privacy and distribution constraints:** Medical data cannot be shared across hospitals due to privacy regulations, limiting model diversity and robustness.
- **Non-IID data across institutions:** Variations in imaging devices, acquisition protocols, and patient populations create distribution shifts that degrade performance.
- **Lack of interpretability:** Clinicians require transparent explanations—such as heatmaps showing fracture lines or diseased lung regions—to trust AI-assisted diagnosis.
- **Efficiency and deployability:** Models must run efficiently, with low communication costs, enabling real-world federated setups across hospitals or clinics.

Therefore, the problem tackled in this work is to design an integrated framework—X-

Fed NeuroScan—that combines federated learning for privacy preservation and multi-institution collaboration, with explainable AI techniques to generate clinically meaningful visual justifications for predictions, while achieving high diagnostic performance across the targeted disease categories.

1.2 Challenges in Medical Image Analysis

Medical imaging plays a fundamental role in modern healthcare, providing clinicians with valuable information for diagnosing, monitoring, and treating a wide range of diseases. Imaging modalities such as chest X-rays and Computed Tomography (CT) scans are routinely used to assess abnormalities within the thoracic region, including pneumonia, tuberculosis, pulmonary edema, and lung cancer. Despite the significant advancements in imaging technologies, the interpretation and analysis of medical images remain highly challenging tasks.

Medical image analysis traditionally relies on the expertise of radiologists and physicians who must examine large volumes of imaging data and identify subtle patterns that may indicate disease. The complexity of anatomical structures, variability among patients, and limitations of human perception can make accurate diagnosis difficult. These challenges can affect diagnostic accuracy, increase workload, and potentially delay clinical decision-making.

Complex Anatomical Structures

The human thoracic cavity contains numerous overlapping anatomical structures, including the lungs, heart, ribs, blood vessels, diaphragm, and surrounding soft tissues. In chest X-ray images, these structures are projected onto a two-dimensional plane, causing different tissues to overlap and potentially obscure important pathological findings.

The presence of overlapping structures makes it difficult for clinicians to accurately distinguish between normal anatomical variations and disease-related abnormalities. Small lesions, nodules, or early-stage infections may be hidden behind ribs or other dense structures, increasing the likelihood of missed diagnoses.

Subtle Appearance of Early-Stage Diseases

Many thoracic diseases initially manifest as subtle visual changes that are difficult to detect through routine examination. Early-stage lung cancer, for example, may appear as a small pulmonary nodule that is barely distinguishable from surrounding tissue. Similarly, mild pneumonia or early tuberculosis infections may produce only faint abnormalities within the lung fields.

Detecting these subtle findings requires significant expertise and experience. Even highly trained radiologists may encounter difficulties when abnormalities are small, poorly defined, or located in anatomically complex regions.

Variability in Disease Presentation

The same disease can appear differently across patients due to factors such as age, disease stage, genetic characteristics, and the presence of other medical conditions. Furthermore, different diseases may exhibit similar radiological appearances, making differential diagnosis challenging.

For example, pulmonary infections, inflammatory conditions, and certain types of lung cancer can produce similar opacities in chest X-rays. Likewise, benign and malignant lung nodules may share similar characteristics in CT scans. This variability increases diagnostic complexity and may lead to uncertainty during image interpretation.

Dependence on Radiologist Expertise

Accurate medical image interpretation requires extensive education, training, and clinical experience. Radiologists must possess detailed knowledge of anatomy, pathology, imaging physics, and disease progression in order to make reliable assessments.

However, diagnostic performance can vary among clinicians depending on their level of expertise and familiarity with specific conditions. Less experienced practitioners may overlook subtle findings or misinterpret complex cases. As a result, diagnostic consistency can differ between healthcare professionals and institutions.

Inter-Observer and Intra-Observer Variability

One of the well-known challenges in medical imaging is the variability in interpretation among radiologists. Different specialists may provide different assessments when evaluating the same image, a phenomenon known as inter-observer variability.

Additionally, the same radiologist may occasionally reach different conclusions when reviewing an image at different times, which is referred to as intra-observer variability. Such inconsistencies can affect diagnostic reliability and complicate treatment planning, particularly in borderline or ambiguous cases.

Large Volume of Imaging Data

Modern healthcare systems generate enormous quantities of medical images on a daily basis. Radiologists are often required to review hundreds of imaging studies during a single workday, creating significant workload pressures.

CT examinations present an additional challenge because a single scan may contain hundreds of image slices that must be carefully reviewed. Thorough examination of such large datasets is time-consuming and can contribute to fatigue, potentially increasing the risk of oversight or diagnostic errors.

Time Constraints in Clinical Practice

Healthcare professionals frequently operate under strict time constraints, particularly in busy hospitals and emergency departments. Rapid diagnosis is often essential for initiating timely treatment and improving patient outcomes.

The need to balance speed with diagnostic accuracy can be challenging, especially when dealing with complex imaging studies. Under time pressure, subtle abnormalities may be overlooked, and difficult cases may require additional consultation or follow-up examinations.

Image Quality Limitations

The diagnostic quality of medical images can be affected by several factors, including patient movement, improper positioning, low contrast, imaging noise, and equipment-related artifacts. Poor image quality may obscure important pathological findings and make interpretation more difficult.

In chest X-ray imaging, underexposure or overexposure can reduce visibility of anatomical details. In CT imaging, motion artifacts caused by breathing or patient movement can degrade image clarity. Such limitations may reduce diagnostic confidence and necessitate repeat imaging procedures.

Diagnostic Errors and Missed Findings

Despite advances in medical imaging technology, diagnostic errors remain an unavoidable challenge. Errors may occur due to perceptual limitations, cognitive biases, fatigue, distraction, or the inherently complex nature of certain cases.

Missed lung nodules, overlooked fractures, and undetected early-stage diseases represent some of the most common diagnostic challenges in thoracic imaging. Because delayed diagnosis can significantly impact patient outcomes, minimizing such errors remains a major objective within radiology.

Challenges in Lung Cancer Detection

Lung cancer detection presents particularly significant difficulties due to the diverse appearance of pulmonary nodules. Nodules may vary greatly in size, shape, texture, density, and location. Small malignant nodules can closely resemble benign lesions or normal anatomical structures.

Furthermore, certain tumors may be partially obscured by blood vessels, ribs, or surrounding tissues. Accurate identification and characterization of suspicious nodules often require careful examination across multiple CT slices and comparison with previous studies, making the diagnostic process both complex and time-intensive.

Increasing Demand for Radiological Services

The global demand for medical imaging services continues to grow due to population growth, aging demographics, and increased utilization of diagnostic imaging technologies. However, many healthcare systems face shortages of trained radiologists and imaging specialists.

This imbalance between demand and available expertise can lead to increased workloads, longer reporting times, and reduced access to timely diagnoses. The challenge is particularly pronounced in developing regions and underserved healthcare environments.

1.3 Background and Motivation

The rapid growth of medical imaging data, coupled with increased clinical workloads, has made conventional diagnostic workflows increasingly insufficient. Although automated analysis using computer vision holds considerable promise in improving diagnostic accuracy and efficiency, practical implementation is hindered by challenges such as patient privacy restrictions, limited data set diversity, and substantial computational and data transfer demands. Therefore, a thorough understanding of both the potential benefits and inherent limitations of AI in medical imaging is essential for the design of systems that are accurate, interpretable, and scalable in heterogeneous clinical environments.

Rise of Computer Vision in Automated Analysis

The rapid digitization of healthcare has led to a dramatic increase in medical imaging data, including X-ray, CT, MRI, and ultrasound scans. Radiologists now face the dual challenges of handling large patient loads and interpreting large volumes of images quickly and accurately. Computer vision has emerged as a crucial tool for addressing these demands by enhancing diagnostic efficiency and accuracy.

To understand why computer vision is now widely adopted in clinical workflows, consider the following advantages:

- **High sensitivity to subtle patterns:** AI can detect faint fractures or early-stage tumors.
- **Rapid processing:** Thousands of images can be analyzed within seconds.
- **Consistency:** Predictions remain stable regardless of fatigue or emotional state.
- **Adaptability:** Deep models can generalize to various imaging modalities.

The effectiveness of computer vision in medical imaging is reinforced by major achievements in the field. These advancements primarily arise from deep learning architectures such as CNNs and Vision Transformers. In practice, computer vision supports several core tasks:

1. Classification of abnormal vs. normal scans.

- 2. Localization of suspicious areas using object detection.
- 3. Segmentation of organs, lesions, and fracture regions.
- 4. Identification of patterns that deviate from healthy anatomy.

A concise summary of computer vision benefits in medical imaging is outlined in Table 1.1.

Table 1.1: Key Advantages of Computer Vision in Medical Image Automation

Computer Vision Capability	Role in Medical Imaging
Feature extraction	Detects subtle abnormalities beyond human perception
Automation of routine tasks	Reduces workload and radiologist fatigue
High-speed inference	Enables rapid triage in emergencies
Generalization across cases	Handles variations in anatomy and imaging conditions

Limitations of Centralized Machine Learning

Despite the promise of AI-based diagnostic systems, models trained in a centralized manner face several constraints. These limitations arise mainly from restrictions on data availability, data diversity, and practical constraints in medical settings.

Centralized learning typically suffers from:

- **Restricted data sharing:** Patient data cannot be freely exchanged due to privacy laws.
- **Lack of representation:** Single-hospital datasets fail to capture global patient diversity.
- **High storage and transfer requirements:** Medical images are large and expensive to move.
- **Data imbalance:** Rare conditions remain underrepresented, harming model performance.

These issues do not simply degrade model accuracy—they introduce significant risks into clinical AI deployment. The consequences can be better understood through the following observations:

- 1. Models trained in isolation often misclassify images from new machines or demographics.
- 2. Performance can fluctuate between institutions due to imaging protocol differences.

3. Bias can emerge when the training set fails to include certain patient groups.

Table 1.2 summarizes these core limitations in a structured manner.

Table 1.2: Challenges of Centralized Machine Learning in Healthcare

Limitation	Impact
Privacy restrictions	Prevent hospitals from sharing sensitive patient data
Limited dataset diversity	Reduces generalization to different populations or imaging devices
Large data transfer requirements	Increases cost and delays in model development
Imbalanced condition frequency	Degrades performance on rare diseases or subtle abnormalities

Centralized machine learning, while powerful, therefore struggles to meet the scalability, diversity, and privacy needs of real-world medical AI deployment. These challenges motivate the development of more robust learning paradigms capable of overcoming data fragmentation and institutional isolation.

1.4 Suggested Solution

The challenges identified in the medical imaging domain—including subtle abnormalities, limited dataset diversity, privacy constraints, and variability in imaging standards—necessitate a more advanced diagnostic support system. The proposed project aims to address these limitations by integrating three complementary technologies: Computer Vision, Explainable AI, and Federated Learning. Together, these components form an intelligent system capable of assisting clinicians in identifying fractures, tumors, infections, and other abnormalities more reliably and transparently.

The core idea behind the project is to design a computer vision pipeline capable of analyzing X-ray and other medical imaging modalities with high accuracy. Deep learning models will detect patterns that may be difficult for clinicians to recognize quickly, especially in emergency or high-stress situations. By augmenting image interpretation with AI-based classification and region highlighting, the system serves as a powerful second-opinion tool, reducing oversight and improving diagnostic confidence.

To ensure that the system remains trustworthy and clinically interpretable, Explainable AI techniques are incorporated to reveal how the model arrives at its predictions. This transparency allows medical professionals to understand the rationale behind the AI's outputs, compare them with clinical knowledge, and detect any mistakes or biases early.

Moreover, Federated Learning is utilized to overcome the limitations of centralized training. Instead of pooling patient data into a single server—a process that is often

legally and ethically restricted—hospitals collaboratively train the model without exposing sensitive information. This approach significantly increases dataset diversity while maintaining strict compliance with privacy regulations.

In summary, the proposed solution seeks to:

- Improve diagnostic accuracy through advanced computer vision algorithms.
- Increase clinician trust via transparent and interpretable model outputs.
- Overcome data-sharing barriers by enabling collaborative, privacy-preserving training.

These three components work together to form a robust, scalable, and clinically meaningful AI-based diagnostic support framework.

Explainable AI (XAI)

Explainable Artificial Intelligence (XAI) refers to a set of methods and frameworks aimed at increasing the transparency, interpretability, and trustworthiness of machine learning models. As modern AI systems grow in complexity, their internal decision-making processes often become opaque, creating what is commonly known as the “black-box” problem. In safety-critical domains such as medicine, where clinical decisions directly affect patient outcomes, this lack of transparency is a major barrier to adoption. XAI methods attempt to address this challenge by providing human-understandable insights into how and why a model arrives at a specific prediction.

In medical imaging, explainability is particularly important due to the high stakes of diagnosis, the need for clinician trust, and the requirement for regulatory compliance. Rather than simply outputting a prediction, explainable models are capable of highlighting meaningful visual patterns, quantifying the contribution of specific features, or generating human-interpretable reasoning. These explanations serve as an additional layer of evidence that supports radiologists and clinicians during diagnostic workflows.

XAI supports clinical decision-making by:

- Identifying the precise image regions or features that contribute most strongly to the model’s prediction.
- Ensuring that the model’s focus aligns with clinically relevant structures rather than spurious artifacts.
- Facilitating the detection of potential model errors, such as overfitting, dataset bias, or misinterpretation of anatomical features.
- Enabling clinicians to verify, critique, or challenge AI-generated predictions using domain expertise.
- Improving trust and accountability by providing a transparent rationale that can be communicated in clinical reports.

A variety of XAI techniques are widely used in medical imaging research and practice. Visual explanation methods such as Class Activation Maps (CAM), Gradient-weighted CAM (Grad-CAM), and attention-based heatmaps highlight the spatial regions that

influence a model's output. Pixel-level saliency maps capture fine-grained sensitivity to image perturbations. Beyond visual methods, some approaches provide concept-based explanations, feature attribution scores, or even natural-language descriptions that mimic clinical reasoning. Together, these techniques help bridge the gap between the sophisticated internal representations learned by deep neural networks and the need for interpretable, clinically meaningful insight within medical environments.

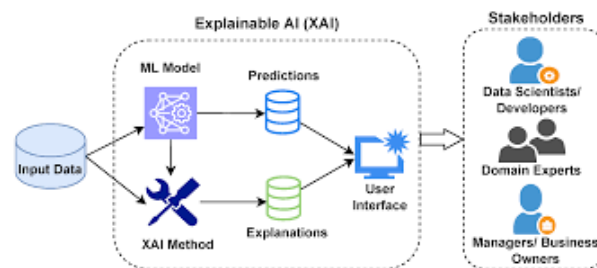


Figure 1.1: Explainable AI in action

Federated Learning

Federated Learning (FL) is a decentralized machine learning paradigm designed to enable collaborative model training without the need to share raw data between institutions. Unlike traditional centralized approaches, where data must be gathered into a single repository, federated learning allows each participating site to keep its data local while only exchanging model parameters or gradients. This architecture directly addresses major concerns in modern healthcare environments, including patient privacy, data security, legal constraints, and the challenges of aggregating large, distributed medical datasets.

In the medical field, data collected across hospitals, clinics, and imaging centers is often heterogeneous and siloed due to regulatory frameworks such as GDPR and HIPAA. Federated Learning provides a solution by ensuring that sensitive patient information never leaves its source institution. Instead, the global model is iteratively improved by aggregating updates computed locally at each site. This setup allows institutions to benefit from the collective data diversity and scale—key factors for building robust AI models—while preserving strict privacy and confidentiality guarantees.

Federated Learning benefits clinical AI development by:

- Allowing organizations to collaboratively train high-performing models without sharing **protected health information (PHI)**.
- Leveraging diverse datasets from multiple geographic regions, improving the model's generalizability across patient populations.
- Reducing the risk of data breaches, since sensitive information remains stored on secure local servers.
- Supporting compliance with ethical and legal requirements governing medical data usage.
- Enabling continuous model improvement as new data becomes available in real clinical environments.

Several FL strategies have been developed to support medical imaging applications. The most widely used is *Federated Averaging (FedAvg)*, where each institution trains a local version of the model and periodically sends its parameter updates to a central aggregator. More advanced techniques—such as personalized federated learning, secure aggregation, and differential privacy—address challenges like domain shifts between institutions, communication efficiency, and protection against inference attacks. In addition, hybrid approaches integrate federated learning with techniques such as transfer learning or multimodal fusion, enabling complex models to be trained across distributed datasets while retaining strong privacy guarantees.

By enabling collaborative model development across institutions without compromising data confidentiality, Federated Learning plays a crucial role in the advancement of trustworthy and scalable AI systems in healthcare.

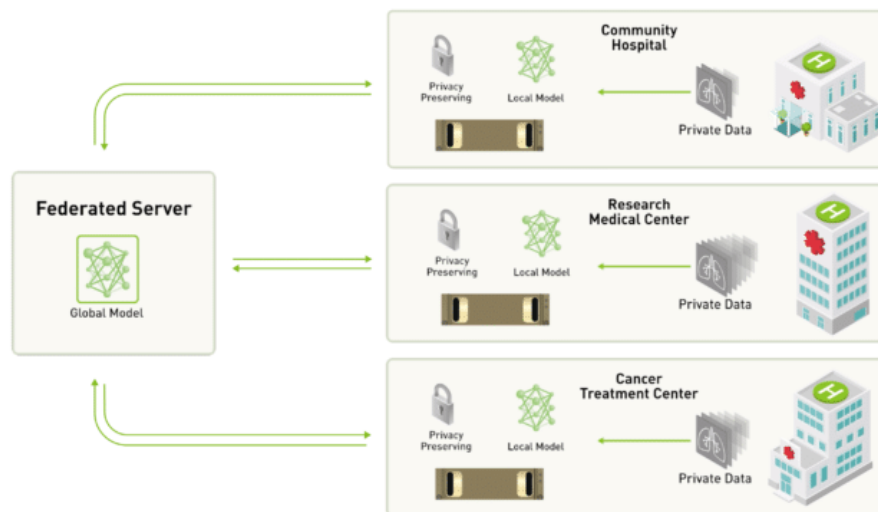


Figure 1.2: Centralized server approach to federated learning

1.5 Project Scope

The scope of this project centers on developing a comprehensive medical image analysis system capable of diagnosing two primary categories of conditions: bone fractures and chest tumors. These conditions were selected due to their prevalence in clinical settings, their potential for severe medical consequences if missed, and the proven effectiveness of computer vision techniques in detecting abnormalities within skeletal and thoracic imaging. The system aims to support clinicians by providing fast, reliable, and interpretable diagnostic assistance.

The core of the project focuses on designing and training deep learning models that can accurately analyze X-ray images. For chest diseases and tumors, the pipeline will identify abnormal masses or opacities that may indicate malignancies or other critical conditions. The emphasis is on achieving high sensitivity and specificity while ensuring that the model remains robust across diverse imaging conditions.

To enhance usability and ensure accessibility for non-technical users, the diagnostic engine will be wrapped within a web-based application. This interface will allow clinicians, technicians, and medical staff to upload images, visualize model predictions, and interact with diagnostic outputs without requiring specialized technical knowledge. Although the interface will play an important role in facilitating real-world use, the primary focus of the project remains on the development, training, and evaluation of the underlying diagnostic models.

The system will include the following high-level functionalities:

- Automated detection of bone fractures in X-ray images.
- Identification of suspicious chest masses or tumor-like abnormalities.
- Visualization of diagnostic results and highlighted regions of interest.
- A user-friendly web interface that integrates the underlying AI engine.

The project does *not* aim to replace professional medical judgment. Instead, it seeks to provide a reliable, efficient, and interpretable tool that supports clinicians in making faster and more accurate assessments. Future extensions—such as expanding the system to additional diseases, modalities, or institutions—fall outside the current scope but remain promising directions for continued development.

1.6 Project Objectives

The objectives of this project are centered around creating a reliable, accurate, and accessible diagnostic support system for bone-fracture and chest tumor identification. The project aims to ensure that the system not only performs well in terms of machine learning metrics but also provides practical value to clinicians through usability, efficiency, and cost-effectiveness. Achieving clinical-level performance requires a well-defined set of goals that align both with technical constraints and real-world medical needs.

A primary objective is to develop a computer vision model capable of achieving high diagnostic performance. In particular, the system is designed to reach an accuracy above 90% across its targeted conditions while maintaining a recall rate that minimizes the risk of missed fractures or tumors. Since false negatives can lead to severe clinical consequences, optimizing recall is essential for ensuring the safety and reliability of the system.

In addition to predictive performance, another major objective is to deliver an intuitive and engaging web interface that enables medical personnel to interact with the system easily. The interface should allow users to upload images, view results, and interpret highlighted regions without requiring technical expertise. Ease of use is critical for facilitating adoption in real clinical workflows, where time efficiency and clarity are important.

Operational efficiency is also a key objective. The system must be designed in a manner that minimizes computational overhead and reduces ongoing costs. This includes optimizing model size, using lightweight inference strategies where possible, and ensuring that the web application remains responsive even with limited hardware resources.

To summarize, the project seeks to meet the following objectives:

- Achieve a diagnostic accuracy exceeding 90% on bone-fracture and chest tumor detection tasks.
- Maintain a high recall rate to reduce the likelihood of missed abnormalities.
- Provide a user-friendly, accessible, and engaging web interface suitable for non-technical clinical staff.
- Develop an efficient system that minimizes operational and computational costs.

Collectively, these objectives define the criteria by which the success of the project will be evaluated and guide the design decisions throughout development.

1.7 Project Challenges

Despite the remarkable progress achieved in recent years, medical image analysis remains a challenging field. The complexity of medical data, the critical nature of clinical decision-making, and the variability of imaging procedures introduce numerous obstacles that must be addressed to develop reliable and clinically applicable systems. The following subsections discuss the major challenges associated with medical image analysis, particularly in the context of chest disease detection using X-ray images and lung cancer detection using CT scans.

Limited Availability of High-Quality Medical Data

One of the most significant challenges in medical image analysis is the limited availability of large, high-quality, and well-annotated datasets. Deep learning models generally require substantial amounts of labeled data to achieve high performance and robust generalization. However, obtaining medical images with accurate annotations is often difficult and expensive.

Medical image annotation requires the expertise of trained radiologists or medical specialists who must carefully examine each image and identify pathological findings. This process is time-consuming and resource-intensive. Furthermore, patient privacy regulations and ethical considerations often restrict the sharing of medical data between healthcare institutions, limiting the availability of publicly accessible datasets.

In the context of chest X-ray analysis, some diseases may occur infrequently, leading to severe class imbalance within datasets. Similarly, for lung cancer detection from CT scans, obtaining sufficiently large collections of confirmed malignant and benign cases can be challenging. These limitations can negatively affect model training and reduce diagnostic performance.

Variability in Image Acquisition

Medical images are acquired using different imaging devices, protocols, and settings across hospitals and healthcare facilities. As a result, images may exhibit significant varia-

tions in quality, resolution, contrast, brightness, noise levels, and anatomical positioning.

Chest X-ray images obtained from different institutions may vary due to differences in machine manufacturers, patient positioning techniques, exposure settings, and image preprocessing procedures. Likewise, CT scans may differ in slice thickness, reconstruction algorithms, scanning protocols, and radiation dose levels.

Such variability can lead to distribution shifts between training and deployment environments. A model trained on data from one institution may experience performance degradation when applied to images acquired in another clinical setting. Ensuring robustness across diverse imaging conditions remains a major challenge for medical image analysis systems.

Image Noise and Artifacts

Medical images frequently contain noise and artifacts that can obscure important anatomical structures and pathological findings. These imperfections may arise from patient movement, hardware limitations, acquisition errors, or environmental factors during image capture.

In chest X-ray imaging, artifacts such as medical devices, implants, external objects, and improper positioning can interfere with disease detection. CT scans may contain motion artifacts, beam-hardening effects, metal artifacts, and reconstruction-related distortions.

The presence of noise and artifacts can significantly reduce image quality and negatively affect the performance of automated detection systems. Consequently, effective preprocessing techniques, including image enhancement, denoising, and artifact removal, are often required before analysis.

Complexity of Disease Manifestations

Many thoracic diseases exhibit highly variable visual characteristics, making automated detection particularly challenging. Diseases may present differently depending on their severity, progression stage, patient demographics, and coexisting medical conditions.

For example, pneumonia, tuberculosis, pulmonary edema, and COVID-19 may produce overlapping radiographic features in chest X-rays. Similarly, lung nodules observed in CT scans can vary considerably in size, shape, texture, density, and location. Some malignant nodules closely resemble benign abnormalities, increasing the difficulty of accurate classification.

Additionally, certain diseases may manifest as subtle changes that are difficult to detect even for experienced radiologists. Developing models capable of recognizing both obvious and subtle pathological patterns remains a critical research challenge.

Class Imbalance in Medical Datasets

Medical datasets often suffer from class imbalance, where healthy cases significantly outnumber disease-positive cases or where some diseases occur much less frequently than others. This imbalance can lead machine learning models to favor majority classes during training.

For instance, in chest X-ray datasets, normal scans may constitute a large proportion of the available data, while rare diseases may only be represented by a small number of samples. Similarly, in lung cancer screening datasets, malignant nodules may be considerably less common than benign findings.

Class imbalance can result in poor sensitivity for detecting rare but clinically important conditions. Addressing this issue often requires specialized techniques such as data augmentation, oversampling, synthetic data generation, weighted loss functions, and balanced training strategies.

Interpretability and Explainability

Healthcare professionals require transparency and trust when using artificial intelligence systems in clinical practice. Many state-of-the-art deep learning models function as black-box systems, producing predictions without providing clear explanations for their decisions.

In medical diagnosis, explainability is particularly important because clinicians must understand the reasoning behind automated recommendations before incorporating them into patient care decisions. A prediction without justification may not be sufficient in high-stakes medical environments.

To address this challenge, researchers employ explainable AI techniques such as saliency maps, Grad-CAM visualizations, attention mechanisms, and feature attribution methods. These approaches help identify image regions that contribute most strongly to model predictions, thereby improving transparency and clinical confidence.

Generalization and Robustness

A medical image analysis system must perform reliably across different patient populations, imaging devices, healthcare institutions, and geographical regions. However, models trained on specific datasets may inadvertently learn dataset-specific characteristics rather than clinically meaningful features.

This issue can lead to reduced performance when models encounter unseen data during real-world deployment. Factors such as demographic differences, variations in disease prevalence, and differences in imaging protocols can all affect model generalization.

Ensuring robustness requires diverse training datasets, rigorous validation procedures, external testing across multiple institutions, and continuous performance monitoring after deployment.

Three-Dimensional Nature of CT Data

Unlike chest X-rays, which provide two-dimensional projections, CT scans generate three-dimensional volumetric data. While this additional information improves diagnostic capabilities, it also introduces substantial computational and analytical challenges.

CT volumes may contain hundreds of slices for a single patient examination. Processing such large datasets requires considerable memory, storage capacity, and computational resources. Furthermore, accurately identifying lung nodules often requires analyzing spatial relationships across multiple slices.

Advanced techniques such as 3D convolutional neural networks, volumetric segmentation algorithms, and multi-view learning approaches are commonly employed to address these challenges. However, these methods generally demand greater computational complexity and longer training times.

Accurate Segmentation of Anatomical Structures

Segmentation is a fundamental step in many medical image analysis pipelines. It involves identifying and isolating relevant anatomical structures such as lungs, airways, lesions, tumors, and nodules.

Accurate segmentation is particularly challenging when boundaries are unclear, tissues overlap, or pathological regions exhibit irregular shapes. In chest X-ray images, overlapping anatomical structures can complicate lung segmentation. In CT scans, tumors may have poorly defined margins or be attached to surrounding tissues.

Segmentation errors can propagate through subsequent analysis stages and adversely affect disease detection and classification performance. Therefore, robust segmentation algorithms remain an important area of ongoing research.

Clinical Validation and Regulatory Requirements

Developing an accurate machine learning model is only one step toward clinical adoption. Medical image analysis systems must undergo extensive validation to demonstrate safety, reliability, and effectiveness in real-world healthcare environments.

Clinical validation typically involves testing models on large and diverse patient populations, comparing performance against expert radiologists, and assessing their impact on clinical workflows. Furthermore, healthcare technologies are subject to strict regulatory requirements and quality standards before deployment.

Meeting these regulatory and clinical validation requirements can be a lengthy and resource-intensive process, but it is essential for ensuring patient safety and maintaining trust in AI-assisted diagnostic systems.

Ethical and Privacy Considerations

Medical image analysis systems operate on highly sensitive patient information. Protecting patient privacy and ensuring compliance with healthcare regulations are critical responsibilities throughout the development lifecycle.

Issues such as data security, informed consent, data ownership, and algorithmic bias must be carefully addressed. Models trained on non-representative datasets may exhibit reduced performance for certain demographic groups, potentially leading to inequitable healthcare outcomes.

Consequently, ethical AI development requires responsible data management practices, fairness assessments, transparency, and continuous evaluation to ensure that diagnostic systems benefit all patient populations.

Computational Resource Requirements

Modern deep learning architectures used in medical image analysis often contain millions of trainable parameters and require substantial computational resources for training and inference. High-resolution chest X-ray images and volumetric CT scans further increase computational demands.

Training advanced models frequently requires powerful Graphics Processing Units (GPUs), large memory capacities, and extensive training times. These requirements may limit accessibility for smaller healthcare institutions and research groups with limited computational infrastructure.

Optimizing model efficiency while maintaining diagnostic accuracy remains an important challenge, particularly for deployment in resource-constrained clinical environments.

1.8 Summary

This chapter introduced the motivation and context behind the development of an intelligent medical image analysis system designed to support clinicians in diagnosing bone fractures and chest tumors. The rapid growth of medical imaging data, combined with the increasing complexity and subtlety of diagnostic tasks, has created a pressing need for automated tools capable of enhancing accuracy and reducing oversight. Traditional interpretation methods are challenged by factors such as faint abnormalities, variations in imaging quality, and the diverse presentation of conditions across patients and imaging devices.

To address these challenges, the chapter examined the rise of computer vision in healthcare and its effectiveness in automating image-based diagnostic tasks. While machine learning has demonstrated considerable promise in detecting complex patterns within medical images, centralized approaches face several limitations including restricted data sharing, lack of dataset diversity, and substantial computational burdens. These constraints hinder the performance and generalizability of AI models, especially in a

domain where reliability is critical.

The chapter also presented the rationale for the proposed solution, which integrates advanced computer vision techniques with Explainable AI and Federated Learning. This combination aims to improve diagnostic accuracy, enhance interpretability, and overcome data privacy barriers that limit large-scale collaborative model training. By incorporating interpretability mechanisms, the system ensures that clinicians can understand and verify model decisions, while federated learning enables multi-institutional training without compromising patient confidentiality.

Furthermore, the scope and objectives of the project were clearly defined. The system aims to achieve high accuracy and recall in detecting bone fractures and chest tumors, provide a user-friendly interface suitable for non-technical users, and maintain operational efficiency to minimize computational costs. The chapter also outlined the challenges likely to arise during development, including data limitations, model robustness, explanation quality, system efficiency, and the complexities associated with implementing federated learning securely.

In summary, the introduction established the need for an intelligent, interpretable, and privacy-preserving diagnostic platform and laid the foundation for the system proposed in this project. The subsequent chapters build upon this foundation by detailing the methodologies, design choices, experiments, and evaluations that guide the development of the final system.

CHAPTER 2

LITERATURE REVIEW

Chapter 2 Literature Review

In this chapter, a literature review is provided in project-related areas. To build the framework (X-Fed NeuroScan: Federated Learning with Explainable AI for Medical Image Classification), we need to get familiar with foundations of medical image analysis, deep learning-based classification techniques, and the challenges associated with handling sensitive clinical data. We also examine the principles and methodologies of Federated Learning (FL) as a privacy-preserving approach for multi-institutional model training, as well as Explainable AI (XAI) techniques that enhance model transparency and clinical trust. Additionally, this chapter reviews existing commercial medical imaging applications, related academic research, and publicly available datasets. Finally, we evaluate the tools, platforms, and frameworks that support the development of deep learning models, federated learning workflows, and explainable AI components.

The chapter is organized as follows:

- Section 2.1: Medical image history and challenges, with subsections on bone fractures and chest diseases.
- Section 2.2: Explainable AI (XAI) in medical image detection.
- Section 2.3: Federated Learning (FL) in medical image detection.
- Section 2.4: Combined approaches integrating detection with XAI and FL.
- Section 2.5: Summary of related papers.
- Section 2.6: Similar commercial applications.
- Section 2.7: Summary of the chapter.

2.1 Medical Image Diagnosis

Medical imaging is the cornerstone of modern diagnosis, providing non-invasive visualization of the body's interior using techniques like X-rays, Computed Tomography (CT), and Ultrasound. Due to the high volume and complexity of medical scans, interpretation is often time-consuming and subject to human variability. This process is increasingly enhanced by **Artificial Intelligence (AI)**, particularly **Computer Vision (CV)** and Deep Learning (DL) methods.

While traditional Deep Learning (DL) approaches have achieved high accuracy in radiological analysis for disease detection and bone fracture localization, their clinical deployment is significantly hindered. The primary issues stem from two critical needs: the strict requirement for **data privacy** and the **black-box nature** of AI decision-making. These barriers prevent the widespread, trustworthy adoption of powerful AI models in healthcare settings.

Consequently, the research focus has shifted toward specialized solutions: **Federated**

Learning (FL), a privacy-preserving paradigm, and **Explainable AI (XAI)**, a necessary component for model transparency. This work provides a systematic review of the state-of-the-art in these domains. It begins by examining advanced deep learning architectures, analyzing the role of FL in data collaboration, and reviewing XAI techniques.

We will first detail the related work in chest X-ray diagnosis and bone fracture detection. Subsequently, it will explore the use of XAI and FL specifically within these two application areas, identifying the gap that this research aims to bridge.

2.1.1 Lung Cancer Detection

Lung cancer is one of the most prevalent and life-threatening malignancies worldwide and remains a leading cause of cancer-related mortality. According to global cancer statistics, lung cancer accounts for a significant proportion of newly diagnosed cancer cases each year and is responsible for more deaths than several other major cancers combined. The high mortality rate associated with lung cancer is largely attributed to the fact that the disease is frequently diagnosed at advanced stages, when treatment options become more limited and patient prognosis deteriorates considerably. Consequently, early detection has become one of the most important objectives in modern thoracic oncology, as timely diagnosis can substantially improve treatment outcomes and survival rates.

Lung cancer originates from the uncontrolled proliferation of abnormal cells within lung tissues. Over time, these abnormal cells can form tumors that interfere with normal respiratory function and may eventually spread to other parts of the body through a process known as metastasis. The disease can affect various regions of the lungs, including the bronchi, bronchioles, and alveolar structures. Depending on the cellular origin and pathological characteristics, lung cancer is generally classified into two major categories: Small Cell Lung Cancer (SCLC) and Non-Small Cell Lung Cancer (NSCLC).

Non-Small Cell Lung Cancer represents approximately 85% of all diagnosed lung cancer cases and includes several subtypes such as adenocarcinoma, squamous cell carcinoma, and large cell carcinoma. Adenocarcinoma is the most common subtype and frequently develops in the peripheral regions of the lungs. Squamous cell carcinoma often arises near the central airways and is strongly associated with smoking history. Large cell carcinoma is less common but tends to exhibit aggressive growth characteristics. Small Cell Lung Cancer, although less prevalent, is generally more aggressive and is characterized by rapid growth and early metastasis, making timely diagnosis particularly important.

The development of lung cancer is associated with multiple risk factors. Cigarette smoking remains the most significant risk factor and is responsible for the majority of lung cancer cases worldwide. Tobacco smoke contains numerous carcinogenic compounds capable of inducing genetic mutations that disrupt normal cellular regulation. However, lung cancer can also occur in non-smokers due to factors such as exposure to secondhand smoke, environmental pollutants, occupational hazards, radon gas, asbestos exposure, genetic predisposition, and chronic pulmonary diseases. The multifactorial nature of lung cancer highlights the importance of comprehensive screening and diagnostic strategies that can identify disease manifestations at their earliest stages.

One of the major challenges associated with lung cancer is that many patients remain

asymptomatic during the early stages of disease progression. Tumors may develop and grow gradually without producing noticeable symptoms, allowing the disease to advance undetected. As a result, a substantial proportion of patients receive a diagnosis only after the cancer has reached a locally advanced or metastatic stage. This clinical characteristic underscores the importance of imaging-based screening programs and computer-aided diagnostic systems capable of identifying subtle radiological abnormalities before the onset of severe symptoms.

As the disease progresses, patients may begin to exhibit a variety of respiratory and systemic manifestations. One of the most commonly reported symptoms is a persistent cough that does not resolve over time or exhibits a noticeable change in character. Patients with chronic coughs may experience increased frequency, severity, or duration of symptoms compared to their baseline condition. In many cases, persistent coughing serves as one of the earliest clinical indicators prompting further medical investigation.

Another important symptom is hemoptysis, which refers to the presence of blood in sputum or respiratory secretions. Hemoptysis may occur when the tumor invades nearby blood vessels or causes irritation within the respiratory tract. Although not all patients experience this symptom, its presence often warrants immediate clinical evaluation due to its association with serious pulmonary conditions, including malignancy.

Shortness of breath, medically referred to as dyspnea, is also frequently observed among lung cancer patients. Dyspnea may result from airway obstruction caused by tumor growth, reduced lung capacity, pleural effusion, or impaired gas exchange within affected pulmonary tissues. As tumors increase in size, respiratory function may progressively decline, leading to noticeable breathing difficulties during physical activity or even at rest.

Chest pain represents another common manifestation of lung cancer, particularly when tumors extend into surrounding structures such as the pleura, chest wall, or mediastinum. Patients often describe the pain as persistent, localized, and exacerbated by deep breathing, coughing, or physical movement. The severity of chest pain may increase as the disease progresses and adjacent tissues become involved.

In addition to respiratory symptoms, patients frequently experience systemic manifestations that reflect the broader physiological impact of malignancy. Unexplained weight loss is one of the most significant warning signs and may occur even in the absence of dietary changes or increased physical activity. Cancer-related metabolic alterations often contribute to progressive weight reduction and muscle wasting. Similarly, persistent fatigue and generalized weakness are common among patients and can substantially impair quality of life and daily functioning.

Advanced lung cancer may also produce symptoms associated with metastatic spread. For example, metastasis to the skeletal system may cause bone pain and pathological fractures, while brain metastases can lead to neurological symptoms such as headaches, dizziness, seizures, or cognitive impairment. Liver involvement may result in jaundice and abdominal discomfort. The presence of such symptoms often indicates advanced disease and is generally associated with a less favorable prognosis.

Figure 2.1 summarizes several common clinical manifestations associated with lung

cancer.

Placeholder for Figure: Common Symptoms and Signs of Lung Cancer

Suggested illustration showing a human body diagram highlighting: persistent cough, hemoptysis, chest pain, dyspnea, fatigue, unexplained weight loss, hoarseness, and metastatic symptoms.

Figure 2.1: Illustration of common symptoms and clinical signs associated with lung cancer.

From a diagnostic perspective, medical imaging plays a central role in lung cancer detection, characterization, staging, and treatment planning. Imaging techniques enable clinicians to visualize pulmonary abnormalities that may not be detectable through physical examination alone. Among the available imaging modalities, chest radiography and computed tomography are the most commonly utilized for initial evaluation and diagnostic assessment.

Chest X-ray imaging is frequently used as a first-line diagnostic tool due to its accessibility, low cost, and widespread availability. Radiographic findings suggestive of lung cancer may include pulmonary nodules, masses, hilar enlargement, pleural effusion, or areas of abnormal opacity. However, chest radiographs possess limited sensitivity for detecting small lesions and early-stage tumors. Consequently, subtle abnormalities may remain undetected, particularly when located behind anatomical structures such as ribs or the mediastinum.

Computed Tomography (CT) has become the gold standard imaging modality for lung cancer assessment because of its ability to generate high-resolution cross-sectional images of thoracic anatomy. CT scans provide significantly greater anatomical detail than conventional radiographs and enable the visualization of small pulmonary nodules, irregular lesion margins, tissue density variations, and regional lymph node involvement. Furthermore, CT imaging facilitates volumetric analysis and supports accurate tumor localization, measurement, and staging.

Radiologists typically evaluate several imaging characteristics when assessing potential lung malignancies. These characteristics include lesion size, shape, margin irregularity, internal texture, density patterns, calcification, spiculation, and growth behavior over time. Malignant nodules often exhibit irregular or spiculated borders, heterogeneous density patterns, and progressive enlargement across sequential imaging studies. The interpretation of these imaging features requires considerable expertise and experience, motivating the development of computer-aided diagnostic systems capable of assisting clinicians in image analysis.

Figure 2.2 illustrates representative examples of CT findings commonly associated

with lung cancer.

Placeholder for Figure: CT Imaging Examples of Lung Cancer

Suggested image panel containing: (a) Normal chest CT scan, (b) Benign pulmonary nodule, (c) Malignant pulmonary nodule, (d) Advanced lung tumor with surrounding tissue involvement. Lesions should be annotated with arrows or bounding markers.

Figure 2.2: Representative CT imaging findings associated with lung cancer.

Recent advances in artificial intelligence and deep learning have significantly transformed the field of medical image analysis. Convolutional Neural Networks (CNNs) and related deep learning architectures have demonstrated remarkable capabilities in automatically extracting discriminative imaging features directly from raw medical images. In the context of lung cancer detection, deep learning models can learn complex patterns associated with pulmonary nodules, tumor morphology, and malignancy-related characteristics without requiring handcrafted feature engineering.

The application of deep learning to lung cancer diagnosis offers several advantages. Automated systems can process large volumes of imaging data efficiently, assist radiologists in identifying subtle abnormalities, reduce inter-observer variability, and potentially improve diagnostic consistency. Furthermore, modern explainability techniques enable visualization of image regions contributing to model predictions, thereby enhancing transparency and supporting clinical decision-making.

Despite these advances, several challenges remain. Lung lesions exhibit substantial variability in size, shape, texture, and anatomical location. Additionally, imaging protocols may differ across institutions, resulting in variations in image quality and acquisition parameters. Addressing these challenges requires robust model design, diverse training datasets, and rigorous validation procedures. Nevertheless, artificial intelligence continues to represent a promising direction for improving early lung cancer detection and supporting healthcare professionals in clinical practice.

In summary, lung cancer is a highly prevalent and potentially fatal disease whose prognosis depends heavily on early diagnosis. Because symptoms often emerge only after significant disease progression, imaging-based screening and diagnostic methods play a crucial role in patient management. Computed tomography has become the preferred modality for lung cancer detection due to its superior anatomical detail and sensitivity to early-stage abnormalities. Recent developments in deep learning have further enhanced the ability of computer-aided diagnostic systems to identify malignant findings, making AI-assisted lung cancer detection an increasingly important area of research within medical

image analysis.

2.1.2 Chest Diseases Detection

The chest, encompassing the thoracic cavity, houses vital organs such as the lungs, heart, and major blood vessels, making it susceptible to various respiratory diseases that can be detected through imaging techniques like chest X-rays (CXR). CXR is a non-invasive, cost-effective diagnostic tool widely used for identifying abnormalities in lung tissue. We focus on four key categories for multiclassification: pneumonia, tuberculosis (TB), COVID-19, and normal (healthy) cases.

- **Pneumonia** is an inflammatory condition of the lung alveoli, often caused by bacterial, viral, or fungal infections, leading to fluid-filled sacs visible as opacities on CXR.
- **Tuberculosis** is a bacterial infection primarily caused by *Mycobacterium tuberculosis*, characterized by granulomas, cavities, or nodular infiltrates in the lungs, which can be chronic and contagious.
- **COVID-19**, caused by the SARS-CoV-2 virus, presents with ground-glass opacities, consolidations, and bilateral lung involvement on CXR, often progressing rapidly.
- **Normal cases** exhibit clear lung fields without pathological signs, serving as a baseline for comparison.

Accurate detection of these diseases via deep learning (DL) models on CXR images is crucial for timely intervention, especially in resource-limited settings [1] [2].

2.1.3 Target Diseases and Their Radiological Characteristics

This study focuses on three major respiratory diseases that can be detected through chest X-ray imaging: pneumonia, COVID-19, and tuberculosis. These diseases represent significant public health concerns worldwide and are among the most commonly investigated conditions in computer-aided diagnostic systems. Although all three diseases affect the respiratory system, each exhibits distinct pathological characteristics and radiographic features that can assist clinicians in diagnosis. Understanding their symptoms and appearance in chest X-ray images is essential for accurate interpretation and effective disease detection.

Pneumonia

Pneumonia is an inflammatory condition of the lungs that primarily affects the alveoli, the small air sacs responsible for gas exchange. The disease is commonly caused by bacterial, viral, or fungal infections, leading to the accumulation of fluid or pus within the affected lung tissue. Pneumonia remains one of the leading causes of respiratory-related morbidity and mortality, particularly among young children, elderly individuals, and immunocompromised patients.

Common symptoms of pneumonia include:

- Persistent cough, often accompanied by mucus production.
- Fever and chills.
- Shortness of breath.
- Chest pain, especially during breathing or coughing.
- Fatigue and general weakness.
- Rapid breathing in severe cases.

In chest X-ray images, pneumonia typically appears as areas of increased opacity, commonly referred to as consolidations or infiltrates. These opacities occur due to the accumulation of inflammatory fluids within the alveoli, reducing the normal air content of the lungs. Depending on the severity and type of infection, the abnormalities may be localized to a single lobe (lobar pneumonia) or spread across multiple regions of the lungs. The affected areas generally appear whiter than healthy lung tissue, which normally appears dark due to the presence of air.

COVID-19

Coronavirus Disease 2019 (COVID-19) is a highly contagious respiratory illness caused by the SARS-CoV-2 virus. Since its emergence, COVID-19 has posed substantial challenges to healthcare systems worldwide due to its rapid transmission and potentially severe respiratory complications.

The clinical presentation of COVID-19 varies considerably among patients, ranging from asymptomatic infections to severe respiratory failure. Common symptoms include:

- Fever.
- Dry cough.
- Fatigue.
- Shortness of breath.
- Loss of taste or smell.
- Sore throat.
- Muscle and body aches.
- Headache.

Chest X-ray findings associated with COVID-19 often include bilateral lung abnormalities, particularly in the peripheral and lower lung regions. The disease commonly manifests as patchy or diffuse opacities that may affect both lungs simultaneously. As the infection progresses, these opacities may increase in size and density, resulting in widespread pulmonary involvement. Severe cases can demonstrate extensive bilateral infiltrates that significantly reduce the visibility of normal lung structures. Although chest X-rays are useful for evaluating disease progression, early-stage COVID-19 infections may not always produce clearly visible abnormalities.

Tuberculosis

Tuberculosis (TB) is a chronic infectious disease caused by the bacterium *Mycobacterium tuberculosis*. The disease primarily affects the lungs, although it can also spread to other organs. Tuberculosis remains a major global health concern, particularly in regions with limited healthcare resources.

The symptoms of pulmonary tuberculosis often develop gradually and may include:

- Persistent cough lasting several weeks.
- Coughing up blood or blood-stained sputum.
- Chest pain.
- Fever.
- Night sweats.
- Unexplained weight loss.
- Fatigue and weakness.
- Loss of appetite.

Tuberculosis exhibits a wide range of radiographic appearances depending on the stage and severity of infection. In chest X-ray images, the disease frequently affects the upper lung regions and may appear as patchy infiltrates, nodular lesions, or areas of consolidation. Advanced cases can produce cavitary lesions, which are hollow spaces formed by tissue destruction within the lungs. Fibrosis and calcified granulomas may also be observed in chronic or healed infections. The diverse presentation of tuberculosis can sometimes complicate diagnosis, as certain imaging features overlap with those of other pulmonary diseases.

Comparative Radiological Characteristics

Although pneumonia, COVID-19, and tuberculosis all affect the respiratory system and produce abnormalities in chest X-ray images, their radiographic patterns often differ. Pneumonia commonly presents as localized consolidations caused by fluid-filled alveoli, whereas COVID-19 typically produces bilateral and diffuse opacities that frequently involve the peripheral lung regions. Tuberculosis often demonstrates upper-lung involvement, nodular infiltrates, and cavitary lesions, particularly in advanced stages of the disease. Recognizing these characteristic imaging patterns is crucial for differentiating between conditions and supporting accurate clinical diagnosis.

2.1.4 Related Work in Lung Cancer Classification

The rapid advancement of medical imaging technologies and artificial intelligence techniques has led to significant progress in the field of automated lung cancer classification. Over the past decade, researchers have increasingly explored the application of machine learning and deep learning methods to assist radiologists in detecting and classifying lung cancer from medical images, particularly computed tomography (CT) scans. The growing availability of publicly accessible datasets, coupled with improvements in computational resources and neural network architectures, has accelerated the development of computer-

aided diagnostic systems capable of identifying malignant pulmonary abnormalities with high accuracy.

Traditional approaches to lung cancer classification primarily relied on handcrafted feature extraction techniques. In these systems, radiological characteristics such as shape, texture, intensity distribution, edge information, and morphological properties of pulmonary nodules were manually engineered and subsequently provided as input to machine learning classifiers. Algorithms such as Support Vector Machines (SVMs), Random Forests, k-Nearest Neighbors (k-NN), and Artificial Neural Networks (ANNs) were widely employed to distinguish between benign and malignant lesions. Although these methods demonstrated promising results, their performance was often constrained by the quality and representativeness of the manually designed features.

The emergence of deep learning fundamentally transformed the landscape of lung cancer classification research. Convolutional Neural Networks (CNNs) enabled the automatic extraction of hierarchical image features directly from raw imaging data, eliminating the need for extensive manual feature engineering. By learning discriminative representations from large-scale datasets, deep learning models demonstrated superior performance compared to many traditional machine learning approaches. Consequently, an increasing number of studies began investigating the use of CNN-based architectures for pulmonary nodule detection, malignancy classification, tumor segmentation, and cancer staging.

In addition to standard convolutional architectures, researchers have explored various advanced techniques to improve classification performance and clinical applicability. These include transfer learning from large natural image datasets, ensemble learning strategies, attention mechanisms, three-dimensional convolutional networks, hybrid machine learning frameworks, and explainable artificial intelligence methods. Such approaches aim not only to improve diagnostic accuracy but also to enhance model robustness, interpretability, and generalization across diverse patient populations and imaging protocols.

Despite the considerable progress achieved in recent years, lung cancer classification remains a challenging research problem. Pulmonary nodules exhibit substantial variability in size, shape, texture, and anatomical location, while imaging datasets often suffer from class imbalance, limited sample sizes, and variations in acquisition parameters. These challenges continue to motivate the development of more sophisticated algorithms capable of delivering reliable and clinically meaningful predictions.

The following subsection reviews a selection of significant studies from the literature, highlighting the methodologies employed, datasets utilized, performance metrics achieved, and key contributions made toward the advancement of automated lung cancer classification systems. Through this review, the strengths and limitations of existing approaches can be identified, thereby providing a foundation for understanding the motivation and design decisions underlying the proposed framework.

- Artificial Intelligence for lung cancer detection using CT scan images aims to improve early and accurate classification of pulmonary nodules while reducing dependency on manual radiological assessment. To address limitations in traditional machine learning and standalone deep learning models, the paper proposes a hybrid Deep Learning–Support Vector Machine (DL-SVM) framework, where a deep learning

model is used to extract high-level features from CT scan images and a Support Vector Machine (SVM) is used for final classification between benign and malignant nodules. The system is evaluated on the LIDC-IDRI dataset, which contains annotated lung CT scans widely used for benchmarking lung cancer detection methods. Experimental results show that the proposed DL-SVM model achieves an accuracy of approximately 94%, outperforming several comparative baseline approaches including traditional machine learning models (such as classical SVM variants and statistical classifiers) and standalone deep learning methods, which achieved lower performance in comparison. The results confirm that the hybrid combination of deep feature extraction and SVM classification improves diagnostic accuracy, robustness, and generalization, making the model more reliable for assisting radiologists in early lung cancer detection. [3]

- Artificial Intelligence-based lung cancer detection systems aim to improve early and accurate diagnosis from CT scan images by automatically classifying lung nodules as benign or malignant. However, conventional deep learning models often suffer from performance degradation due to high-dimensional feature spaces and suboptimal optimization of CNN parameters. To address these limitations, the paper proposes a hybrid framework that combines a deep convolutional neural network (CNN) with the Ebola Optimization Search Algorithm (EOSA) for optimized weight and bias selection during training. The CNN is used to extract deep features from CT images, while EOSA is applied to optimize the model parameters and improve classification performance. The system is evaluated on the IQ-OTH/NCCD lung cancer CT scan dataset. Experimental results show that the proposed EOSA-CNN model achieves a classification accuracy of approximately 93.21%, outperforming other comparative methods including GA-CNN (91%), WOA-CNN (90%), MVO-CNN (89%), SBO-CNN (88%), and standard CNN models with lower accuracy. Additionally, the model demonstrates improved sensitivity and specificity across benign and malignant classes, confirming that the use of metaheuristic optimization significantly enhances CNN performance. The study concludes that integrating deep learning with Ebola Optimization Search Algorithm provides a more accurate and robust solution for lung cancer detection from CT images, improving diagnostic reliability and supporting clinical decision-making. [4]
- The study aims to improve the accuracy and robustness of classifying lung cancer subtypes from CT scan images by combining radiomic analysis with deep learning methods. However, single-source approaches often suffer from limited feature representation and reduced generalization across different imaging conditions. To address these limitations, the paper proposes a hybrid framework that integrates a Deep Convolutional Neural Network (DeepCNN) with attention mechanisms and radiomic feature extraction to enhance lung cancer subtype classification. The model extracts radiomic features such as texture, shape, and statistical descriptors using PyRadiomics, while the DeepCNN learns deep spatial representations from CT images. An attention mechanism is incorporated within the CNN architecture to focus on the most diagnostically relevant tumor regions, improving feature relevance and reducing noise from irrelevant areas. The extracted radiomic and deep features are combined and evaluated using machine learning classifiers across a dataset of 2,725 CT images from multiple healthcare centers covering different lung cancer subtypes. Experimental results show that the proposed hybrid model achieves a classification

accuracy of approximately 92.47%, with an AUC of 93.99% and sensitivity of 92.11%, outperforming traditional machine learning models and standalone deep learning approaches. Additional experiments show that feature fusion significantly improves performance compared to using radiomic or deep features alone. The study concludes that combining attention-based deep learning with radiomic feature engineering provides a more reliable and reproducible framework for lung cancer subtype classification, improving diagnostic consistency and clinical decision support. [5]

- Artificial Intelligence-based lung cancer detection aims to improve early and accurate diagnosis by classifying CT scan images into benign and malignant cases. However, traditional deep learning methods often suffer from redundant features, overfitting, and weak generalization on diverse medical datasets. To address these issues, the paper proposes a radiomics-based machine learning framework that extracts handcrafted CT image features such as texture, shape, and intensity, followed by feature selection to keep only the most relevant attributes. These optimized features are then used to train machine learning classifiers for lung cancer detection. The model is tested on public lung CT datasets and compared with multiple baseline approaches using different feature selection and classification combinations. The proposed method achieves F1-score (95.24%), AUC (98.00%), specificity (93.17%), accuracy (94.40%), precision (95.80%), and sensitivity (94.69%). with improved sensitivity and specificity, and consistently outperforms standard radiomics and baseline machine learning models in lung cancer classification performance. [6]

2.1.5 Related Work in Chest Diseases Multi-class Classification

Interpreting Chest X-rays (CXRs) is vital for diagnosing various thoracic diseases like pneumonia and tuberculosis. Due to the subtlety of findings and high image volume, **Deep Learning (DL)** models have been widely adopted to automate diagnosis. These models, trained on large datasets, primarily focus on **multi classification** of different chest diseases. The following research examples showcase the success of various **Convolutional Neural Network (CNN)** architectures in achieving high accuracy in CXR analysis.

- The researchers Develop an automated Deep Neural Network (CNN) to classify Pneumonia, COVID-19, Tuberculosis, and No-findings from chest X-rays, addressing limitations in individual disease detection and small/biased datasets in prior studies. they Created a custom CNN architecture with 5 convolutional blocks (16-256 channels, ReLU activation, max-pooling), trained on split data (80% training, 20% testing). They Addressed heterogeneous data sources by preprocessing images (resizing to 300x300, denoising, normalization, filtering) to create a coherent dataset of 21,581 images from multiple Kaggle sources they achieve accuracy 98.72% (using validation metrics) , Outperformed SOTA schemes like Venkataramana et al. (96.6%) and Bhandari et al. (94.54%), The study proved that a large diverse dataset is essential to achieve high multiclass accuracy and prevent overfitting, though class imbalance slightly reduced COVID-19 performance (96.27% recall) [7].
- The researchers Develop a custom CNN model with soft attention to classify chest

X-ray images into normal, COVID-19, pneumonia, and tuberculosis, while analyzing misclassifications through handcrafted features and statistical methods to improve clinical interpretability and diagnostic reliability. , Utilized transfer learning with pre-trained CNN models (VGG-19, ResNet-50, EfficientNet, DenseNet), trained on a balanced dataset split into training (70%), validation (15%), and testing (15%) sets. They Preprocessed images via normalization, data augmentation (rotation, flipping, zooming), and resizing; applied SMOTE at the feature level to balance the originally imbalanced dataset from heterogeneous sources Best Modelis ResNet-50 with Accuracy: 97% Compared to VGG-19 (95%), DenseNet (91%), EfficientNet (94%) [8].

- This study proposed a multi-level classification model using CNNs to differentiate COVID-19 from Tuberculosis (TB) and other Pneumonias based on CXR images. The primary challenge was addressing the severe data imbalance, specifically the scarcity of COVID-19 samples. The novelty involved combining data augmentation with the SMOTE technique to artificially balance the minority class. The system employed a two-stage process: first, classifying TB versus general Pneumonia, and second, sub-classifying the Pneumonia cases into COVID-19 and viral/bacterial types. This multi-level approach was designed to significantly reduce misdiagnosis between these similar lung diseases. The model achieved a high accuracy of 97.4% for the TB/Pneumonia stage. The second stage achieved 88% accuracy for bacterial/viral/COVID-19 classifications, demonstrating an overall 8–10% improvement over existing methods [9].
- Additionally, a unified DL framework like CXR-MultiTaskNet used ResNet50 for joint disease localization and classification, incorporating thoracic diseases like pneumonia and TB, with an F1-score of 96.5% through shared feature extraction [10].
- Another deep-learning pipeline classified chest X-ray images into normal, pneumonia, TB, and COVID-19 using a sequential CNN approach. Preprocessing steps such as grayscale conversion and normalization improved robustness, while transfer learning enhanced performance on limited data. VGG-16 achieved 95.9%, DenseNet-161 reached 98.9% for distinguishing COVID-19 from pneumonia, and ResNet-18 obtained 76% for COVID-19 severity classification. The study demonstrates the effectiveness of multi-stage CNN models for automated COVID-19 screening from CXR images[11].
- A knowledge-distillation deep-learning model classified CXR images into COVID-19, pneumonia, TB, and normal using the RDD4 dataset. A deep CNN teacher network transferred its knowledge to a lightweight student model built in Keras/TensorFlow, preserving accuracy while reducing complexity. Data imbalance was addressed through augmentation. The distilled student model achieved 96.08% accuracy, with high precision and recall, showing strong suitability for real-time, low-resource deployment[12].
- They develop a custom CNN model with soft attention to classify imbalanced chest X-ray images into normal, COVID-19, pneumonia, and tuberculosis, the approach Designed COV-X-net19, a 16-layer CNN architecture (4 conv layers, 6 max-pooling, soft attention, dropout, dense with SoftMax), trained on split data

(70% training, 10% validation, 20% testing) with ablation studies for hyperparameter optimization (ReLU, Nadam optimizer, LR=0.001, batch=32), the technique used preprocessed images via text removal (OCR+inpaint), morphological opening (3x3 kernel), CLAHE (ClipLimit=5.0, TileGridSize=3x3), histogram equalization. Best Model: COV-X-net19 with accuracy: 95.18% Compared to 8 transfer learning models (e.g., MobileNet at 82.96%, VGG16 at 62.46%), base CNN (89.96%), and SOTA studies with similar multi-class setups (e.g., accuracies below 95% in imbalanced scenarios) [13].

- A CNN-based system classified chest X-ray images into COVID-19, pneumonia, TB, and normal using over 21,000 images from three public datasets. The model used five convolutional blocks with preprocessing steps such as resizing, denoising, and normalization to handle heterogeneous sources. It achieved a high validation accuracy of 98.72%, with strong per-class performance across all categories. The study shows that a carefully designed CNN can effectively support joint diagnosis of major chest diseases from CXR images [14].

2.2 Explainable AI (XAI) in Medical Image Diagnosis

Explainable AI (XAI) refers to techniques that make the decision-making process of AI models transparent and interpretable, addressing the "black-box" nature of DL models by providing insights into why a prediction was made. In this project, XAI is integrated using methods like Gradient-weighted Class Activation Mapping (GradCAM), which generates heatmaps highlighting regions in medical images (e.g., fracture lines in bone X-rays or opacities in CXR) that influenced the model's classification. This linkage enhances trust in diagnostics for both bone fractures and chest diseases, allowing clinicians to verify AI outputs against radiological features, reducing misdiagnosis risks, and facilitating model debugging.

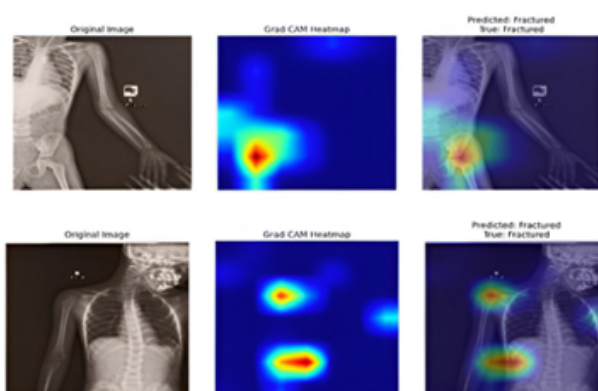


Figure 2.3: Heatmap for bone

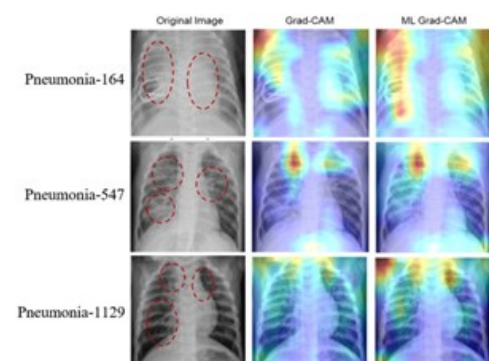


Figure 2.4: Heatmap for pneumonia detection

Related Papers on XAI For Lung Cancer Classification

- This paper introduces LCxNet, a novel, custom-designed convolutional neural network (CNN) framework for the early and accurate computer-aided diagnosis of lung cancer using CT scans, which leverages Explainable AI (XAI) techniques such as Grad-CAM for decision visualization and t-SNE for feature space analysis to ensure clinical transparency. By training and evaluating the model on the IQ-OTH/NCCD lung CT dataset—comprising benign, malignant, and normal classes—and comparing learning behaviors across five optimizers (SGD, RMSProp, Adam, AdamW, and NAdam), the researchers demonstrated that LCxNet outperforms state-of-the-art models, sensitivity of 99.39% is paramount, ensuring that the risk of missing a malignant tumor (false negative) is minimized, which is a top priority in clinical diagnostics. Coupled with its high accuracy (99.39%), precision (99.40%), F1-score (99.40%), and low generalization loss (0.17), AdamW offers the most robust and clinically reliable performance for aiding in the early and accurate detection of lung cancer. While Adam and NAdam are also excellent choices, AdamW's slight edge in specificity (99.89%) and Cohen's Kappa (99.01%). [15]
- The study introduces a deep learning framework that utilizes a customized VGG16 convolutional neural network combined with Grad-CAM for the autonomous and transparent classification of lung cancer from CT scans. By training and evaluating the model on the IQ-OTH/NCCD dataset—which consists of 1190 CT images categorized into normal, benign, and malignant classes—the researchers applied data augmentation and transfer learning to achieve an impressive overall classification accuracy of 96%. The model demonstrated perfect 100% accuracy and specificity for healthy cases, alongside strong 98% precision and 97% recall for identifying malignant tumors, though it exhibited a slightly lower precision of 81% for benign cases due to overlapping features with malignant lesions. To bridge the gap between complex AI predictions and clinical trust, the study successfully employed Grad-CAM to generate visual heatmaps that accurately highlight the critical tissue regions influencing the model's decisions, ultimately providing radiologists with a highly effective, accurate, and interpretable decision-support tool for early lung cancer diagnosis. [16]

Related Papers on XAI For Chest Disease Multi-class Classification

- This research introduces an Explainable AI (XAI) framework for the automatic detection, interpretation, and explanation of pulmonary diseases from chest radiographs (CXRs). The framework uses a transfer learning approach with a fine-tuned ResNet50 Convolutional Neural Network (CNN) trained on the COVID-CT and COVIDNet datasets. The Local Interpretable Model-agnostic Explanations (LIME) technique is employed to visually highlight the key regions in the CXR images that contribute most to the prediction. The developed system demonstrated strong performance, achieving high classification accuracy, reaching up to 97% on the evaluated datasets. Validation confirmed that the model accurately focuses on clinically important features, thus assisting radiologists in early diagnosis and clinical decision-making[17].

- They develop XAI-TRANS, using inception-based transfer learning to classify CXR images for COVID-19, bacterial/viral pneumonia, and tuberculosis, addressing overlapping radiological features and black-box AI limitations by integrating improved U-Net segmentation and XAI tools the approach used is Utilized transfer learning with four pre-trained models (ResNet50, VGG19, VGG16, Inceptionv3), combined with an improved U-Net for lung segmentation to extract features, followed by XAI-TRANS for classification and explanation using Grad-CAM visualizations Technique Preprocessed images via resizing to 256*256 and normalization between 0 and 1fer learning models (e.g., MobileNet at 82.96%, VGG16 at 62.46%), base CNN (89.96%), and SOTA studies with similar multi-class setups (e.g., accuracies below 95% in imbalanced scenarios) the Best Model is XAI-TRANS (with and Grad-CAM) with accuracy: 97% (overall classification accuracy) and 98% precision, outperforming traditional DL methods by 4.75% through XAI integration [18].
- A survey reviewed over 200 papers on XAI in DL-based medical image analysis, emphasizing visual explanations for chest radiography to build clinician confidence[19].
- Additionally, an XAI-integrated DL framework for multiclass lung disease classification, targeting five classes: viral pneumonia, bacterial pneumonia, COVID-19, tuberculosis, and normal. It evaluates various DL architectures, including CNNs, hybrids, ensembles, transformers, and Big Transfer, with hyperparameter tuning and stratified 5-fold cross-validation to ensure robust performance. The Xception model, enhanced with transfer learning, emerges as the top performer, achieving 96.21% accuracy by focusing on discriminative features in lung region[20].
- They built a lightweight CNN (5 convolutional layers) to classify chest X-rays into four classes: COVID-19, Pneumonia, TB, Normal. They applied Grad-CAM, LIME, and SHAP to generate visual explanations of model decisions, validating them with radiologists. The dataset was 7,132 public CXR images from three sources, with class imbalance handled by stratified batching. The model achieved a 10-fold cross-validation average test accuracy of $94.31\% \pm 1.01\%$, and they reported strong precision, recall, and F1[21].

2.3 Federated Learning

Federated Learning (FL) is a decentralized ML approach where models are trained collaboratively across multiple devices or institutions without sharing raw data, preserving privacy through local updates aggregated centrally. In medical imaging like bone X-rays and CXR classification, FL is used to train on distributed datasets from hospitals, mitigating data silos and complying with regulations like GDPR or HIPAA. It involves clients training on local data, sending model weights to a server for aggregation (e.g., via FedAvg), and is quickly implemented for real-time applications, reducing communication overhead while handling heterogeneous data.

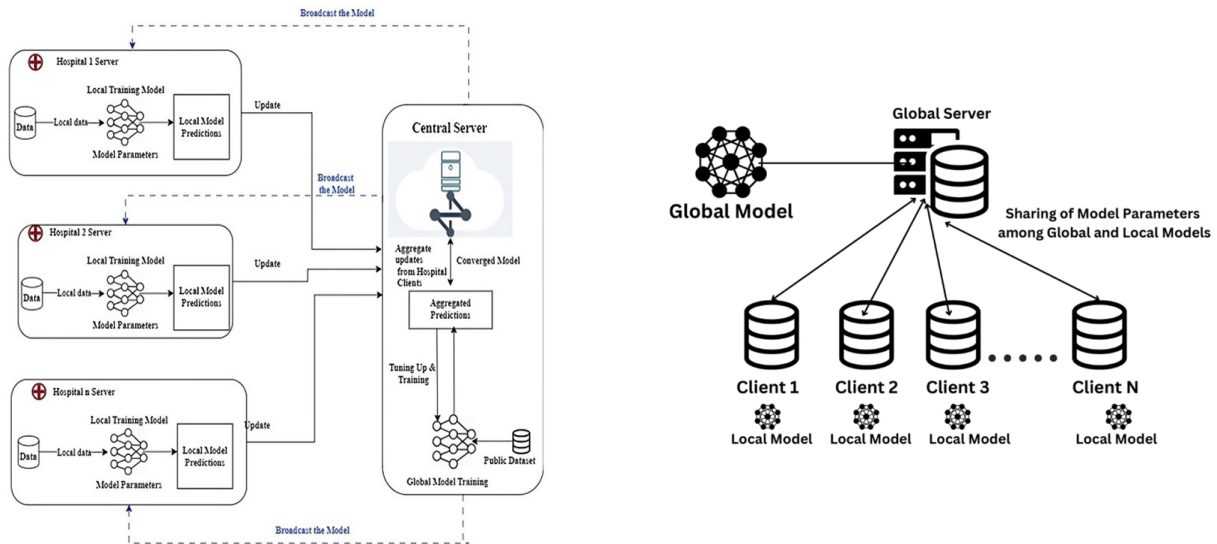


Figure 2.5: FL-Based efficient framework for smart healthcare sector

Related Papers on FL For Lung Cancer Classification

- This paper introduces an ensemble Federated Learning (FL) approach to address the limitations of centralized data processing and single machine learning models in the early detection and multi-order classification of lung cancer. By transitioning from a traditional client-server architecture to a decentralized federated network, the proposed system customizes local neural networks to enable faster, broader training across diverse datasets while strictly preserving patient data privacy and security through techniques like federated averaging and differential privacy. Utilizing a Kaggle cancer dataset, the distributed framework focuses on collective decision-making to reliably classify lung cancer cases into normal, benign, and malignant categories. Experimental evaluations demonstrate that this decentralized approach effectively mitigates the computational load overheads and data accumulation anomalies associated with centralized servers, ultimately achieving a classification accuracy of 89.63% and offering improved generalization and diagnostic reliability compared to traditional standalone machine learning models. [22]
- The study presents a privacy-preserving Federated Learning (FL) framework designed to classify lung cancer severity from pulmonary CT scans into benign, malignant, and normal categories without centralizing sensitive patient data. By utilizing datasets from the IQ-OTH/NCCD collection distributed across different simulated clients, the study evaluates the performance of the CatBoost algorithm alongside the deep learning Inception V3 architecture using both transfer learning and fine-tuning techniques. After applying crucial preprocessing steps, including median blur filtering for noise removal and BGR-to-RGB color space transformation, the experiments revealed that the fine-tuned Inception V3 model significantly outperformed the CatBoost model, particularly in managing complex medical imaging data and consistently detecting malignant cases. Ultimately, the proposed decentralized approach successfully achieved an accuracy of 89.0% using the fine-tuned Inception V3 model, demonstrating that federated learning can effectively bridge the gap between building robust, highly accurate collaborative diagnostic tools and ensuring

strict compliance with healthcare data privacy regulations such as HIPAA and GDPR. [23]

- This paper introduces FBCLC-Rad, a novel framework designed to securely and accurately detect and categorize lung cancer from chest CT images by integrating deep learning, blockchain technology, and Federated Learning (FL). To overcome the privacy and trust challenges associated with sharing sensitive medical data across global healthcare institutions, the system employs blockchain for data authentication and FL to collaboratively train a global model without requiring data to be centralized. The methodology utilizes Capsule Networks (CapsNets) for local classification and leverages Transfer Learning (TL) to maximize diagnostic accuracy while significantly reducing training times, alongside a specific data normalization approach to manage the variability of CT scans from different scanners. Evaluated extensively using Python libraries on datasets including the Cancer Imaging Archive (CIA), Kaggle Data Science Bowl (KDSB), LUNA 16, and a custom local dataset, the decentralized FBCLC-Rad technique achieved an exceptional 99.69% accuracy with minimal categorization error. Ultimately, this privacy-preserving approach provides medical professionals with a fast, highly reliable, and automated diagnostic tool to manage the growing volume of lung cancer patients globally while ensuring strict organizational anonymity. [24]
- This paper introduces "FVCM-Net," a novel, privacy-preserving deep learning framework that integrates federated learning, attention mechanisms, and ensemble learning to achieve highly accurate and interpretable lung cancer detection from CT scan images. To overcome the challenges of data scarcity and sensitive patient data security, the model utilizes federated learning to enable collaborative training across decentralized medical institutions without directly sharing records, while incorporating a Convolutional Block Attention Module (CBAM) with a VGG-16 architecture to extract crucial imaging features. Evaluated on diverse datasets, including LIDC-IDRI and IQ-OTH/NCCD, the decentralized approach utilizes a majority voting ensemble strategy to further enhance diagnostic robustness, achieving an exceptional average accuracy of 98.26% and an F1-score of 97.37%. To ensure clinical transparency and build trust in the AI's diagnostic logic, the framework seamlessly employs Explainable AI (XAI) techniques, specifically SHAP and High-Resolution Class Activation Mapping (HiResCAM), effectively visualizing the critical regions influencing the model's predictions and providing radiologists with a secure, highly reliable, and interpretable decision-support tool. [25]
- This research paper presents a comprehensive deep learning approach for detecting lung nodules in medical CT and PET-CT images to improve early lung cancer diagnosis. The authors developed a preprocessing pipeline that filters and segments CT images to isolate regions of interest (ROI), then trained and evaluated seven different deep learning architectures—CNN, DCNN, 3DCNN, VGG19, ResNet18, Inception V1, and Inception-ResNet—on a dataset of 220 cases containing 560 nodules from Harbin's imaging data center. Among all tested models, VGG19 demonstrated superior performance, achieving 98.65% accuracy with high precision (98.80%), sensitivity (98.55%), specificity (98.70%), and F1-score (98.60%), while maintaining the lowest false positive rate (1.05%), making it the most reliable model for automated lung nodule detection despite being trained on a limited dataset. [26]

- This paper proposes a deep learning framework for segmentation and classification of lung cancer histopathological images to improve early detection and diagnosis. The approach uses U-Net for segmentation of lung cancer histopathological images to identify regions of interest, followed by classification models including ResNet-50, VGG-16, EfficientNet-B5, and an Ensemble model of these three to categorize images into three types: lung adenocarcinoma, squamous cell carcinoma, and normal lung tissue. Trained on 15,000 histopathological images from the LC15000 dataset (9000 training, 3000 validation, 3000 testing), the U-Net segmentation model achieved 81% accuracy, while the classification models progressively improved performance—ResNet-50 achieved 91% accuracy, VGG-16 94%, and EfficientNet-B5 97%. Notably, the Ensemble model combining all three classifiers significantly outperformed individual models, achieving 99% accuracy with perfect precision, recall, and F1-score metrics (0.99), demonstrating the effectiveness of ensemble approaches for automated lung cancer histopathological image analysis. [27]
- This paper presents MCAT-Net, a novel lung nodule segmentation model based on the transformer architecture with multiple thresholds and coordinate attention designed to address limitations in capturing edge information, semantic features, and long-range dependencies. The model integrates three key components: (1) a Multi-threshold Feature Separation (MFS) module that extracts edge and texture features at different intensity levels using four optimal thresholds (T7, T6, T5, T4) to enrich input features without excessive computational cost, (2) a Coordinate Attention (CA) mechanism that preserves spatial position information and enables cross-regional feature integration by encoding features in horizontal and vertical dimensions, and (3) a Transformer module applied at the fourth skip connection to capture global long-range dependencies and enhance feature integration. Evaluated on the LIDC-IDRI dataset (888 CT images with 1186 lung nodules) and LNDdb dataset (294 CT images), MCAT-Net achieved Dice Similarity Coefficient (DSC) values of 88.29% and 78.51% respectively, with sensitivities of 86.33% and 75.05%, significantly outperforming traditional segmentation methods (U-Net, U-Net++, SegNet, nn-UNet) while demonstrating superior performance on complex nodule types including ground-glass nodules, partially solid nodules, and wall-adjacent nodules. [28]
- This paper proposes SSLKD-UNet, a semi-supervised learning and knowledge distillation model for lung nodule segmentation that addresses the challenge of obtaining large-scale, high-quality annotated medical datasets. The model employs a hybrid CNN-Transformer architecture that combines a U-Net encoder-decoder with a Swin-Transformer in the bottleneck layer to capture both local spatial details and global context information. The training employs a novel teacher-student framework where a teacher model (UNet, ResUNet, AttUNet, SwinUNet, or UTransformer) is trained on coarse-grained annotations to generate pseudo-labels for unlabeled CT images, and a student model (SSLKD-UNet) learns from both pseudo-labeled and fine-grained annotated data. Evaluated on the publicly available LUNA16 dataset (888 CT scans with 1186 nodules) and a private LC183 dataset (183 lung cancer patients from Dalian Medical University), the model demonstrates superior segmentation performance compared to baseline models including UNet, ResUNet, and AttUNet, effectively handling varying nodule sizes and complex anatomical structures. The approach successfully reduces reliance on expensive fine-grained

annotations by leveraging coarse annotations and large amounts of unlabeled data through the teacher-student training paradigm. [29]

- This paper presents YOLOV11-EAR, an improved YOLO-based pulmonary nodule detection framework that addresses limitations in capturing spatial context and small object localization. The approach introduces two primary innovations: (1) a Spatial Squeeze-and-Excitation (SSE) module that combines channel-wise and spatial attention mechanisms to enhance discriminative features for nodules by processing global context and local spatial information in parallel, and (2) an Explicit Aspect Ratio Penalty IoU (EAPIoU) loss function that directly penalizes squared differences in aspect ratios between predicted and ground-truth bounding boxes to refine localization, particularly for small and irregularly shaped nodules. Evaluated on three diverse datasets—LUNA16 (888 CT scans with 1186 nodules), NODE21 (4882 chest X-rays with 1476 annotated nodules), and LungCT (1152 CT slices)—the YOLOV11-EAR framework consistently outperforms baseline YOLO variants and mainstream detectors (Faster R-CNN, RetinaNet), achieving precision of 0.781 with mAP@50 of 75.5% on LUNA16, mAP@50 of 92.4% on LungCT, and mAP@50 of 89.1% on NODE21, while maintaining computational efficiency with only 2.5M parameters and 6.3 GFLOPs.[30]
- This paper proposes SwiF-YOLO, an improved lung nodule detection method based on the YOLOx framework that addresses challenges in detecting small, morphologically similar nodules in CT images. The model incorporates three key improvements: (1) replacing the YOLOx-M backbone network with a Swin Transformer to leverage its hierarchical feature extraction and linear computational complexity through shifted window-based self-attention mechanisms, (2) adopting Adaptively Spatial Feature Fusion (ASFF) to dynamically weight and combine multi-scale features for improved scale invariance and reduction of inconsistent information across feature pyramid levels, and (3) replacing the standard IoU loss function with Generalized IoU (GIoU) loss to better handle cases where bounding boxes do not overlap and to provide more informative gradient signals for bounding box regression. Trained and evaluated on the LUNA16 dataset (888 CT scans with 1186 annotated nodules, split 70% training, 15% validation, 15% testing), SwiF-YOLO achieves precision of 84.54%, recall of 81.37%, and mAP of 83.17%, outperforming baseline YOLOx-M (83.78% precision, 79.78% recall, 81.77% mAP) and comparing favorably to other mainstream detection methods including Faster R-CNN, Mask R-CNN, YOLOv5, YOLOv7, and SSD. [31]
- This paper proposes a novel hybrid deep learning model combining CNN and YOLOv8 for automated real-time detection of lung nodules from CT images to address challenges in early lung cancer diagnosis. The approach employs a two-stage framework: (1) in the first stage, a 2D Convolutional Neural Network is trained on preprocessed DICOM images from the LUNA16 dataset after extensive image processing (including Hounsfield Unit scaling, min-max normalization, unsharp masking, median filtering, Laplacian filtering, and morphological operations) to detect nodule regions and achieve 94% training accuracy over 150 epochs, and (2) in the second stage, the detected nodule coordinates are combined with expert radiologist annotations and fed into YOLOv8 (tested across five scale variants: YOLOv8n, YOLOv8s, YOLOv8m, YOLOv8l, YOLOv8x) for fine-tuning and real-time detection.

Evaluated on the LUNA16 dataset (888 CT scans generating 551,065 PNG images), the proposed model achieves an overall accuracy of 92.3%, sensitivity of 88.5%, and mean average precision (mAP) of 53.5%, with YOLOv8m configuration demonstrating superior performance (mAP50: 53.5%, mAP50-95: 40.0%), outperforming several comparable approaches in the literature while maintaining computational efficiency. [32]

- This paper proposes a novel lung nodule detection method based on YOLOv3 that addresses the limitations of traditional convolution neural networks in capturing rich feature information and the inaccurate bounding box positioning of standard YOLO approaches. The method introduces two primary innovations: (1) integration of Inception ResBlocks into the YOLOv3 feature extraction network (creating an "Inception-DarkNet" architecture with 44 convolutional layers and 4 Inception ResBlocks) to extract multi-scale features more precisely through asymmetric convolution kernels while reducing overfitting and enhancing non-linear expression capability, and (2) a novel Generalized Distance and Intersection over Union (GDIoU) loss function that combines the GIoU metric with the Euclidean distance between bounding box centers to provide more sensitive and accurate bounding box regression, particularly when boxes do not overlap. The model uses K-means clustering to determine 9 optimal anchor sets tailored to the lung nodule dataset and is trained on a combination of the public LUNA16 dataset (888 CT scans with 1186 nodules) and 80 collected patient CT images (total 1266 nodules), with preprocessing including Hounsfield Unit scaling, morphological operations, and lung parenchyma extraction. The proposed method achieves 83.5% Average Precision and 92.6% sensitivity with a detection time of 31.3ms per image, outperforming original YOLOv3 (78.6% AP, 88.3% sensitivity) and competing favorably with Faster R-CNN variants while maintaining faster inference speed. [33]
- This paper proposes AutoLungDx, a comprehensive end-to-end hybrid deep learning framework for early lung cancer diagnosis through automated segmentation, detection, and classification of lung nodules from CT images. The framework integrates three complementary architectures: (1) a novel 3D Res-U-Net for lung parenchyma segmentation that incorporates residual connections to improve feature flow and gradient propagation, particularly in challenging juxta-pleural regions, (2) YOLOv5s for efficient slice-level nodule detection on segmented lung tissue that reduces background noise and false positives, and (3) a Vision Transformer (ViT) for malignancy classification that leverages self-attention mechanisms to capture global context and long-range dependencies. The framework was developed and evaluated on the LUNA16 dataset (888 CT scans with 1,186 nodules $\geq 3\text{mm}$, augmented with LIDC-IDRI malignancy annotations), with comprehensive preprocessing including Hounsfield Unit scaling (-1200 to 600 HU), normalization, binarization, resampling to 1mm isotropic resolution, and resizing to $256 \times 256 \times 256$ voxels. The integrated system achieves state-of-the-art performance across all stages: 98.82% Dice score for lung segmentation (exceeding prior methods including R2U-Net's 98.80%), 0.76 mAP@50 for nodule detection with competitive efficiency, and 96.29% accuracy for malignancy classification (4.25% improvement over prior SOTA), with clinically relevant attention maps demonstrating focus on lesion regions and excellent interpretability through ROC-AUC of 0.989. [34]

Related Papers on FL For Chest X-ray Disease Multi-class Classification

- This study addresses the primary challenge in applying Federated Learning to chest X-ray analysis, which is the significant non-IID data distribution between adult and pediatric cohorts from different institutions. Analyzing over 400,000 radiographs, the research showed that conventional FL struggles to maintain consistent performance, particularly with the challenging pediatric dataset. The authors proposed enhancing FL by integrating general-purpose Self-Supervised Learning representations, using the DINOv2 model as a robust initialization. The new SSL-equipped framework proved highly effective, achieving an improvement exceeding 20% in the AUC metric for pneumonia diagnosis in the difficult pediatric cases. This validates that readily available SSL models provide a practical and scalable solution for enabling FL in diverse healthcare settings while preserving data privacy [35].
- This study focuses on using federated learning (FL) for the accurate and privacy-preserving diagnosis of pediatric pneumonia via chest X-ray classification. Traditional machine learning approaches face significant challenges related to centralizing sensitive medical data, which risks patient privacy and security. FL addresses this by enabling collaborative model training across different institutions while ensuring the raw patient data remains local and secure. The researchers specifically tackled the common problem of data heterogeneity, which typically compromises classification accuracy in real-world clinical settings. They proposed an improved FL method utilizing "two-end control variables," modifying the loss function on the client side and adding momentum on the server. This novel approach successfully enhanced performance, achieving an average accuracy of 98% and demonstrating an approximate 2% improvement over existing federated learning algorithms. [35].
- Artificial Intelligence for lung disease detection is limited by major privacy concerns due to the need to centralize sensitive patient data (X-ray and CT scans). To overcome this, the paper proposes a decentralized Federated Transfer Learning (FTL) system, which enables effective collaborative model training without ever pooling raw data. This approach leverages federated learning's privacy features and the efficiency of transfer learning to train models on limited local datasets. The study evaluated four methods using the ResNet-50 model, confirming FTL as the most effective method for high accuracy. Specifically, the resulting accuracies were: Centralized (87.95%), Transfer (88.04%), Federated (87.55%), and FTL (89.96%) [36].
- A federated learning framework distinguished COVID 19 from four other chest diseases (lung cancer, TB, pneumothorax, pneumonia) plus normal using chest X rays from multiple hospitals. They used three pre trained models (DenseNet 169, VGG 16, VGG 19) and applied SMOTE to mitigate class imbalance. The aggregated system (using DenseNet 169) achieved 98.45% accuracy, Grad CAM was used for model explainability, and their decision making client selection strategy reduced communication overhead significantly [37].

2.4 Combined Approaches: Medical Image Diagnosis with XAI and Federated Learning

Integrating XAI and FL in medical image diagnosis combines privacy with interpretability, enabling collaborative training on bone X-rays and CXR while explaining predictions, with potential for broader medical domains. This section details key papers that merge these elements.

Related Papers on Combined Approaches for Chest Diseases Detection with XAI and Federated Learning

- A Federated Learning Based Real Time Ensemble Model with Explainable AI Framework for an Efficient Diagnosis of Lung Diseases. It is FLEM-XAI, a privacy-preserving FL framework with XAI for real-time lung disease diagnosis from CXR, using FedAvg for decentralized training across hospitals without data sharing. It ensembles pre-trained models (InceptionV3, Conv2D, VGG16, ResNet-50) trained locally, aggregated centrally, and incorporates XAI via SHAP for feature importance and GradCAM for heatmaps highlighting affected lung areas. Results show ResNet-50 achieving 95.5% overall accuracy (97.72% COVID-19, 95.42% pneumonia, 93.17% TB), outperforming centralized models in bandwidth efficiency and maintaining convergence [38].
- This study introduces FLPneXAINet, a framework that securely predicts pneumonia from Chest X-ray (CXR) images by integrating Federated Learning (FL), Deep Learning (DL), and Explainable AI (XAI) to overcome data privacy issues. The method utilized an 8,402-image Kaggle dataset, which was augmented using a CycleGAN network, and then features were extracted via pre-trained DL models and four Ensemble DL (EDL) models. Feature optimization was performed using RFE, ANOVA, and RF to select the most relevant features, which were then classified by various machine learning models in an FL environment. The EDL network achieved the best results in the FL environment, with a high accuracy of 97.61%, and was validated using XAI techniques (LIME and Grad-CAM). The full performance metrics were: F1 score 98.36%, recall 98.13%, and precision 98.59%, offering a robust solution for healthcare professionals [39].
- A secure lung cancer prediction used mapreduce blockchain FL with XAI, ensuring interpretability in federated settings [40].

2.5 Summary of Related Papers

Table 2.1: Summary of related papers.

Focus Area	Sub-category	Model	Dataset	Result	Ref. Num.
Bones	General	Custom CNN and transfer learning	FracAtlas	The custom CNN achieved superior performance with 95.96% accuracy, significantly outperforming the transfer learning models (EfficientNetB0, MobileNetV2, ResNet50) which only managed between 65% and 67% accuracy.	[1]
Bones	General	YOLOv11 was developed and benchmarked against Faster R-CNN and SSD.	—	YOLOv11 achieved superior performance with 96.8% mAP and 95.14% accuracy, significantly outperforming Faster R-CNN (87.5% mAP) and SSD (82.9% mAP) in both real-time fracture detection and localization.	[1]
Bones	General	YOLOv11 was developed and benchmarked against Faster R-CNN and SSD.	—	YOLOv11 achieved superior performance with 96.8% mAP and 95.14% accuracy, significantly outperforming Faster R-CNN (87.5% mAP) and SSD (82.9% mAP) in both real-time fracture detection and localization.	[1]
Bones	General	EfficientNet-B6	FracAtlas	The EfficientNet-B6 model achieved a 96.83% testing accuracy	[3]
Bones	General	Custom CNN	—	The model achieved a classification accuracy of 92.44%	[4]
Bones	General	SVM, k-NN, Naïve Bayes	—	Support Vector Machine (SVM) classifier achieved the highest accuracy, with a classification rate of 98%.	[5]
Bones	General	Custom CNN		Accuracy: 89.865% (Approx. 90%), Specificity: 89.90%	[6]
Bones	General	Custom CNN	MURA dataset	Testing Accuracy: 99.79% (at epoch 13).	[7]

Bones	XAI	MobileNetV3 style convolutional blocks with self-attention layers.	NEU Surface Defect Database, MVTec AD (Anomaly Detection), Severstal Steel Defect Dataset	proposed EdgeViT-Slim achieved 96.0% accuracy, 0.953 F1 score, and 85 FPS on industrial defect datasets, outperforming baselines with far fewer parameters	[18]
Bones	XAI	ResNet-50, Faster R-CNN, and Mask R-CNN with Grad-CAM explainability	MURA	achieving 83% classification accuracy (ResNet-50)	[19]
Bones	XAI	Random Forest, SVM, and KNN	MURA	Random Forest classifier achieved the best performance with 95% accuracy, 93% precision, and 0.96 F1-score, outperforming SVM (91%) and KNN (88%)	[20]
Bones	XAI	Logistic Regression, Random Forest, XGBoost, AdaBoost, Light-GBM, and Gradient Boosting	custom multi-center clinical osteoporosis dataset (1,250 records)	ensemble model achieved the best performance with 90.82% accuracy, 100% precision, 81.63% recall, and 89.99% F1-score	[21]

Bones	XAI	(CNN, VGG, DenseNet) and found that Ensemble-2 (combining DenseNet121 and DenseNet169)	MURA dataset	The Ensemble-2 model achieved the highest accuracy of 85% and an F1 score of 0.79. Additionally, the integration of Explainable AI (Grad-CAM and Saliency Maps)	[22]
Bones	XAI	YOLO, Faster R-CNN,	——	YOLOv5m (Best Performer) Achieved a Pointing Game Accuracy of > 85%, meaning Grad-CAM heatmaps consistently aligned with the ground truth fracture locations.	[23]
Bones	FL	DenseNet161		Federated Learning (FedAvg): The 5-node FedAvg model achieved the best performance with 91% accuracy. Centralized Training: The centralized DenseNet161 model achieved 89.6% accuracy.	[29]
Bone	FL	Local ML Models (Client-side), Federated Learning Frameworks (Server-side)	—	Highest accuracy of 94.4% (FL+VGG16 with augmentation) for binary classification.	[30]
Bones	FL	EfficientNet V2 (implemented within an Adaptive Federated Learning framework)	—	federated methods lagged slightly behind the 96–98% accuracy achieving near-centralized performance of up to 98.6% using EfficientNetV2	[31]

Bones	Combined	SemFedXAI (a framework combining Semantic Web technologies and Federated Learning)	—	The SemFedXAI framework achieved the highest predictive accuracy at 85.3%, outperforming standard Federated Learning (73.5%) and other baselines. It also demonstrated superior explainability, scoring 0.85 in feature identification accuracy	[36]
Chest	General	Custom CNN	Kaggle & IEEE	The model achieved 98.72% overall accuracy across all classes. Recall per class: Pneumonia 99.66%, No findings 99.35%, Tuberculosis 98.10%, COVID-19 96.27%.	[10]
Chest	General		Kaggle	Accuracy: 97% (using Resnet-50) VGG-19 (95%), DenseNet (91%), EfficientNet (94%)	[11]
Chest	General	Custom CNN	Kaggle	TB vs Pneumonia classification: 97.4% accuracy; COVID-19 vs viral/bacterial pneumonia vs (other pneumonia types) classification: 88.0% accuracy	[12]
Chest	General	Custom CNN	VinDr-CXR	Classification Accuracy = 96.8% (from micro-F1 $\approx$ 0.968) Localization Accuracy $\approx$ 85% (IoU $\approx$ 0.851)	[13]
Chest	General	VGG16, DenseNet-161 and ResNet-18	Custom clinical dataset collected and labeled	Normal/Pneumonia/TB/COVID-19 classification (via VGG-16): 95.9% . Normal/Pneumonia/COVID-19 classification (via DenseNet-161): 98.9% . COVID-19 severity classification (via ResNet-18): 76.0% .	[14]
Chest	General	Custom CNN	Kaggle	97% average accuracy (with precision $\approx$ 94% and recall $\approx$ 97%) for the full classification task.	[15]

Chest	General	Custom CNN and VGG-16	Kaggle & public dataset from Github	The COV-X-net19 achieves 95.18% , MobileNet : 82.96%, VGG16 : 62.46% and base CNN : 89.96% .	[16]
Chest	General	DenseNet201, ResNet152V2	IEEE	The model achieved superior performance with a test accuracy of 98.87%, outperforming the individual base models (DenseNet201 and ResNet152V2) and demonstrating robust multi-class classification capabilities.	[17]
Chest	XAI	Explainable Framework based on DenseNet-201	–	The proposed explainable DenseNet-201 model achieved a high classification accuracy of 98.08%, while also providing visual explanations (Grad-CAM) to build trust and interpretability for clinical diagnosis.	[24]
Chest	XAI	CNN and Vision Transformer	Kaggle	XAI-TRANS achieves accuracy: 97% and 98% precision	[25]
Chest	XAI	Review of various Deep Learning models and XAI techniques (e.g., Grad-CAM, LIME)	Survey of medical image analysis studies		[26]
Chest	XAI	Custom CNN with Grad-CAM and LIME	–	The proposed Custom CNN model achieved a high classification accuracy of 98.41%, with XAI techniques successfully highlighting the critical image regions used for diagnosis, thereby improving model interpretability.	[27]

Chest	XAI	Ensemble of DenseNet201, Inception-ResNetV2, Xception with Grad-CAM++	Kaggle & IEEE	The proposed deep learning ensemble achieved a superior average classification accuracy of 98.97%, with XAI visualizations providing clinically plausible explanations for the model's decisions across all disease classes.	[28]
Chest	FL	Federated Learning (FedAvg) with a ResNet-based backbone and a novel age de-biasing loss function.	Multi-institutional chest X-ray datasets	The proposed de-biased FL model achieved an AUC of 89.7%, outperforming the standard FL model (AUC of 86.2%) and demonstrating significantly improved fairness and accuracy across different age groups.	[32]
Chest	FL	Federated Averaging (FedAvg) algorithm with a DenseNet-121 architecture as the core CNN model.	Chest X-ray images from multiple clients	The federated DenseNet-121 model achieved a test accuracy of 96.7% and an F1-score of 95.8%, matching the performance of a model trained on centralized data while preserving privacy.	[33]
Chest	FL	Federated Transfer Learning using a pre-trained ResNet-50 model, fine-tuned collaboratively across sites.	Collection from Kaggle	The federated transfer learning model achieved a classification accuracy of 96.3% for detecting various lung diseases, significantly outperforming models trained on single, isolated datasets.	[34]

Chest	FL	A custom Federated Learning framework using a specialized Composite-DenseNet	–	The DMFL_Net framework achieved a high overall classification accuracy of 98.21% and a precision of 98.14%, demonstrating superior performance for multi-disease classification in a federated setting.	[35]
Chest	Combined	A Federated Learning-based Real-Time Ensemble Model combining multiple CNNs (VGG16, ResNet) with XAI techniques like Grad-CAM.	Kaggle	The FLEM-XAI ensemble model achieved accuracy of 99.4% for multi-class lung disease classification.	[37]
Chest	Combined	A Federated Deep Learning framework using a custom CNN architecture integrated with XAI methods such as LIME and SHAP for model interpretation.	Chest X-ray images from distributed clients	The proposed FLPneX-AINet framework achieved a high accuracy of 98.7% in detecting pneumonia from chest X-rays	[38]

Chest	Combined	A Secure Framework using MapReduce, Blockchain-based Federated Learning, and XAI with a DenseNet-201.	Multi-source CT scan datasets	The secure federated learning model achieved an exceptional accuracy of 99.8% for lung cancer prediction, leveraging blockchain for security and XAI for providing transparent, interpretable results.	[39]
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2.6 Datasets Utilized in the Project

Table 2.2: Summary of Datasets Used in the Project

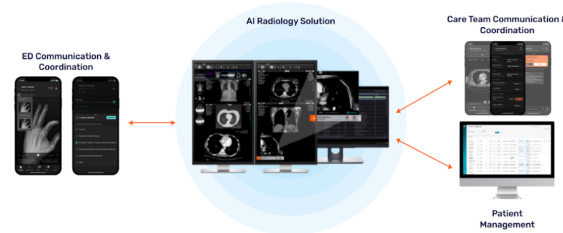
Dataset Name	Domain	Class Distribution / Description	Total Images
FracAtlas	Bone	Fractured (1707), Hand (3487), Hardware (210), Hip (796), Leg (5293), Shoulder (796)	9,383
COVID-19 & Pneumonia & Tuberculosis with Normal & Non-X-ray	Chest	COVID-19 (3,616), Normal (5,688), Pneumonia (4,861), Tuberculosis (3,453)	17,618
Chest X-ray Dataset for Lung Segmentation	Chest	Annotated chest X-ray images for lung segmentation tasks	6,106 images with corresponding 6,106 masks
Random Images for Image Classification	General	Unlabeled random images used for general image classification experiments	9,958
MURA	Bone	Elbow (1,912), Finger (2,110), Hand (2,185), Humerus (727), Forearm (1,010), Shoulder (3,015), Wrist (3,697)	40,561 images (14,656 studies)

2.7 Similar Commercial Applications

Several commercial applications leverage AI for medical image diagnosis, providing tools for clinicians to detect conditions like bone fractures and chest diseases. These applications often use DL models for classification but may lack advanced XAI or FL features for interpretability and privacy. Below, we discuss some notable examples.

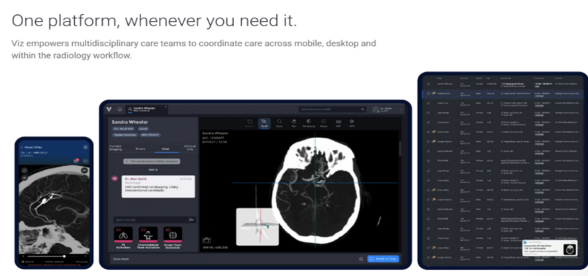
2.7.1 Aidoc (aiOS™)

Aidoc is a leading clinical AI software company that utilizes its proprietary aiOS™ platform to deliver high-performance AI solutions for medical imaging interpretation and workflow optimization. The platform automatically analyzes medical scans, prioritizing critical findings like stroke, pulmonary embolism, and intracranial hemorrhage, and then notifies care teams in real-time. By integrating seamlessly with existing systems like EHR and PACS, Aidoc's solutions help radiologists streamline workflows, enhance diagnostic accuracy, and accelerate time-to-treatment for patients with acute conditions across radiology, cardiology, and neurovascular specialties.



2.7.2 Viz.ai (Viz.ai One®)

Viz.ai is a leading AI-powered care coordination platform focused on transforming time-sensitive medical care by accelerating diagnosis and treatment decisions. The Viz.ai One® platform features multiple FDA-cleared AI algorithms that analyze imaging data, including CT scans and EKGs, to detect suspected diseases like stroke and pulmonary embolism. Crucially, it then uses real-time alerts and communication tools to instantly connect multidisciplinary care teams on mobile and desktop devices, ensuring the patient receives the necessary intervention as quickly as possible.



2.7.3 Qure.ai (qXR, qCT)

Qure.ai specializes in developing deep learning algorithms for affordable and accessible diagnostics, with a strong presence in global health initiatives. Their core products, qXR (for chest X-rays) and qER/qCT (for head and lung CTs), use AI to detect and quantify abnormalities, including tuberculosis, lung nodules, and critical trauma findings. By analyzing scans with high accuracy in minutes, Qure.ai's solutions assist radiologists in early disease detection, risk stratification, and provide essential diagnostic support, especially in areas with limited access to specialists.



2.7.4 Harrison.ai

Harrison.ai, through its partnership with the I-MED Radiology Network (Annalise.ai), offers comprehensive, decision-supported AI tools designed to raise the standard of patient care and combat capacity limits. Their advanced AI solutions for Chest X-ray (CXR) and Non-contrast Head CT (CTB) are designed to detect a massive number of potential findings (often over 100) within seconds. By acting as a 'co-pilot' for clinicians, their AI solutions flag emergent and incidental findings, helping to reduce stress from chaotic worklists and significantly improving both diagnostic accuracy and efficiency.

2.7.5 Lunit (Lunit INSIGHT, Lunit SCOPE)

Lunit is a global leader in medical AI software with a singular mission to conquer cancer through cutting-edge technology. The company's solutions, including the Lunit INSIGHT suite for cancer screening (especially mammography and chest X-ray) and the Lunit SCOPE platform for precision oncology, transform complex clinical data into clear, actionable insights. By delivering highly accurate diagnostic results and extracting intelligence from tissue images to predict treatment response, Lunit empowers earlier detection and more precise treatment decisions across the entire cancer care continuum.

2.8 Summary

In this chapter, we presented the background on bone fractures and chest diseases as initial focus areas for medical image diagnosis, emphasizing multiclassification. We discussed similar commercial applications, reviewed related works on detection using DL for each domain, explainable AI techniques like GradCAM for interpretability (with potential

image insertions for heatmaps), federated learning for privacy-preserving training, and combined approaches integrating these elements. We also evaluated tools for development. These insights form the foundation for our extensible project’s development, highlighting gaps in interpretability and data privacy that our system aims to address for bone fracture detection, chest disease detection, and future expansions.

CHAPTER 3

PROPOSED MODELS

Chapter 3 Proposed Models

This chapter presents the proposed end-to-end solution developed for the automated detection of chest diseases and bone fractures from medical imaging data. The chapter aims to provide a comprehensive description of the overall system architecture as well as a detailed explanation of the individual models that constitute the proposed solution.

The chapter begins by outlining the solution architecture, which describes the complete pipelines of the system, including the high-level data flow and processing in the DL pipeline, and the backend processes and structure. This section illustrates how both the DL pipeline and the system's backend operate cohesively to form a unified diagnostic framework.

Subsequently, the chapter discusses the transfer learning models used in the proposed system. This section explains the pre-trained architectures employed, and their advantages in medical image analysis.

The medical image routing model is then introduced, which serves as an intelligent intermediary responsible for directing input images to the appropriate diagnostic model based on anatomical or modality-specific characteristics. This component plays a critical role in enabling the downstream models to act on an expected input image modality and anatomical region, it is also crucial in dropping out-of-distribution input data from being processed by the diagnostic models.

Following this, the bone fracture detection model is presented in detail. This section describes the model architecture, and classification approach used to identify bone fractures from radiographic images.

Then, an in-depth explanation of the chest disease detection model, which focuses on identifying various thoracic conditions from chest X-ray images, is presented. The section covers the model design, feature extraction process, and classification methodology employed to achieve robust disease detection.

Finally, the chapter ends with a summary to conclude all the information presented and an emphasis is put on the key information given in the chapter.

3.1 Solution Architecture

This chapter presents the architecture of the proposed intelligent medical image analysis framework. The system is designed as a hierarchical, modality-aware diagnostic pipeline capable of analyzing heterogeneous medical imaging data while maintaining high levels of reliability, interpretability, and scalability. The architecture addresses a practical challenge frequently encountered in real-world clinical environments: different diseases are diagnosed using different imaging modalities, and therefore a single monolithic model

may not be the most effective solution. Instead, the proposed framework decomposes the overall diagnostic task into a sequence of specialized stages that collectively form an end-to-end decision-making pipeline.

The framework has been designed to support the automated analysis of chest radiographs and computed tomography (CT) scans. More specifically, the system is capable of detecting COVID-19, Tuberculosis, and Pneumonia from chest X-ray images, while also performing lung cancer detection from CT scans. In addition to disease classification, the framework incorporates Explainable Artificial Intelligence (XAI) mechanisms to improve transparency and trustworthiness, as well as Federated Learning (FL) capabilities to facilitate privacy-preserving collaborative model development across multiple healthcare institutions.

The overall architecture follows a two-level classification paradigm. At the first level, an intelligent routing model determines the modality of the incoming image and verifies whether it belongs to one of the supported image categories. At the second level, modality-specific diagnostic models perform disease prediction. This hierarchical structure enables the system to leverage specialized classifiers that are optimized for their respective tasks while reducing the complexity associated with training a single generalized model.

Figure 3.1 presents a high-level overview of the proposed framework.

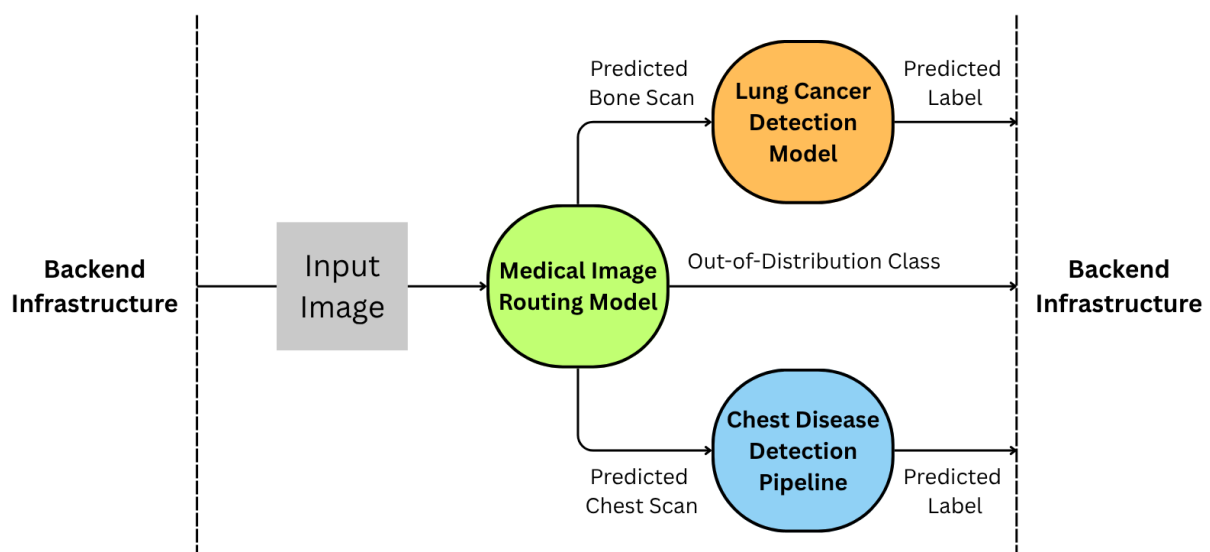


Figure 3.1: High-level architecture of the proposed framework. The system consists of a modality routing model, specialized disease classifiers, Explainable AI modules, and a Federated Learning infrastructure connecting multiple healthcare institutions.

3.1.1 Architectural Design Philosophy

The proposed framework was developed according to several architectural principles that were established at the beginning of the system design process. These principles were intended to ensure that the resulting framework would not only achieve strong predictive performance but would also remain practical for real-world deployment in healthcare environments.

One of the most important design considerations was modality awareness. Medical images generated by different imaging technologies exhibit fundamentally different visual characteristics and diagnostic properties. Chest radiographs provide projection-based representations of thoracic structures, whereas CT scans generate detailed cross-sectional views that reveal significantly more anatomical information. These differences influence the types of features that neural networks must learn in order to perform accurate diagnosis. A single generalized model trained across multiple modalities may struggle to simultaneously learn optimal representations for all imaging types. Consequently, the proposed architecture explicitly separates modality identification from disease diagnosis. This design allows each diagnostic model to focus exclusively on data originating from the modality for which it was designed.

Another fundamental design principle was task specialization. The diseases targeted by this work differ substantially in terms of pathological manifestations, imaging characteristics, and diagnostic requirements. COVID-19, Tuberculosis, and Pneumonia are primarily identified using chest radiographs, whereas lung cancer diagnosis often relies on the higher anatomical resolution available in CT imaging. Rather than attempting to train a single network to recognize all diseases across all modalities, the proposed framework distributes these tasks among specialized models. This decomposition reduces task complexity and allows each model to learn more discriminative feature representations relevant to its specific diagnostic objective.

Reliability and patient safety also played a significant role in the architectural design. Medical AI systems must operate in environments where unexpected inputs are inevitable. Hospital information systems frequently contain diverse imaging studies, and a deployed diagnostic model may encounter unsupported image types, corrupted files, incomplete scans, or non-medical images. If such inputs are processed without validation, the resulting predictions may be unreliable and potentially dangerous. To address this issue, the proposed framework incorporates an explicit out-of-distribution detection mechanism within the routing stage. This component enables the system to identify unsupported inputs before they reach the diagnostic models.

Transparency represents another critical requirement for healthcare AI systems. Clinical practitioners often require evidence supporting automated predictions before integrating them into their decision-making processes. While deep neural networks have demonstrated remarkable predictive capabilities, they are frequently criticized for their lack of interpretability. The proposed architecture therefore integrates explainability mechanisms directly into the diagnostic workflow. These mechanisms provide visual explanations that highlight image regions contributing to the final prediction, thereby allowing clinicians to better understand and validate the reasoning process of the model.

Finally, the architecture was designed with scalability and long-term deployment in mind. Healthcare institutions are often unable to share patient data due to privacy regulations, legal restrictions, and ethical concerns. Consequently, the framework incorporates Federated Learning capabilities that enable multiple institutions to collaboratively improve model performance without exchanging raw patient information. This design promotes the creation of a continuously improving diagnostic ecosystem while preserving patient confidentiality.

3.1.2 System Workflow

The complete workflow of the proposed framework begins when a medical image is submitted to the system. Following acquisition, the image undergoes preprocessing operations intended to standardize the input format and improve compatibility with the downstream neural networks. These operations may include resizing, normalization, intensity standardization, and other transformations required by the underlying deep learning models.

Once preprocessing is completed, the image is forwarded to the first level of the architecture, namely the modality routing model. The purpose of this stage is to determine the nature of the input image and identify the appropriate diagnostic pathway. The routing decision governs all subsequent processing stages and therefore serves as a critical component of the framework.

After the routing decision is generated, the image is directed toward a specialized diagnostic model. Images identified as chest radiographs are analyzed by the chest disease classifier, while CT scans are analyzed by the lung cancer detection model. Images identified as out-of-distribution are excluded from automated diagnosis and may instead be flagged for manual review by healthcare personnel.

Following disease prediction, explainability modules generate visual interpretations of the model output. These interpretations provide insight into the image regions that most strongly influenced the prediction process. Finally, the prediction and associated explanation are presented to the clinician. During model training, Federated Learning mechanisms facilitate collaborative model improvement across multiple participating institutions.

Figure 3.2 illustrates the classification pathway implemented by the proposed framework.

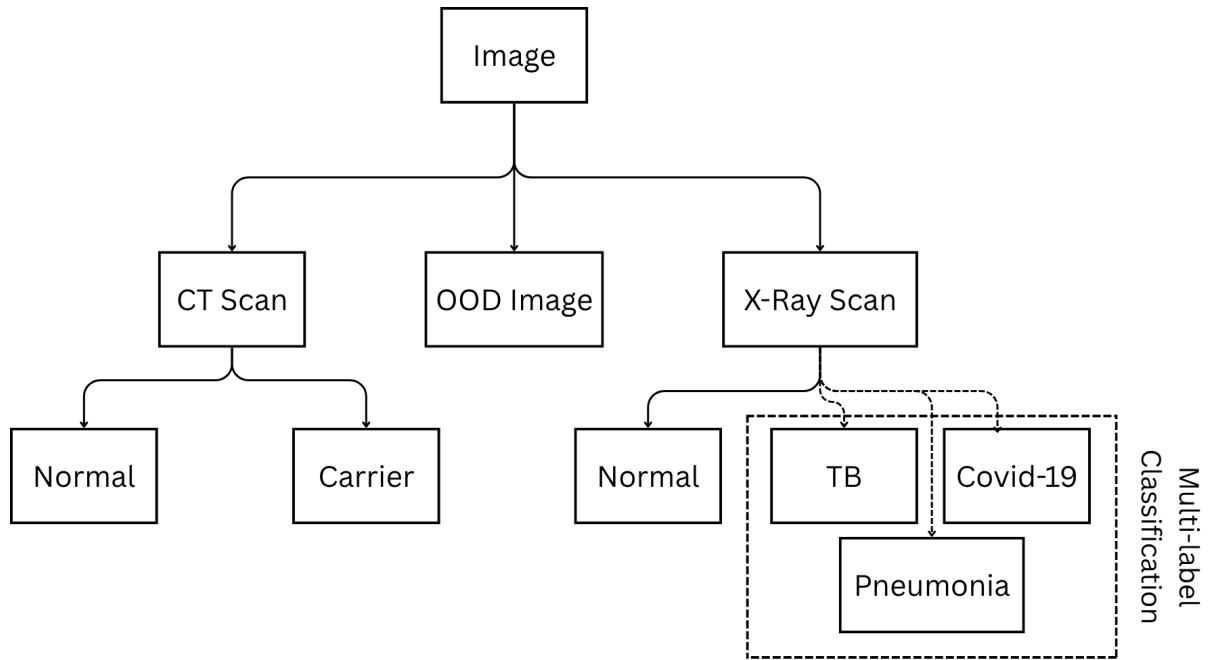


Figure 3.2: Classification tree of the proposed two-level framework. The routing model first determines the image modality before directing the image toward the appropriate disease-specific classifier.

3.1.3 Level One: Modality Routing and Out-of-Distribution Detection

The first level of the proposed architecture functions as an intelligent routing mechanism responsible for identifying the modality of the incoming image. Although this component performs a relatively simple classification task compared to disease diagnosis, its importance should not be underestimated. Every subsequent prediction generated by the framework depends directly on the correctness of the routing decision.

Formally, let I denote an input image. The routing model $R(\cdot)$ maps the image into one of three categories:

$$R(I) \in \{\text{X-ray}, \text{CT}, \text{OOD}\} \quad (3.1)$$

where OOD represents an out-of-distribution image.

The motivation for introducing a dedicated routing stage arises from the substantial differences between X-ray and CT imaging modalities. Chest radiographs represent two-dimensional projection images that summarize anatomical information along a viewing direction. In contrast, CT scans provide detailed cross-sectional information that enables the visualization of internal structures with considerably greater precision. These differences imply that the features required for effective diagnosis are modality-dependent. Routing images to specialized models therefore allows the framework to exploit modality-specific representations rather than relying on a generalized feature extraction strategy.

From a computational perspective, routing also improves efficiency. Without this stage, every image would need to be processed by every available diagnostic model, resulting in increased computational cost and unnecessary inference overhead. By ensuring that each image is evaluated only by the most relevant classifier, the routing model reduces resource consumption and improves scalability.

An equally important function of the routing stage is out-of-distribution detection. In practical deployment scenarios, healthcare systems may receive images that differ significantly from those used during training. Examples include magnetic resonance imaging studies, ultrasound images, pathology slides, non-medical photographs, or improperly formatted files. Diagnostic models are generally incapable of producing reliable predictions for such inputs because they fall outside the distribution of the training data.

To mitigate this problem, the routing model incorporates a dedicated OOD category. Images assigned to this category are prevented from entering the diagnostic pipeline. This safeguard reduces the likelihood of erroneous predictions and contributes to the overall reliability of the framework. From a clinical perspective, this mechanism functions as a preliminary quality-control layer that ensures only supported image types proceed to automated analysis.

3.1.4 Level Two: Chest Disease Detection Model

Images identified as chest radiographs are forwarded to the chest disease classification model. This model is responsible for detecting three clinically significant respiratory diseases: COVID-19, Tuberculosis, and Pneumonia.

The selection of these diseases was motivated by both clinical relevance and public health importance. COVID-19 remains one of the most extensively studied respiratory illnesses in recent years and has highlighted the importance of rapid imaging-based screening techniques. Tuberculosis continues to represent a major global health challenge, particularly in developing countries where early detection plays a crucial role in disease management and transmission control. Pneumonia remains one of the most common respiratory infections worldwide and is associated with substantial morbidity and mortality across multiple age groups.

A key characteristic of the proposed chest disease model is its formulation as a multi-label classification problem rather than a conventional multi-class classification problem. This distinction is essential because the targeted diseases are not mutually exclusive. In clinical practice, a patient may simultaneously exhibit radiological findings associated with multiple respiratory conditions. For example, viral infections may be accompanied by secondary bacterial pneumonia, and complex pathological interactions may lead to overlapping imaging manifestations.

Traditional multi-class classification assumes that each sample belongs to exactly one category. Such an assumption would force the model to select a single disease label even when multiple diseases are present. This formulation would therefore fail to accurately represent many real-world clinical scenarios.

To address this limitation, the proposed architecture employs independent disease

predictions. Let

$$\mathbf{y} = [y_{\text{COVID}}, y_{\text{TB}}, y_{\text{Pneumonia}}] \quad (3.2)$$

denote the output vector generated by the classifier. Each element corresponds to the estimated probability that a specific disease is present within the image.

Unlike a softmax-based multi-class architecture, the output layer uses independent sigmoid activations. This allows the model to assign high probabilities to multiple diseases simultaneously whenever appropriate. The final prediction for each disease is obtained by applying a predefined threshold:

$$\hat{y}_i = \begin{cases} 1 & y_i \geq \tau \\ 0 & y_i < \tau \end{cases} \quad (3.3)$$

where τ denotes the classification threshold.

This formulation better reflects the realities of clinical diagnosis and provides greater flexibility when interpreting model outputs. Instead of receiving a single categorical prediction, clinicians are presented with independent disease probabilities that can be incorporated into broader diagnostic assessments.

3.1.5 Level Two: Lung Cancer Detection Model

Images classified as CT scans are forwarded to the lung cancer detection model. The decision to utilize CT imaging for lung cancer analysis is motivated by the superior anatomical detail provided by computed tomography compared to conventional radiography.

Lung cancer frequently manifests through subtle structural abnormalities such as pulmonary nodules, irregular masses, and localized tissue changes that may not be visible in chest radiographs. The high spatial resolution of CT imaging enables these features to be observed with significantly greater clarity, making CT the preferred modality for many lung cancer screening and diagnostic procedures.

The objective of the lung cancer model is to determine whether cancer-related findings are present within the input scan. The classifier can be represented mathematically as

$$P_{\text{Cancer}} = C(I) \quad (3.4)$$

where $C(\cdot)$ denotes the lung cancer detection network and P_{Cancer} represents the estimated probability of malignancy.

By focusing exclusively on CT-derived information, the model can dedicate its representational capacity to learning highly discriminative cancer-related features. This

specialization improves the model's ability to identify subtle pathological patterns that may be difficult to detect using a generalized classifier.

3.1.6 Explainable Artificial Intelligence Integration

The increasing adoption of artificial intelligence within healthcare has generated significant interest in model interpretability. Although deep learning systems are capable of achieving remarkable predictive performance, their internal decision-making processes often remain difficult to understand. This lack of transparency can hinder clinical adoption, particularly when predictions influence critical healthcare decisions.

To address this challenge, Explainable Artificial Intelligence techniques are integrated directly into the proposed architecture. Rather than treating explainability as an optional post-processing step, it is considered a fundamental component of the diagnostic workflow.

Following prediction generation, the explainability module produces visual explanations illustrating the regions of the image that most strongly influenced the model's decision. These visualizations may take the form of saliency maps, activation maps, attention maps, or gradient-based explanations depending on the selected implementation.

The integration of explainability serves several purposes. First, it increases clinician confidence by providing visual evidence supporting the model's prediction. Second, it facilitates model validation by enabling researchers to verify that the network is focusing on medically meaningful regions rather than irrelevant artifacts. Third, it assists in identifying potential biases or failure modes that may otherwise remain hidden within the network. Ultimately, explainability strengthens the collaborative relationship between clinicians and AI systems by transforming predictions into interpretable decision-support information.

3.1.7 Federated Learning Infrastructure

A distinguishing feature of the proposed framework is its support for Federated Learning. This capability was incorporated to address one of the most significant challenges facing medical artificial intelligence: limited access to large and diverse datasets.

Traditional deep learning systems typically rely on centralized training approaches in which data from multiple institutions are collected and stored in a single repository. While effective from a machine learning perspective, this strategy introduces substantial privacy, security, and regulatory concerns. Healthcare organizations are often prohibited from sharing patient data due to legal and ethical restrictions.

Federated Learning offers an alternative paradigm in which training occurs locally at each institution. Rather than transmitting patient data, participating hospitals train local models using their own datasets and share only model parameters or parameter updates. These updates are subsequently aggregated to produce a global model.

Let w_i denote the model parameters learned by institution i , and let n_i denote the number of training samples available at that institution. Using the Federated Averaging

algorithm, the global model is computed as

$$w_{\text{global}} = \sum_{i=1}^K \frac{n_i}{\sum_{j=1}^K n_j} w_i \quad (3.5)$$

where K represents the total number of participating institutions.

Beyond privacy preservation, Federated Learning provides several additional benefits. Because participating institutions often serve different patient populations, the resulting datasets contain greater demographic, geographic, and clinical diversity. Training across such heterogeneous data sources can improve model generalization and reduce susceptibility to institution-specific biases. Furthermore, the collaborative nature of Federated Learning enables continuous improvement of the framework as new institutions join the network and contribute additional knowledge.

3.2 Transfer-Learning Models Used

This section presents the transfer learning models employed in the proposed system and discusses the rationale behind their selection. Transfer learning is adopted to leverage pre-trained deep neural networks that have been trained on large-scale datasets, enabling effective feature extraction and improved generalization in medical imaging tasks with limited labeled data. The section provides an overview of the selected architectures and highlights their suitability for different components of the proposed diagnostic pipeline.

3.2.1 ResNet50 Model

Residual Networks (ResNets) were introduced to address the degradation problem observed in deep convolutional neural networks, where increasing network depth leads to saturation or decline in training accuracy. This issue emerges due to optimization difficulties such as vanishing gradients. ResNet addresses this limitation through the concept of residual learning, which allows the training of substantially deeper networks by enabling more effective gradient propagation.

The key idea of residual learning is to reformulate the learning objective of a neural network block. Instead of directly learning a desired underlying mapping $H(x)$, a residual block learns a residual function $F(x)$ such that:

$$H(x) = F(x) + x$$

where x represents the input to the residual block and $F(x)$ represents the output of the stacked convolutional layers within the block. The identity shortcut connection that performs the addition allows gradients to flow directly through the network during backpropagation, thereby minimizing vanishing gradient effects and improving training stability.

As illustrated in Figure 3.3, the ResNet50 architecture begins with an initial convolutional layer preceded by zero padding, followed by batch normalization, a ReLU activation

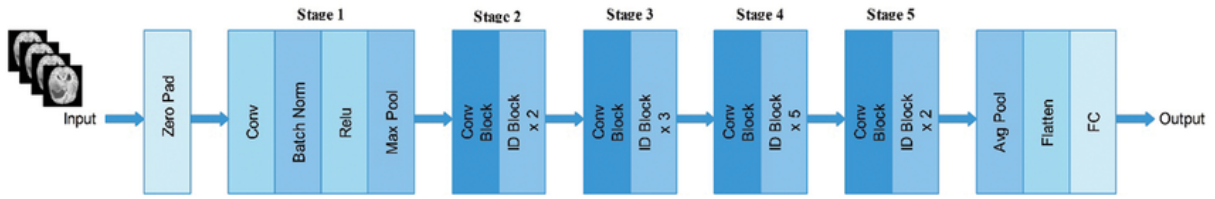


Figure 3.3: Architecture of the ResNet50 illustrating the successive stages of the model.

function, and a max pooling operation. This initial stage is responsible for extracting low-level visual features from the input image. The network then proceeds through five sequential stages composed of convolutional blocks and identity blocks, each incorporating residual connections that bypass multiple convolutional layers.

ResNet50 employs a bottleneck design within its residual blocks, consisting of a sequence of 1×1 , 3×3 , and 1×1 convolutional layers. The 1×1 convolutions are used to reduce and subsequently restore feature dimensionality, thereby decreasing computational complexity while preserving representational power. This design enables ResNet50 to achieve substantial depth without incurring prohibitive computational costs.

Compared to shallower variants such as ResNet18 and ResNet34, ResNet50 offers increased representational capacity and improved ability to capture complex and abstract visual patterns. While deeper variants such as ResNet101 and ResNet152 further extend network depth, they introduce higher computational overhead and an increased risk of overfitting, particularly in medical imaging applications where annotated data is often limited. Consequently, ResNet50 represents a balanced trade-off between depth, efficiency, and performance within the ResNet family.

As shown in Figure 3.3, the final stages of ResNet50 consist of a global average pooling layer, followed by feature flattening and a fully connected layer. In transfer learning scenarios, this classification head can be replaced or fine-tuned to adapt the network to domain-specific medical tasks.

In the context of medical image analysis, ResNet50 is well-suited for extracting both low-level structural features and high-level semantic representations relevant to pathological patterns. The residual connections promote feature reuse and robust optimization, which is particularly important for detecting subtle abnormalities in medical images such as fractures or thoracic diseases. These characteristics make ResNet50 a reliable and widely adopted backbone for medical image classification tasks.

3.2.2 DenseNet121 Model

DenseNet121 is a deep convolutional neural network architecture belonging to the family of Densely Connected Convolutional Networks (DenseNets), which were introduced by Huang *et al.* in 2017 to address several limitations associated with traditional deep neural networks. DenseNet architectures were designed to improve feature propagation, encourage feature reuse, alleviate the vanishing gradient problem, and reduce the number of trainable parameters while maintaining high classification performance. Since its introduction, DenseNet121 has become one of the most widely adopted architectures in

medical image analysis due to its ability to learn rich hierarchical representations from relatively limited datasets.

The fundamental idea behind DenseNet differs significantly from that of conventional convolutional neural networks. In traditional architectures, each layer receives information only from the immediately preceding layer and passes its output solely to the next layer. Although this sequential information flow is effective, it may lead to difficulties in gradient propagation as network depth increases. Furthermore, many learned feature maps become redundant because similar patterns are repeatedly relearned at different layers throughout the network.

To address these limitations, DenseNet introduces direct connections between layers. Instead of receiving input from only the previous layer, each layer receives the feature maps generated by all preceding layers within the same dense block. Consequently, information can flow directly from earlier layers to later layers without repeatedly passing through intermediate transformations. This dense connectivity pattern enables more efficient feature reuse and facilitates the training of very deep networks.

Figure 3.4 illustrates the overall architecture of DenseNet121.

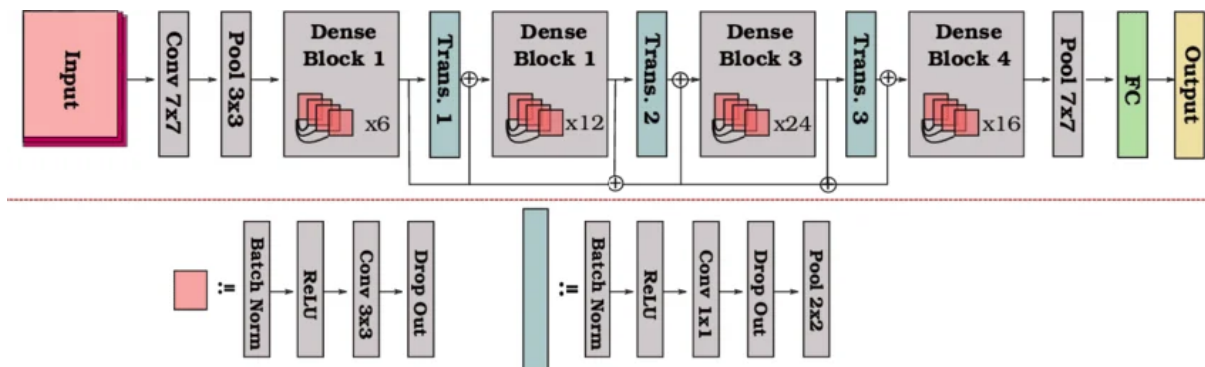


Figure 3.4: Overall architecture of DenseNet121 showing the arrangement of dense blocks, transition layers, and the final classification stage.

Motivation Behind Dense Connectivity

As neural networks become deeper, the flow of information and gradients through the network becomes increasingly difficult. One of the most common challenges encountered in deep architectures is the vanishing gradient problem, where gradients become progressively smaller as they are propagated backward through many layers. When gradients become extremely small, early layers receive little useful learning signal, slowing convergence and potentially reducing overall performance.

Residual Networks (ResNets) partially addressed this issue through skip connections that allow gradients to bypass certain transformations. DenseNet extends this concept further by establishing direct connections between every pair of layers within a dense block. Instead of merely adding outputs from previous layers, DenseNet concatenates them. This strategy preserves information from all preceding layers and makes it directly accessible to subsequent layers.

The dense connectivity mechanism provides several important benefits. First, it improves gradient flow during training, allowing deeper networks to be optimized more effectively. Second, it encourages feature reuse, reducing redundancy among learned representations. Third, it improves parameter efficiency because layers can leverage previously learned features rather than learning entirely new representations from scratch.

Dense Connectivity Mechanism

The defining characteristic of DenseNet is the dense connection pattern between layers. Let x_0 represent the input to a dense block and let x_l denote the output of the l^{th} layer. Unlike traditional convolutional networks, where the output of layer l depends only on the output of layer $l - 1$, DenseNet computes each layer using the outputs of all preceding layers:

$$x_l = H_l([x_0, x_1, x_2, \dots, x_{l-1}]) \quad (3.6)$$

where:

- $H_l(\cdot)$ denotes a composite transformation.
- $[\cdot]$ represents the concatenation operation.
- $x_0, x_1, \dots, x_{l-1}$ are the feature maps generated by previous layers.

The concatenation operation differs fundamentally from the addition operation used in residual networks. Instead of merging features by summation, DenseNet preserves all previous feature maps and passes them directly to subsequent layers. As a result, each layer has access to both low-level and high-level information simultaneously.

This dense feature propagation mechanism allows later layers to exploit a richer collection of learned representations, improving feature diversity and reducing information loss.

Dense Blocks

The primary computational components of DenseNet121 are the dense blocks. A dense block consists of multiple densely connected layers arranged according to the connectivity pattern described previously.

Within each dense block, every layer receives as input the concatenated outputs of all preceding layers. Consequently, the number of feature maps grows progressively as additional layers are added. This growth is controlled through a parameter known as the *growth rate*.

The growth rate, typically denoted by k , determines the number of new feature maps generated by each layer. If the input to a dense block contains F_0 feature maps and the block contains L layers, the total number of feature maps at the end of the block can be expressed as:

$$F = F_0 + kL \quad (3.7)$$

where:

- F_0 is the number of initial feature maps.
- k is the growth rate.
- L is the number of layers within the block.

For DenseNet121, the growth rate is typically set to 32. This means that each layer contributes 32 new feature maps while simultaneously utilizing all previously generated feature maps.

The use of a relatively small growth rate contributes to the parameter efficiency of DenseNet architectures, allowing them to achieve competitive performance with fewer parameters than many alternative architectures.

Composite Function within Dense Layers

Each layer within a dense block consists of a sequence of operations designed to transform incoming feature maps into more informative representations.

The composite transformation $H_l(\cdot)$ generally consists of:

1. Batch Normalization (BN)
2. Rectified Linear Unit (ReLU)
3. Convolution

This sequence is commonly referred to as the BN-ReLU-Conv pattern.

Batch normalization stabilizes the learning process by normalizing intermediate activations. This reduces internal covariate shift and allows higher learning rates to be used during training.

The ReLU activation function introduces non-linearity into the network and is defined as:

$$f(x) = \max(0, x) \quad (3.8)$$

Following activation, convolutional filters extract spatial patterns from the input feature maps. Through repeated application of these operations, the network gradually learns increasingly abstract representations of the input image.

Bottleneck Layers

As the number of feature maps grows within dense blocks, computational complexity can increase significantly. To reduce this cost, DenseNet¹²¹ employs bottleneck layers.

A bottleneck layer consists of a 1×1 convolution applied before the standard 3×3 convolution. The purpose of this operation is to reduce the number of feature channels while preserving important information.

The bottleneck structure can be summarized as:

$$BN \rightarrow ReLU \rightarrow Conv(1 \times 1) \rightarrow BN \rightarrow ReLU \rightarrow Conv(3 \times 3) \quad (3.9)$$

The use of bottleneck layers substantially reduces computational requirements while maintaining representational capacity.

Transition Layers

Because feature map dimensions increase continuously within dense blocks, DenseNet introduces transition layers between consecutive blocks to control network complexity.

A transition layer performs two primary functions. First, it reduces the number of feature maps through a 1×1 convolution. Second, it decreases spatial resolution using average pooling.

The transition layer architecture typically follows:

$$BN \rightarrow ReLU \rightarrow Conv(1 \times 1) \rightarrow AvgPool(2 \times 2) \quad (3.10)$$

These operations prevent excessive growth in memory consumption while preserving the most informative features.

Transition layers also contribute to hierarchical feature extraction by gradually reducing spatial dimensions as network depth increases.

Architecture of DenseNet¹²¹

DenseNet¹²¹ consists of four dense blocks separated by three transition layers. The architecture begins with an initial convolution layer and concludes with global average pooling followed by a fully connected classification layer.

The distribution of layers across dense blocks is as follows:

Table 3.1: Layer distribution within DenseNet121.

Component	Number of Layers
Dense Block 1	6
Dense Block 2	12
Dense Block 3	24
Dense Block 4	16
Total Dense Layers	58

When all convolutional, pooling, and classification layers are included, the network contains 121 trainable layers, which gives the architecture its name.

The input image first passes through an initial 7×7 convolution layer followed by max pooling. Feature extraction then proceeds through the sequence of dense blocks and transition layers. Finally, global average pooling aggregates spatial information into a compact feature vector, which is passed to the final classification layer.

Advantages of DenseNet121 for Medical Image Analysis

DenseNet121 possesses several characteristics that make it particularly suitable for medical image classification tasks.

The dense connectivity pattern encourages extensive feature reuse, allowing the network to learn robust representations even when training datasets are relatively small. This characteristic is especially important in medical imaging, where acquiring large annotated datasets is often challenging.

The improved gradient flow resulting from dense connections facilitates stable optimization and reduces the risk of vanishing gradients. Consequently, deeper representations can be learned without the training difficulties commonly associated with very deep neural networks.

DenseNet121 is also highly parameter-efficient compared to many competing architectures. Despite its depth, the network requires fewer trainable parameters than several traditional convolutional architectures because previously learned features are reused rather than repeatedly regenerated.

Furthermore, the architecture has demonstrated strong performance across a wide range of medical imaging applications, including chest disease classification, pneumonia detection, COVID-19 diagnosis, tuberculosis screening, diabetic retinopathy analysis, and cancer detection. Its proven effectiveness in healthcare applications has made DenseNet121 a popular backbone network for medical image analysis research.

DenseNet121 in the Proposed Framework

In the context of this work, DenseNet121 was selected as the backbone architecture for the first-level routing model due to its strong balance between classification perfor-

mance, computational efficiency, and feature extraction capability. The dense connectivity structure enables the model to capture both low-level radiological patterns and high-level pathological characteristics that are essential for accurate disease recognition.

Additionally, the feature reuse mechanism of DenseNet121 is particularly advantageous for medical datasets, where image variability can be substantial and labeled samples may be limited. By leveraging information from all preceding layers, the network can construct rich hierarchical representations that support robust classification performance.

Overall, DenseNet121 provides a powerful and efficient deep learning architecture that aligns well with the requirements of medical image analysis systems. Its dense connectivity, parameter efficiency, strong gradient propagation, and proven effectiveness in healthcare applications make it an appropriate choice for the disease classification tasks addressed in this project.

3.2.3 YOLOv8 Model

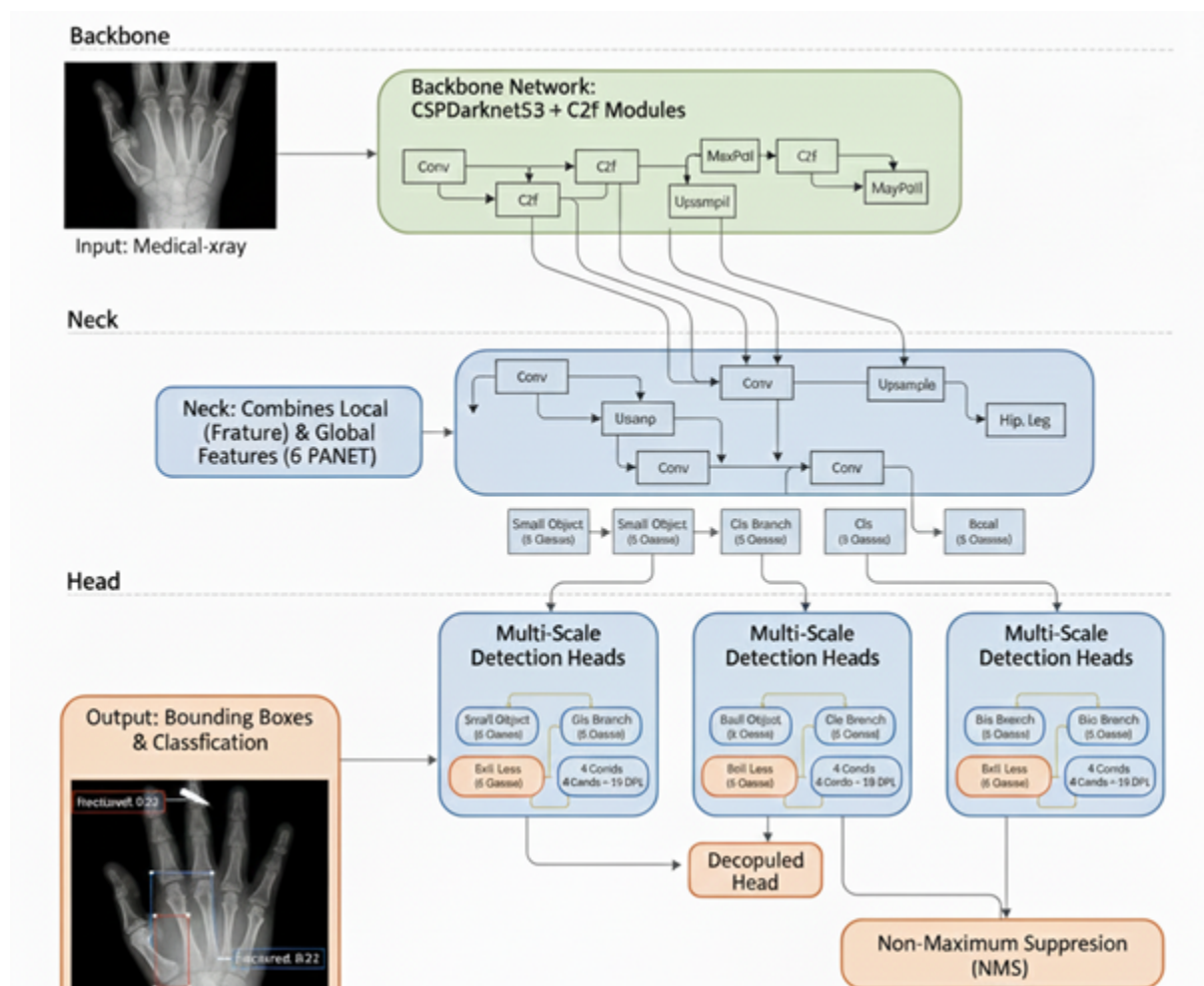


Figure 3.5: Architecture of the YOLOv8 illustrating the successive stages of the model.

As shown in Figure 3.5, the architecture is a "single-stage" pipeline that processes an X-ray in one forward pass, making it fast enough for real-time clinical use. It is composed of three main parts:

- **The Backbone (C2f Module):** This is the model's feature extractor. It uses the C2f (Cross-stage Partial Bottleneck with 2 convolutions) structure to scan the X-ray for textures. It learns to distinguish the high-density, sharp-edged appearance of Hardware (screws/plates) from the organic textures of Hand, Hip, Leg, or Shoulder bones.
- **The Neck (PANet):** This part handles "feature fusion." In medical imaging, context is everything. The Neck combines fine details (like a jagged fracture line) with global shapes (the overall structure of a Hip vs. a Shoulder). This helps the model realize that a specific shadow is a Fractured region and not just a natural bone groove.
- **The Head (Decoupled & Anchor-Free):** Unlike older models, YOLOv8 uses a Decoupled Head, which separates the task of identifying the class (e.g., "Leg") from the task of drawing the box. It is Anchor-Free, meaning it predicts the center of the object directly. This is much better for medical use because fractures and surgical hardware don't come in "standard" box shapes.

During training, the model uses three mathematical "grades" to improve its accuracy:

- **Classification Loss (BCE):** Using Binary Cross-Entropy, the model learns to accurately label its 6 classes. It ensures that 'Hardware' isn't confused with 'Fractured' bone and that the anatomical regions (Hand, Hip, etc.) are correctly identified.
- **Box Localization Loss (CIoU):** This measures how well the predicted box overlaps with the actual object. For a long Leg bone or a small Hardware screw, CIoU adjusts the box's aspect ratio and center point to ensure a tight, perfect fit.
- **Distribution Focal Loss (DFL):** This is critical for hairline fractures. Since the edges of a crack can be blurry on an X-ray, DFL predicts a probability distribution for the box boundaries, allowing the model to be more precise even when the image quality is low.

To ensure the model is both reliable and clinically safe, we evaluate its performance using three primary metrics: Precision, Recall, and mAP50. These indicators provide a comprehensive look at the model's accuracy, its ability to detect every injury, and its overall effectiveness in a real-world medical environment. By analyzing these figures, we can understand how the system balances the need for "Quality Control" with the "Safety Net" required to ensure no fracture goes unnoticed.

Precision acts as the model's primary "Quality Control" mechanism. In the context of medical imaging, high precision ensures that when the model identifies an injury as "Fractured," there is a high statistical certainty that a break actually exists. By focusing on the accuracy of positive predictions, precision prevents doctors from being overwhelmed by "False Alarms," allowing them to trust that the alerts they receive are legitimate and actionable.

While precision focuses on accuracy, Recall serves as the critical "Safety Net" for patient care. It measures the model's ability to find every single instance of a fracture within the dataset. Maintaining high recall is vital in a clinical setting because it ensures that no patient is accidentally sent home with an undiagnosed broken bone. In short, recall prioritizes "catching" every case, minimizing the risk of a dangerous false negative.

The mAP50, or mean Average Precision at a 50% overlap threshold, serves as the comprehensive "Overall Score" for the model's performance. This metric provides the best single indicator of how well the model identifies both the specific anatomy and the underlying injuries simultaneously. By calculating the average precision across all categories at a standard detection threshold, mAP50 offers a balanced view of the system's total effectiveness.

3.3 Medical Image Routing Model

This section presents the proposed medical image routing model employed in the system and discusses the motivation behind the model, its problem formulation, and finally its architecture. This model serves as the decision-making unit of the system, where the destination of each input image is determined and forwarded thereafter.

3.3.1 Motivation

The increasing use of deep learning in medical image analysis has enabled the development of highly specialized models for tasks such as disease detection and diagnosis. In the context of lung cancer detection, chest X-ray and computed tomography (CT) images provide complementary diagnostic information and often require different processing pipelines and model architectures. While chest X-rays offer a low-cost and widely accessible imaging modality for preliminary screening, CT scans provide detailed cross-sectional information that supports more comprehensive assessment of lung abnormalities.

However, in real-world deployment scenarios, diagnostic systems are often exposed to heterogeneous image inputs originating from different imaging modalities, anatomical regions, or acquisition protocols. Applying a modality-specific diagnostic model to an inappropriate image type may lead to unreliable predictions and compromise the robustness and safety of the overall system. Consequently, there is a need for an initial classification mechanism capable of identifying the nature of incoming medical images before further analysis is performed.

This section introduces a medical image classification model designed to function as an image routing component within a larger lung cancer diagnostic pipeline. The model distinguishes between chest X-ray images, chest CT scan images, and an additional out-of-distribution category referred to as "Other." By incorporating this third category, the system is able to explicitly handle images that do not belong to the intended diagnostic classes, thereby reducing the risk of misclassification and improving overall system reliability. Once classified, chest X-ray and CT scan images are forwarded to their corresponding lung cancer diagnostic models, while out-of-distribution images are rejected from further analysis. The following subsections present the motivation for this approach, formally define the classification problem, and describe the model architecture employed.

3.3.2 Problem Statement

The goal of this work is to develop an automated medical image classification model capable of identifying the type of an input image and routing it to the appropriate downstream analysis. The task is formulated as a three-class classification problem, where each input image $X \in \mathbb{R}^{224 \times 224 \times 3}$ must be assigned to one of three categories: chest X-ray images (0), chest CT scan images (1), or out-of-distribution images labeled as *Other* (2). This multi-class formulation ensures that irrelevant or ambiguous inputs are explicitly handled, improving system safety and reducing the risk of inappropriate downstream processing.

The model is designed to output a probability distribution $\mathbf{p} = [p_0, p_1, p_2]$ over the three classes, where $p_k = P(y = k | X)$ and $\sum_{k=0}^2 p_k = 1$. The predicted class is obtained by selecting the class with the highest probability, $\hat{y} = \arg \max_k p_k$. Images classified as chest X-rays are forwarded to the X-ray-based lung cancer detection model, while images classified as CT scans are routed to the CT-based lung cancer detection model. By incorporating the *Other* category, the model avoids forcing all inputs into diagnostic categories and can robustly separate images outside the target distribution, laying a solid foundation for safe integration within a larger medical image analysis pipeline.

3.3.3 Model Architecture

The proposed medical image classification model is designed using a transfer learning approach based on the ResNet50 convolutional neural network. The architecture follows a structured pipeline consisting of input preprocessing, deep feature extraction, feature aggregation, and final classification. This modular design allows the model to benefit from representations learned on large-scale image datasets while adapting effectively to the target medical imaging task. An overview of the complete classification pipeline is illustrated in Figure 3.6.

The input to the model is a medical image $X \in \mathbb{R}^{224 \times 224 \times 3}$, resized to a fixed spatial resolution and represented in RGB color space. Prior to being processed by the pretrained network, the image undergoes a preprocessing step to ensure compatibility with the ResNet50 weights. This step mirrors the preprocessing applied during the original ImageNet training of ResNet50 and is essential for effective transfer learning. Specifically, the input image is first converted to floating-point representation, its color channels are reordered from RGB to BGR, and a fixed channel-wise mean is subtracted. Let X_{rgb} denote the raw input image; the preprocessed image X_{prep} is obtained as $X_{prep} = \text{BGR}(X_{rgb}) - \mu$, where $\mu = [103.939, 116.779, 123.68]$ represents the ImageNet mean for the blue, green, and red channels, respectively. This operation centers the input distribution around zero and aligns it with the statistical properties expected by the pretrained backbone.

Following preprocessing, the image is passed through the ResNet50 network with its original classification layers removed. In this configuration, ResNet50 functions as a fully convolutional feature extractor. For an input tensor of shape $(224, 224, 3)$, the output of the backbone is a high-dimensional feature map of shape $(7, 7, 2048)$. This tensor contains 2048 feature channels, each encoding abstract visual patterns such as anatomical structures, textures, and modality-specific characteristics while preserving

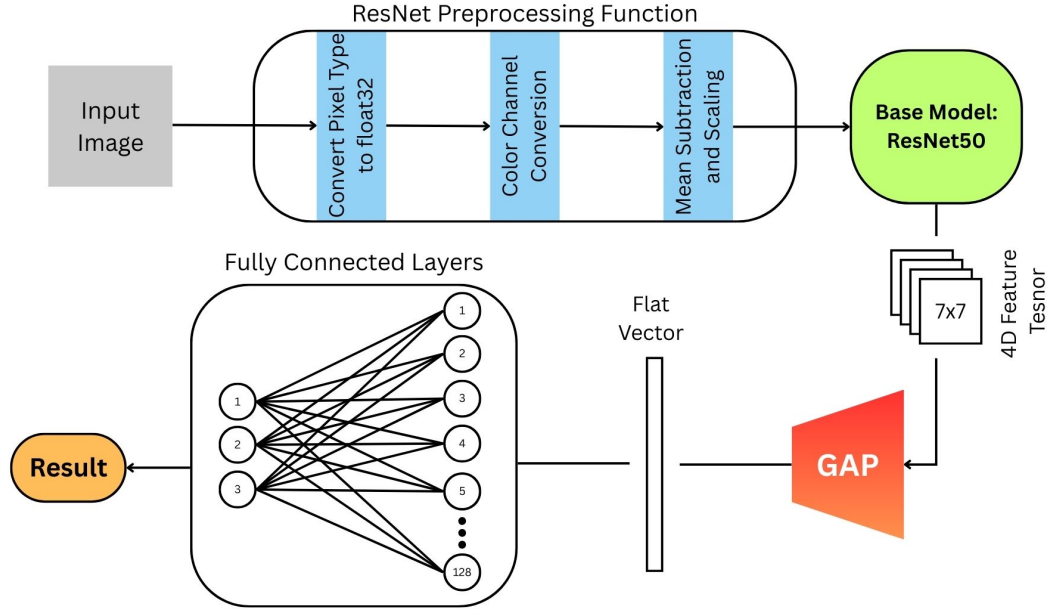


Figure 3.6: Overall architecture of the proposed medical image routing model.

coarse spatial relationships within the image. For a detailed discussion of the ResNet50 model architecture, refer to Subsection 3.2.1.

To transform the spatial feature maps into a compact representation suitable for classification, a global average pooling operation is applied to the output of the ResNet50 backbone. This operation computes the average activation of each feature channel across the spatial dimensions, effectively summarizing the presence of each learned feature over the entire image. Given a feature map $F \in \mathbb{R}^{7 \times 7 \times 2048}$, the pooled feature vector $z \in \mathbb{R}^{2048}$ is computed such that each element z_k corresponds to the mean of the activations in channel k , i.e.

$$z_k = \frac{1}{7 \times 7} \sum_{i=1}^7 \sum_{j=1}^7 F_{i,j,k}$$

This operation reduces the tensor shape from $(7, 7, 2048)$ to (2048) while significantly lowering the number of parameters required in subsequent layers and improving generalization.

The resulting feature vector is then passed to a task-specific classification head composed of fully connected layers. These layers learn discriminative combinations of the extracted features and map them to the target class space. The final layer applies a softmax activation function, producing a probability distribution $\mathbf{p} = [p_0, p_1, p_2]$, where each component represents the predicted probability of the image belonging to the chest X-ray class, chest CT scan class, or the out-of-distribution *Other* class. The predicted label is obtained as $\hat{y} = \arg \max_k p_k$.

3.4 Chest Disease Classification Model

This section presents one of the downstream diagnostic models developed within the project, focusing on automated multi-class classification of chest diseases from chest X-ray (CXR) images. The model is designed as a robust and interpretable pipeline that integrates standardized image preprocessing, lung region isolation, and deep learning-based classification using transfer learning. Emphasis is placed on architectural design choices and methodological foundations rather than performance evaluation, which is discussed in a later chapter.

3.4.1 Motivation

Deep learning has demonstrated strong potential in medical image analysis, yet several challenges remain, including variability in image acquisition, class imbalance, and the need for explainable decision-making. The proposed model addresses these issues through a multi-stage pipeline that enhances image quality, isolates clinically relevant lung regions, and applies a transfer learning-based classifier augmented with explainability techniques.

3.4.2 Problem Statement

The objective of this model is to perform multi-class classification of chest X-ray images into the following diagnostic categories:

$$\mathcal{C} = \{\text{NORMAL}, \text{PNEUMONIA}, \text{COVID-19}, \text{TUBERCULOSIS}\}.$$

Given an input chest X-ray image $I \in \mathbb{R}^{H \times W}$, the task is to learn a mapping

$$f_{\theta} : I \rightarrow \mathbf{p},$$

where $\mathbf{p} \in \mathbb{R}^{|\mathcal{C}|}$ is a probability distribution over disease classes and θ denotes the model parameters.

To improve robustness and clinical relevance, the classification task is constrained to lung parenchyma regions extracted via an explicit segmentation stage. Additionally, the model must provide interpretable outputs to support clinical trust, motivating the use of gradient-based visualization techniques.

3.4.3 Model Architecture

The diagnostic model is structured as a multi-stage pipeline, illustrated conceptually in Figure 3.7. The pipeline consists of three main components: image preprocessing, lung segmentation, and disease classification.

Each chest X-ray image undergoes a standardized preprocessing pipeline to enhance diagnostically relevant features. Images are first converted to grayscale and enhanced using Contrast Limited Adaptive Histogram Equalization (CLAHE). Noise introduced during enhancement is mitigated using Gaussian smoothing with a 5×5 kernel.

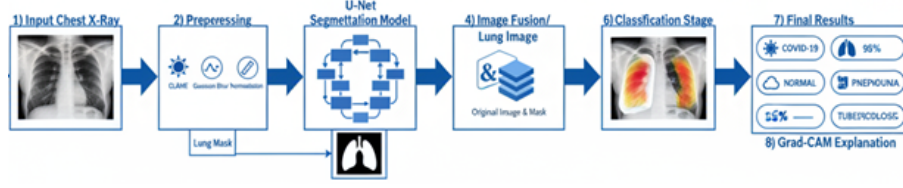


Figure 3.7: High-level overview of the multi-stage chest disease classification pipeline.

Pixel intensities are then normalized using Min–Max normalization: $I_{\text{norm}} = \frac{I - I_{\min}}{I_{\max} - I_{\min}}$ where $I_{\min}$ and $I_{\max}$ represent the minimum and maximum pixel values of the image. Finally, images are converted to RGB format to ensure compatibility with the ResNet50 model. For a discussion of the ResNet50 model, refer to Subsection 3.2.1.

Figure 3.8 illustrates the effect of the preprocessing pipeline on representative CXR images.

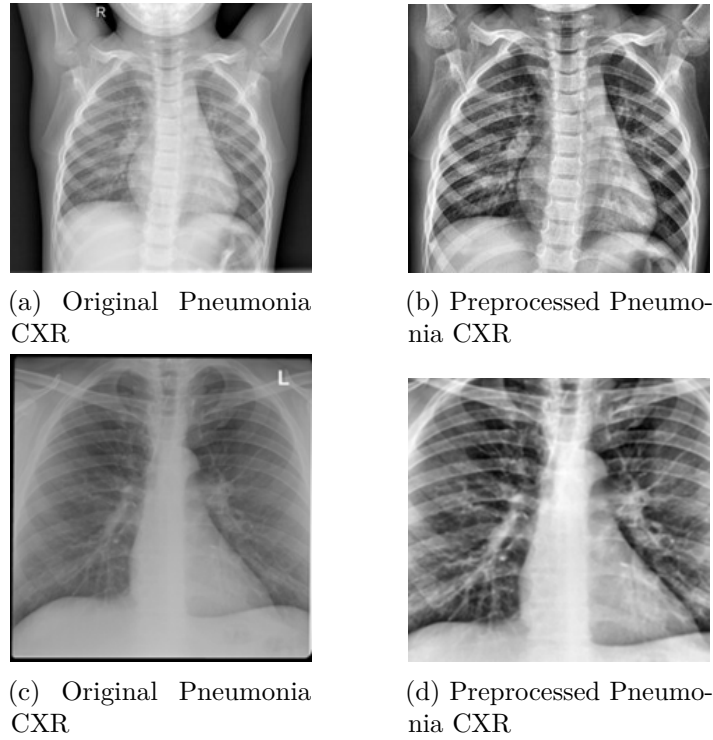


Figure 3.8: Chest X-ray images before and after preprocessing, highlighting improved contrast and reduced noise.

To ensure that classification focuses exclusively on clinically relevant regions, a U-Net–based segmentation model is employed to extract lung masks. The U-Net architecture follows an encoder–decoder structure with skip connections that preserve spatial information, making it well suited for biomedical image segmentation. The U-Net architecture used is shown in Figure 3.9

The segmentation network outputs a binary lung mask $M \in \{0, 1\}^{H \times W}$, which is applied to the preprocessed image using element-wise multiplication:

$$I_{\text{lung}} = I_{\text{norm}} \odot M.$$

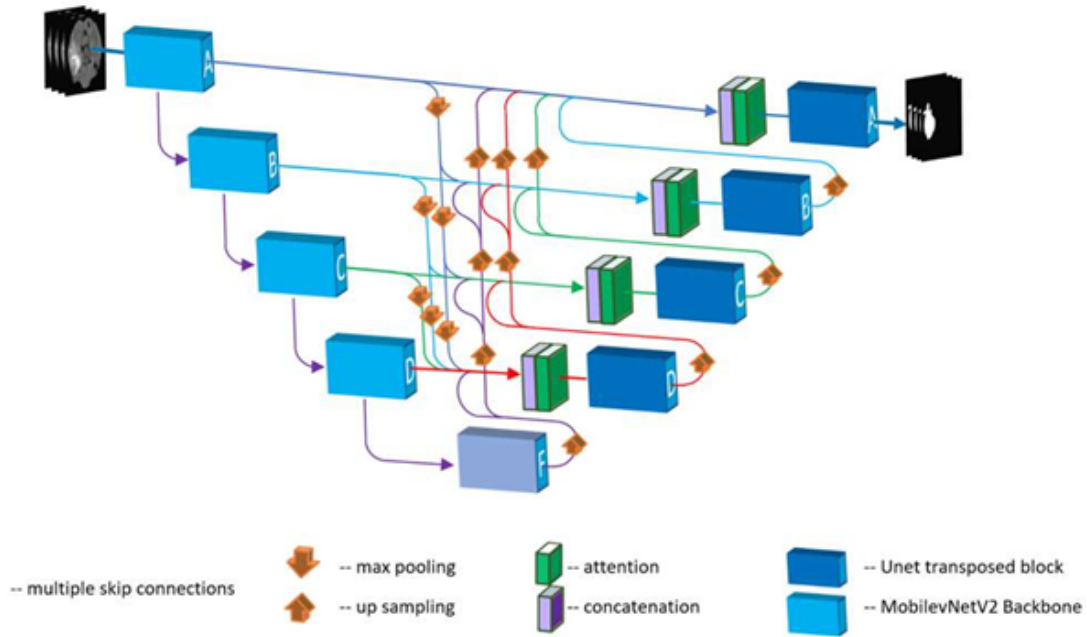


Figure 3.9: The U-Net architecture, illustrating the encoder-decoder structure with skip connections that merge high-resolution features from the contracting path with upsampled features in the expansive path. This design is highly effective for precise medical image segmentation.

Training of the segmentation model is guided by the Dice Loss, derived from the Dice-Sørensen Coefficient (DSC):

$$\text{DSC} = \frac{2|X \cap Y|}{|X| + |Y|},$$

where X and Y denote the predicted and ground-truth masks, respectively. The loss function is defined as

$$\mathcal{L}_{\text{Dice}} = 1 - \text{DSC},$$

which is particularly effective in handling class imbalance between lung and background pixels. The result of using segmentation on CXR images is illustrated in Figure 3.10

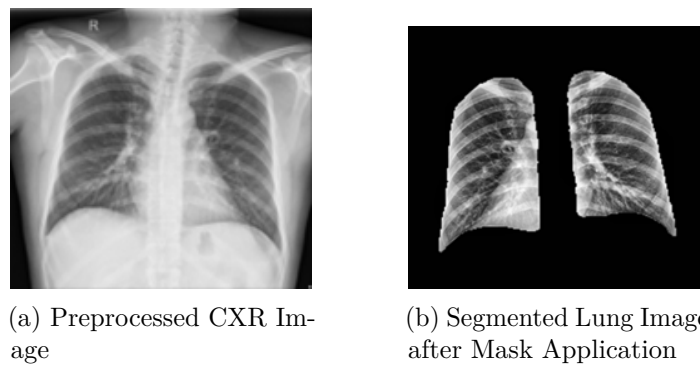


Figure 3.10: Application of the U-Net segmentation mask. The original preprocessed Figure 3.10a is processed to produce the final segmented lung Figure 3.10b, which serves as the input for the classification stage

The final classification stage employs a ResNet50-based transfer learning model. ResNet architectures leverage residual connections to facilitate gradient flow and enable

the training of deep networks. As the ResNet family is discussed in detail earlier in this book, only a brief overview is provided here.

A ResNet50 model pre-trained on ImageNet is used as a feature extractor, with early layers frozen to preserve general visual representations. A custom classification head is appended, consisting of global average pooling, fully connected layers with ReLU activation, dropout for regularization, and a final softmax layer producing class probabilities.

The model is trained using Sparse Categorical Cross-Entropy loss:

$$\mathcal{L}_{\text{CE}} = - \sum_{i=1}^N \log p_{i,c},$$

where $p_{i,c}$ denotes the predicted probability of the true class c for sample i . To address class imbalance, class-weighted loss scaling is applied during training.

To enhance interpretability, Gradient-weighted Class Activation Mapping (Grad-CAM) is employed to visualize the regions most influential to model predictions, an example is shown in Figure 3.11. Grad-CAM computes the gradients of the target class score with respect to the feature maps of the final convolutional layer and produces a coarse localization map highlighting salient regions.

These heatmaps are overlaid on the original chest X-ray images, enabling qualitative assessment of whether the model attends to clinically meaningful lung regions.

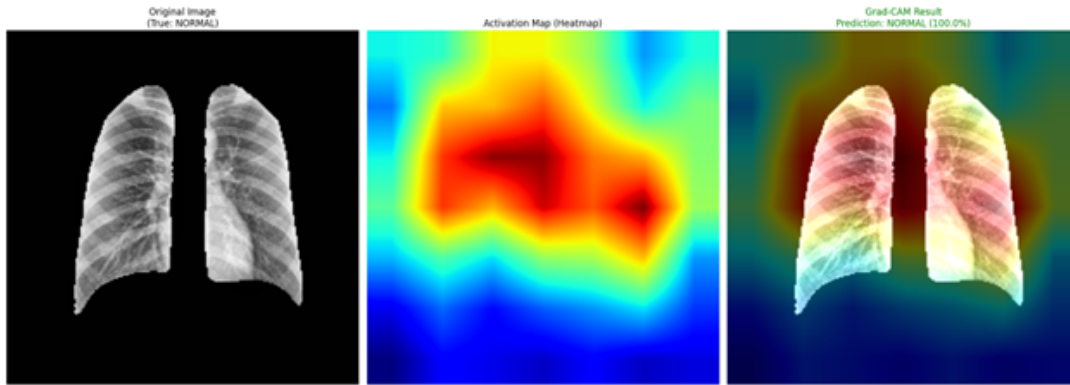


Figure 3.11: Grad-CAM visualizations highlighting regions contributing to model predictions.

3.5 Lung Cancer Detection Model

This section presents the lung cancer and tuberculosis detection model developed within the project, focusing on automated localization and classification of pulmonary abnormalities from chest X-ray (CXR) images[cite: 1, 2]. The model is designed as an end-to-end object detection pipeline that integrates binary mask conversion, bounding box generation, and deep learning-based detection using the YOLOv8 architecture[cite: 3]. Emphasis is placed on the data preparation strategy and architectural design choices that underpin the detection framework[cite: 4].

3.5.1 Motivation

Object detection frameworks offer distinct advantages over image-level classification for medical imaging tasks, as they provide explicit spatial localization of pathological regions rather than a single diagnostic label[cite: 5, 6]. For lung cancer and tuberculosis screening, identifying the precise location and extent of pulmonary lesions or infiltrates is clinically essential, as it directly informs radiological assessment and subsequent clinical decision-making[cite: 7]. The proposed model addresses this requirement by reformulating the diagnostic task as a bounding box regression and classification problem, enabling simultaneous localization and identification of disease regions within a single unified forward pass[cite: 8].

3.5.2 Problem Statement

The objective of this model is to perform multi-class detection of pulmonary abnormalities in chest X-ray images, identifying and localizing instances belonging to the following diagnostic categories[cite: 9, 10]:

$$C = \{\text{tumor, tuberculosis}\}[cite : 11] \quad (3.11)$$

Given an input chest X-ray image $I \in \mathbb{R}^{H \times W}$, the task is to learn a mapping[cite: 12]:

$$f_{\theta} : I \rightarrow \{(b_i, c_i, s_i)\}_{i=1 \dots N}[cite : 13] \quad (3.12)$$

where each tuple (b_i, c_i, s_i) denotes the predicted bounding box coordinates, class label, and confidence score for the i -th detected region, and θ represents the model parameters[cite: 14].

To bridge the gap between pixel-level segmentation annotations (binary masks) and the bounding box representation required by the detection framework, an explicit mask-to-box conversion stage is incorporated as the first step of the pipeline[cite: 15]. Each connected foreground component in the binary annotation mask is converted into a normalized YOLO-format bounding box, enabling the use of richly annotated mask datasets for detection training without manual re-labeling[cite: 16].

3.5.3 Model Architecture

The detection model is structured as a three-stage pipeline consisting of: (1) mask-to-bounding-box conversion and dataset preparation, (2) YOLO-format data organization and configuration, and (3) YOLOv8-based detection model training and evaluation[cite: 17, 18]. Each stage is described in detail below[cite: 19].

Mask-to-Bounding-Box Conversion

Annotation masks for pulmonary lesions are provided as binary PNG images in which foreground pixels (intensity > 127) correspond to pathological regions[cite: 20, 21]. To convert these masks into YOLO-format bounding box annotations, connected component analysis is applied using external contour detection[cite: 22]. For each detected contour,

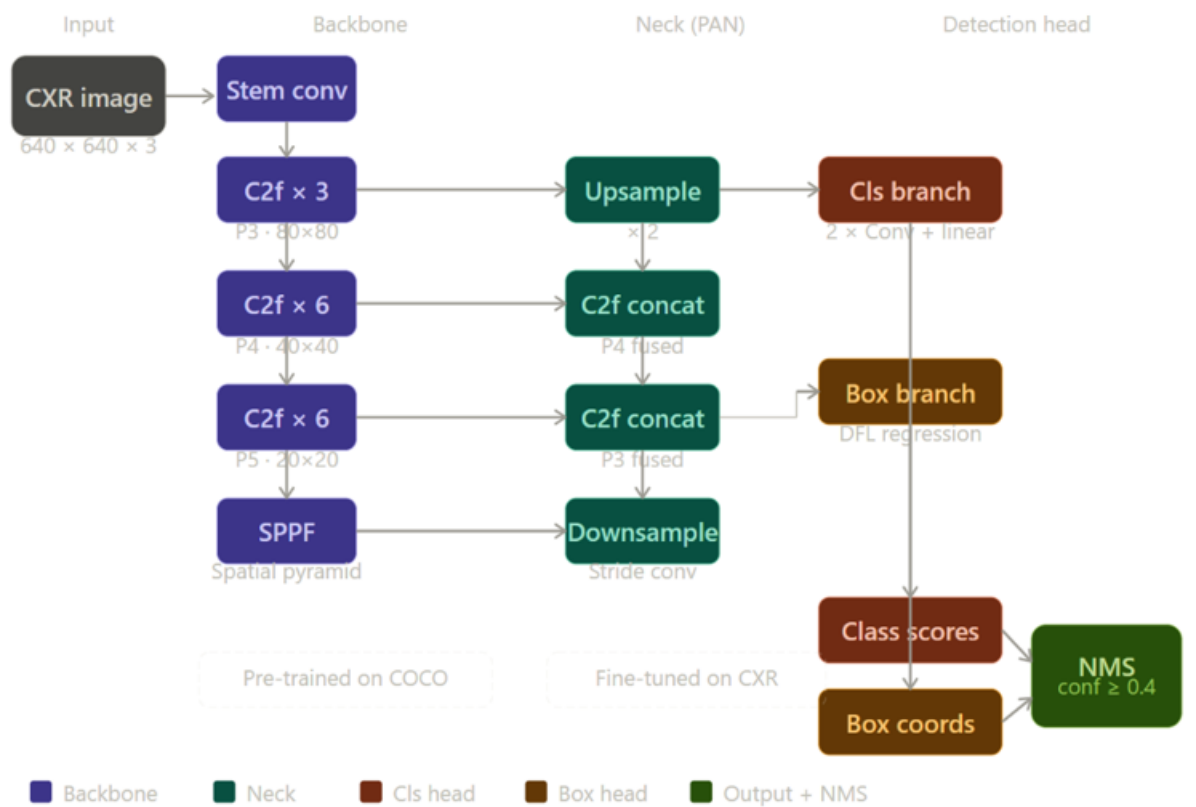


Figure 3.12: YOLOv8 architecture pipeline for automated localization and classification of pulmonary abnormalities.

an axis-aligned bounding box is computed and normalized with respect to the image dimensions, yielding center coordinates and extent values in the range $[0, 1]$ [cite: 23].

Formally, for a contour with pixel-space bounding rectangle (x, y, w, h) within an image of width W and height H , the normalized YOLO representation is[cite: 24]:

$$c_x = \frac{x + w/2}{W}, \quad c_y = \frac{y + h/2}{H}, \quad n_w = \frac{w}{W}, \quad n_h = \frac{h}{H} \text{[cite : 25]} \quad (3.13)$$

Contours enclosing fewer than 5 pixels in either dimension are discarded as noise prior to annotation generation[cite: 26]. All foreground components within a given mask are assigned the class index corresponding to the image-level label (tumor or tuberculosis), since mask annotations are provided per diagnostic category[cite: 27].

Dataset Preparation and Splitting

Following bounding box conversion, the dataset is partitioned into training, validation, and test subsets using stratified random splitting[cite: 28, 29]. The default split proportions are 75% training, 15% validation, and 10% test, with random seed fixed at 42 to ensure reproducibility[cite: 30]. Image files and their corresponding YOLO label files are organized into the standard YOLO directory structure, with separate image and label folders for each split[cite: 31]. A YAML configuration file is generated automatically to specify dataset paths, the number of classes, and class names, serving as the interface between the prepared dataset and the YOLOv8 training framework[cite: 32].

YOLOv8 Detection Architecture

Object detection is performed using YOLOv8, the latest generation of the You Only Look Once (YOLO) family of single-stage detectors[cite: 33, 34]. YOLOv8 follows an anchor-free design and adopts a CSPDarknet backbone for feature extraction, a Path Aggregation Network (PAN) neck for multi-scale feature fusion, and a decoupled detection head that separately predicts classification scores and bounding box offsets[cite: 35]. This design enables real-time inference while maintaining competitive localization accuracy[cite: 36].

A YOLOv8n (nano) variant is employed as the base model, pre-trained on the COCO dataset, and fine-tuned on the prepared CXR detection dataset[cite: 37]. The model is trained for 50 epochs with an input resolution of 640×640 pixels and a batch size of 16[cite: 38]. Transfer learning is leveraged by initializing all network weights from the COCO pre-trained checkpoint, with all layers unfrozen to allow domain-specific adaptation[cite: 38]. The training objective combines binary cross-entropy loss for classification, distribution focal loss (DFL) for bounding box regression, and a complete intersection-over-union (CIoU) penalty term to improve localization precision[cite: 39]:

$$L = \lambda_{\text{cls}} \cdot L_{\text{cls}} + \lambda_{\text{box}} \cdot L_{\text{box}} + \lambda_{\text{dff}} \cdot L_{\text{DFL}} \text{[cite : 40]} \quad (3.14)$$

where λ_{cls} , λ_{box} , and λ_{dff} are scalar weighting coefficients that balance the contribution of each loss component during optimization[cite: 41].

After training, the best-performing checkpoint, selected based on validation mean Average Precision at IoU threshold 0.50 (mAP50), is retained for evaluation[cite: 42]. Model performance is assessed on both the held-out validation and test splits, reporting mAP50, mAP50–95, precision, and recall as primary evaluation metrics[cite: 43]. Qualitative evaluation is conducted by visually inspecting annotated predictions on test set images, examining the spatial correspondence between predicted bounding boxes and known pathological regions[cite: 44].

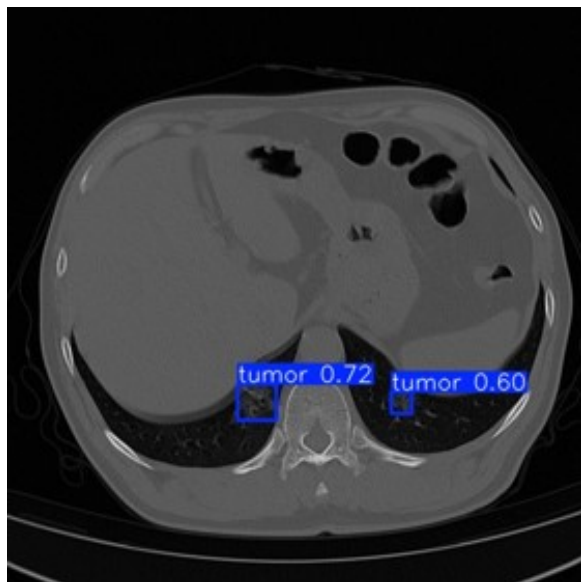


Figure 3.13: Qualitative evaluation showing tumor localization (confidence scores 0.72 and 0.60) on an axial CT slice.

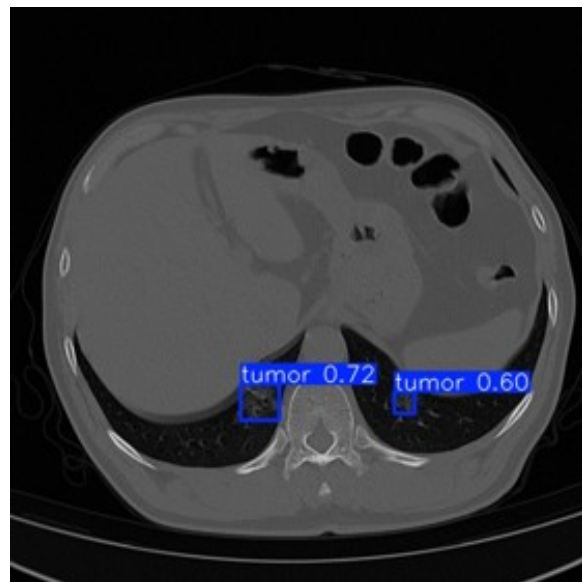


Figure 3.14: Qualitative evaluation showing tumor localization (confidence score 0.87) on an axial CT slice.

3.5.4 Interpretability Considerations

Unlike classification-only models that require post-hoc gradient visualization to identify diagnostically relevant regions, the detection framework provides explicit spatial outputs in the form of bounding boxes and associated confidence scores[cite: 45, 46]. Each prediction directly encodes the model’s estimate of where a pathological region is located and how confident it is in that localization[cite: 47]. This built-in spatial explainability reduces the reliance on auxiliary interpretability methods and aligns more naturally with clinical reporting conventions, in which radiologists annotate specific anatomical regions rather than assigning image-level labels[cite: 48]. Confidence thresholding ($\tau = 0.4$) and non-maximum suppression with an IoU threshold of 0.6 are applied at inference time to filter low-quality or redundant detections, producing a concise set of high-confidence predictions per image[cite: 49].

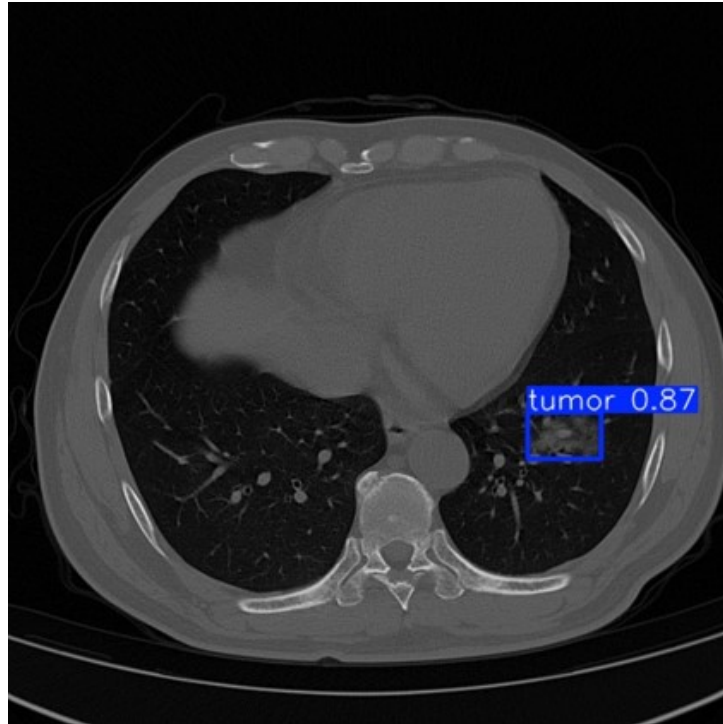


Figure 3.15: Example of an interpretable YOLO detection result on a chest CT image. The model localizes a suspected lesion using a bounding box and reports a confidence score of 0.87. Such spatial predictions provide inherent interpretability by explicitly identifying the anatomical region associated with the model’s decision, reducing the need for post-hoc visualization techniques.

3.6 Federated Learning

3.6.1 Concept and Motivation

Federated learning (FL) is a distributed machine learning paradigm in which a model is trained collaboratively across multiple independent nodes without any raw data leaving its source institution [41]. Rather than centralizing training samples on a shared server, FL brings the computation to the data: each participating node trains a local model on its private dataset and shares only the resulting model parameters with a central aggregation server, which combines the updates into an improved global model. This cycle repeats for a defined number of communication rounds until the global model converges.

The motivation for adopting FL in this work is rooted in the fundamental tension between data-driven model development and patient privacy. Healthcare institutions collect large volumes of annotated medical images, yet regulatory frameworks — including the Health Insurance Portability and Accountability Act (HIPAA) in the United States and the General Data Protection Regulation (GDPR) in the European Union — impose strict constraints on cross-institutional data sharing. Negotiating data transfer agreements is costly, time-consuming, and in many cases legally impractical. FL resolves this tension by enabling institutions to contribute to a shared model without relinquishing data custody, making large-scale collaborative medical AI both legally feasible and technically practical.

3.6.2 Federated Learning in Medical Imaging

The adoption of FL in the medical domain has grown substantially since its introduction, driven by the recognition that many clinical datasets are too small at any single institution to train robust deep learning models, while the combined data across institutions would be sufficient [42]. FL directly addresses this data insufficiency problem without requiring physical data consolidation.

In medical imaging specifically, FL has been applied across a broad range of modalities and anatomical targets. Early work by Sheller *et al.* demonstrated FL for brain tumor segmentation from MRI scans across multiple institutions, achieving performance comparable to centralized training without sharing any patient data [42]. Subsequent studies have extended FL to chest radiograph classification, diabetic retinopathy screening from fundus photography, histopathology slide analysis, and cardiac structure segmentation from echocardiography, among others [43].

This breadth of applicability is a key motivation for the design of the proposed system. Rather than building a solution tied to a specific model architecture or anatomical region, the FL framework developed in this work is designed to be **model-agnostic** and **organ-agnostic**: any classification or segmentation backbone can be substituted as the local model, and the training data can comprise images from any body region or imaging modality. The federated coordination layer — covering client management, weight aggregation, and communication — remains unchanged regardless of the underlying clinical task.

3.6.3 The Federated Averaging Algorithm

The aggregation algorithm that underpins the proposed system is Federated Averaging (FedAvg), introduced by McMahan *et al.* [41] and the most widely adopted FL optimization method to date. FedAvg operates in iterative communication rounds. Let K denote the number of participating clients, $\mathcal{D}_k$ the private dataset of client k , $n_k = |\mathcal{D}_k|$ its size, and $n = \sum_{k=1}^K n_k$ the total number of training samples across all clients. The global model at round t is parameterized by weight vector $\mathbf{w}_t$. Each round consists of three phases:

(1) **Broadcast.** The server distributes the current global model weights to all K clients:

$$\mathbf{w}_t \longrightarrow \{\text{client}_1, \dots, \text{client}_K\}. \quad (3.15)$$

(2) **Local Training.** Each client k independently minimizes its local empirical risk $\mathcal{L}_k$ over E local epochs using a gradient-based optimizer, producing updated local weights:

$$\mathbf{w}_t^k = \mathbf{w}_t - \eta \nabla \mathcal{L}_k(\mathbf{w}_t), \quad (3.16)$$

where η is the local learning rate and $\mathcal{L}_k$ is the task-specific loss function evaluated over the client's local dataset:

$$\mathcal{L}_k(\mathbf{w}) = \frac{1}{n_k} \sum_{(x_i, y_i) \in \mathcal{D}_k} \ell(f_{\mathbf{w}}(x_i), y_i). \quad (3.17)$$

(3) Aggregation. The server collects the locally updated weights from all clients and computes the new global model as a weighted average proportional to each client's dataset size:

$$\mathbf{w}_{t+1} = \sum_{k=1}^K \frac{n_k}{n} \mathbf{w}_t^k. \quad (3.18)$$

This process minimizes the global objective:

$$F(\mathbf{w}) = \sum_{k=1}^K \frac{n_k}{n} \mathcal{L}_k(\mathbf{w}). \quad (3.19)$$

Equation (3.18) assigns greater influence to clients with larger datasets, which is appropriate in the medical setting where institutional dataset sizes vary considerably. Clients contributing more annotated images should exert proportionally greater influence on the global model.

3.6.4 Key Parameters of a Federated Learning System

Configuring a federated learning system involves a distinct set of parameters beyond those of conventional deep learning. These parameters govern the coordination between server and clients and directly affect convergence speed, model quality, communication cost, and privacy. A clear understanding of each parameter is essential for adapting the system to different clinical tasks. Table 3.2 provides an overview of the critical parameters and their implications.

Table 3.2: Key Federated Learning Parameters and Their Clinical Implications

Parameter	Definition	Implication
Number of clients (K)	Total participating institutions	More clients improve data diversity and model generalization but increase coordination overhead.
Communication rounds (T)	Number of server–client training cycles	More rounds improve accuracy at the cost of higher communication volume. Convergence is typically monitored to select T .
Client fraction (C)	Proportion of clients selected per round ($C \in (0, 1]$)	$C = 1.0$ uses all clients every round (synchronous FL). $C < 1.0$ enables asynchronous, scalable training.
Local epochs (E)	Training epochs each client performs before returning weights	Higher E reduces communication rounds but risks <i>client drift</i> — divergence of local models due to non-IID data [44].
Local batch size (B)	Mini-batch size for local gradient computation	Affects local convergence speed, memory footprint, and gradient privacy (smaller batches increase reconstruction attack risk).
Local learning rate (η)	Step size for local gradient updates	Must be tuned per model architecture; too large a value exacerbates client drift in non-IID settings.
Aggregation strategy	Algorithm used to combine client updates (e.g., FedAvg, FedProx, FedNova)	Choice depends on data heterogeneity; FedAvg is the standard baseline; FedProx adds a proximal term to limit client drift.
Data partitioning scheme	How training data is divided across clients (IID vs. non-IID)	Non-IID partitions (realistic in healthcare) slow convergence and reduce final accuracy compared to IID partitions.
Minimum clients for aggregation	Minimum number of client updates required before the server aggregates	Prevents the global model from being updated on insufficient information; critical for stability.

Among these, the number of local epochs E and the aggregation strategy warrant particular attention in the medical imaging context. Increasing E reduces the number of costly communication rounds, which is beneficial in healthcare networks where bandwidth and uptime may be limited. However, when client datasets are statistically heterogeneous — a common condition across hospitals using different scanners, populations, or labeling protocols — high E causes the local models to overfit to their local distributions and

diverge from the global optimum, a phenomenon known as *client drift* [44]. The selection of E therefore requires careful empirical tuning for each deployment scenario.

3.6.5 Privacy and Security

The primary privacy guarantee of federated learning is that raw data never leaves its source institution. The server observes only model weight tensors, which are orders of magnitude smaller than the original imaging data and contain no explicit patient identifiers. This stands in contrast to centralized approaches, where a single data breach exposes the entire training corpus. Table 3.3 summarizes the key differences.

Table 3.3: Privacy and Security: Centralized vs. Federated Learning

Property	Centralized	Federated
Raw data leaves institution	Yes	No
Single point of data breach	Yes	No
HIPAA / GDPR compliance	Difficult	Easier
Shared artifact	Full dataset	Model weights
Supports differential privacy	N/A	Yes
Resilient to server breach	No	Partial

It is important to acknowledge that gradient sharing is not unconditionally private. Gradient inversion attacks [45] have demonstrated that an adversarial server can reconstruct approximate training images from shared weight updates, particularly at small batch sizes. Two complementary mitigations are incorporated into the design of the proposed system:

Batch size selection. Using a sufficiently large mini-batch (e.g., $B \geq 32$) aggregates gradient information across many samples, significantly reducing the fidelity of any reconstruction attempt.

Differential Privacy (DP). For deployments requiring formal privacy guarantees, the system supports (ϵ, δ) -DP [46] by adding calibrated Gaussian noise to client updates before transmission:

$$\tilde{\mathbf{w}}_t^k = \mathbf{w}_t^k + \mathcal{N}(0, \sigma^2 C^2 \mathbf{I}), \quad (3.20)$$

where C is the gradient clipping threshold and σ is the noise multiplier. The privacy budget ϵ quantifies the worst-case privacy loss: smaller ϵ implies stronger privacy at the cost of model utility.

3.6.6 What Makes the Proposed System Distinctive

While federated learning has been applied in various medical imaging studies, the majority of existing implementations are designed around a fixed model architecture and a specific anatomical target, limiting their transferability to new clinical tasks. The proposed framework differentiates itself through the following design principles:

Model Agnosticism. The federated coordination layer — encompassing client management, weight serialization, aggregation, and communication — is fully decoupled from the local model definition. Any differentiable model that can be expressed as a parameter vector (e.g., CNNs for classification, U-Net variants for segmentation, Vision Transformers) can be substituted as the local model without modifying the federated infrastructure. This makes the system immediately applicable to new imaging tasks simply by swapping the model definition on the client side.

Organ and Modality Agnosticism. The system imposes no constraints on the nature of the input images. Training data may originate from any body region (chest, brain, abdomen, retina) and any imaging modality (X-ray, CT, MRI, ultrasound, fundus photography). The federated server is entirely unaware of image content; it operates solely on weight tensors.

Staged Data Integration. The system incorporates a staged data feeding protocol that progressively increases each client’s active training subset across federated rounds. This design reflects the clinical reality that expert-annotated medical imaging datasets grow incrementally over time as radiologists label new cases. Rather than requiring a fully annotated dataset before training begins, the system accommodates continuous data growth, enabling earlier model deployment and iterative improvement as annotation progresses.

Containerized Reproducibility. Each participating node is deployed as an isolated Docker container, ensuring that the federated environment is fully reproducible across different hardware configurations and operating systems. The container-based architecture also simplifies scaling: adding a new institution requires only the deployment of an additional client container with the appropriate data volume mount, with no changes to the server or existing clients.

Framework Choice. The system is implemented using the Flower (flwr) framework [47], selected for its support of multiple ML backends (PyTorch, TensorFlow, JAX), its clean abstraction of the client-server protocol, and its active development for healthcare FL research. The backend-agnostic design of Flower further reinforces the model-agnostic objective of the proposed system.

Collectively, these properties position the proposed FL framework as a general-purpose infrastructure for privacy-preserving collaborative medical image analysis, extensible to a wide range of clinical specialties and diagnostic tasks without architectural redesign.

3.7 Summary

This chapter presented the proposed end-to-end model for automated detection of chest diseases and bone fractures from medical images. The overall solution architecture was introduced, highlighting the integration of a deep learning-based diagnostic pipeline with supporting backend processes to enable efficient and scalable medical image analysis.

The deep learning pipeline was described in detail, illustrating the sequential flow

from image preprocessing and feature extraction to intelligent routing and task-specific inference. Transfer learning was adopted as a core strategy to leverage pre-trained deep neural networks, enabling effective feature representation and improved generalization in the presence of limited labeled medical data.

The chapter also introduced the medical image routing model, which enables dynamic task allocation by directing input images to specialized diagnostic models. This design enhances modularity and allows the system to support multiple medical imaging tasks within a unified framework.

Then, the architecture of the chest disease classification model was discussed in detail. The design was illustrated, and the use of ResNet50 as the main feature extractor was justified. The formulation of the problem and the motivation behind such a model were presented while detailed architectural aspects of the transfer-learning model were omitted to avoid redundancy.

Finally, the bone fracture detection model was presented as a task-specific component of the system, implemented using an unmodified YOLOv8 architecture. The formulation of the fracture detection problem and the motivation for adopting a one-stage object detection approach were discussed, while detailed architectural aspects were referenced to earlier sections to avoid redundancy.

Overall, this chapter established the structural and methodological foundation of the proposed system, providing a clear understanding of its components and design choices. The described models form the basis for the experimental evaluation and performance analysis presented in the subsequent chapter.

CHAPTER 4 SYSTEM IMPLEMENTATION

Chapter 4 System Implementation

The previous chapter presented the conceptual design of the proposed framework, describing the overall solution architecture, its constituent components, and the rationale behind the adoption of a two-level classification pipeline for medical image analysis. While the architectural design establishes the theoretical foundation of the system, the successful realization of such a framework requires careful implementation decisions that translate these concepts into a functional and deployable solution. Therefore, this chapter focuses on the practical aspects of developing the proposed system, providing a detailed description of the methodologies, technologies, and engineering processes employed throughout the implementation phase.

Developing an intelligent medical image analysis framework involves considerably more than selecting suitable machine learning models. The implementation process encompasses a broad range of interconnected tasks, including data acquisition and preprocessing, model development, training pipeline construction, performance optimization, explainability integration, federated learning infrastructure design, and system-level deployment considerations. Each of these components introduces unique technical requirements that must be addressed to ensure that the final system operates reliably, efficiently, and consistently under realistic conditions.

The implementation of the proposed framework was guided by the architectural principles established in the previous chapter. Particular emphasis was placed on maintaining modularity, scalability, and reproducibility throughout the development process. As a result, the system was designed as a collection of loosely coupled components that can be independently developed, evaluated, and improved without requiring significant modifications to the remainder of the framework. Such an approach not only facilitates future enhancements but also supports the integration of additional disease classifiers, imaging modalities, and collaborative learning mechanisms as the system evolves.

A significant portion of the implementation effort was dedicated to constructing a robust data processing pipeline capable of handling heterogeneous medical imaging datasets originating from multiple sources. Medical images frequently exhibit substantial variability in terms of resolution, image quality, acquisition protocols, and annotation standards. These variations introduce challenges that can negatively impact model performance if not properly addressed. Consequently, extensive preprocessing procedures were developed to standardize input data and ensure compatibility across the different classification stages of the framework.

The implementation process also involved the development and training of multiple deep learning models corresponding to the various components of the proposed architecture. These models include the modality routing network responsible for distinguishing between X-ray images, CT scans, and out-of-distribution samples, the multi-label chest disease classifier designed to detect COVID-19, Tuberculosis, and Pneumonia from chest radiographs, and the lung cancer detection model operating on computed tomography

images. The selection of network architectures, optimization strategies, hyperparameter configurations, and training procedures was driven by both empirical evaluation and the specific requirements of each diagnostic task.

In addition to predictive performance, considerable attention was devoted to the interpretability and practical usability of the framework. Explainable Artificial Intelligence (XAI) techniques were integrated into the implementation to provide visual explanations that assist users in understanding model predictions. The incorporation of explainability mechanisms required the development of additional processing pipelines capable of generating meaningful visualizations while maintaining compatibility with the underlying deep learning models.

Another important aspect of the implementation concerns the integration of Federated Learning capabilities. While the primary objective of the classification framework is accurate disease detection, long-term adoption within healthcare environments requires mechanisms that support privacy-preserving collaboration among institutions. Consequently, the system was implemented with the necessary infrastructure to facilitate distributed model training and parameter aggregation without requiring the exchange of sensitive patient data. This component introduces additional technical considerations related to communication, synchronization, and model update management that are discussed throughout this chapter.

As with any complex engineering project, the implementation process presented a variety of technical challenges. These challenges ranged from issues related to dataset heterogeneity and class imbalance to computational limitations, model convergence difficulties, and the integration of multiple independent components into a unified workflow. Addressing these obstacles required iterative experimentation, extensive evaluation, and the adoption of various optimization techniques. Understanding these challenges and the solutions developed to overcome them provides valuable insight into the practical realities of building large-scale artificial intelligence systems for healthcare applications.

Accordingly, this chapter not only describes the final implementation of the proposed framework but also documents the key challenges encountered during development and the engineering decisions that enabled their resolution. By presenting both the implementation details and the problem-solving process behind them, this chapter provides a comprehensive account of how the theoretical architecture introduced in the previous chapter was transformed into a fully operational medical image analysis system.

4.1 Implementation of the First-Level Routing Model

The first component implemented within the proposed framework is the modality routing model. As discussed in Chapter ??, this model serves as the entry point of the entire classification pipeline and is responsible for determining the type of the incoming image before directing it toward the appropriate diagnostic model. Since the subsequent classification stages were specifically designed for particular imaging modalities, ensuring

the accuracy of the routing stage was considered a critical requirement for the reliability of the overall framework.

The routing model was formulated as a three-class image classification problem in which each input image was assigned to one of three categories: chest X-ray images, chest CT scans, or out-of-distribution (OOD) images. The OOD category was intentionally incorporated to improve the robustness of the system during deployment. In practical healthcare environments, medical information systems may contain a wide variety of image types that fall outside the scope of the proposed framework. Without an explicit OOD category, unsupported images could be mistakenly forwarded to disease classification models, potentially resulting in unreliable predictions. The inclusion of this category therefore acts as a safety mechanism that prevents unsupported inputs from entering the diagnostic pipeline.

The final routing dataset was constructed by aggregating images originating from multiple sources. After dataset integration, the X-ray category contained 7,123 images, the CT category contained 2,534 images, and the OOD category contained 9,958 images. Although this aggregation process provided a sufficiently diverse collection of samples, it also introduced a noticeable class imbalance problem. The CT class represented a considerably smaller portion of the dataset compared to the other categories, creating a risk that the trained model would become biased toward the more frequently represented classes.

To mitigate this issue, several balancing strategies were incorporated into the training pipeline. Data augmentation techniques were applied to increase the effective diversity of minority-class samples while simultaneously improving model generalization. Augmentation operations included random rotations, horizontal flipping, zoom transformations, and intensity variations. These transformations were selected because they preserve medically relevant structures while exposing the model to realistic image variations that may occur across different acquisition settings. In addition to augmentation, class balancing procedures were employed during dataset preparation to reduce the influence of class distribution disparities on the learning process. These measures contributed to a more stable training process and improved the model's ability to recognize underrepresented classes.

The complete routing workflow implemented within the framework is illustrated in Figure 4.1. The figure summarizes both the operational behavior of the routing model and the architectures investigated during the model selection process.

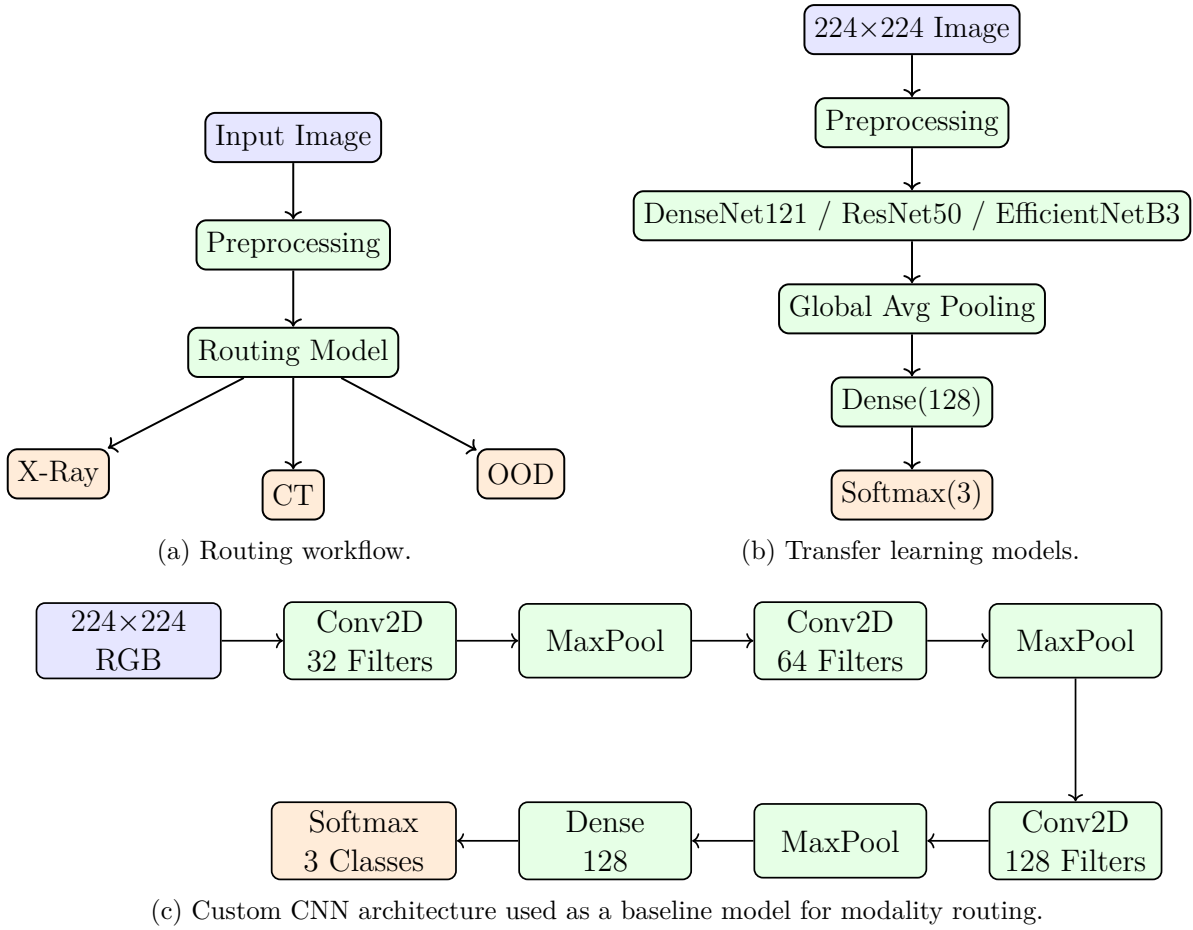


Figure 4.1: Architectures and workflow investigated during the implementation of the routing model. Blue blocks represent inputs, green blocks denote processing stages, and orange blocks correspond to classification outputs.

Rather than committing immediately to a single architecture, several alternative models were implemented and evaluated in order to identify the most suitable solution for the routing task. This decision was motivated by the observation that image modality classification differs substantially from disease classification. While disease detection often requires learning subtle pathological patterns, modality recognition primarily depends on identifying broader structural and acquisition-specific characteristics. Consequently, it was necessary to experimentally determine whether a lightweight custom architecture would be sufficient or whether a deeper pretrained network would provide superior performance.

The first architecture investigated was a custom convolutional neural network developed specifically for the routing task. The model begins with a rescaling layer that normalizes pixel intensities into the range $[0, 1]$. Feature extraction is then performed through three convolutional blocks containing 32, 64, and 128 filters respectively. Each convolutional layer uses a ReLU activation function and is followed by a max-pooling operation that progressively reduces the spatial dimensions of the feature maps while preserving the most salient visual information. After the final convolutional block, the extracted features are flattened and passed through a fully connected layer consisting of 128 neurons. The output layer employs a softmax activation function to generate probability estimates for the three target classes.

Although the custom CNN provided a relatively lightweight solution with low computational requirements, the project also explored the use of transfer learning techniques. Transfer learning has become a widely adopted strategy in medical image analysis because it enables models to leverage feature representations learned from large-scale image datasets. Instead of training a deep architecture entirely from scratch, a pretrained backbone can be used as a feature extractor while only a small classification head is trained on the target dataset.

Three state-of-the-art architectures were evaluated during experimentation: ResNet50, DenseNet121, and EfficientNetB3. For each architecture, the pretrained backbone was initialized using publicly available weights and subsequently frozen during training. Freezing the backbone prevented updates to the pretrained parameters and allowed the network to function as a fixed feature extractor. A Global Average Pooling layer was then applied to aggregate spatial information into a compact feature vector, followed by a fully connected layer containing 128 neurons and a final softmax classification layer producing probabilities for the three routing classes.

The implementation of transfer learning significantly reduced training complexity while benefiting from the rich visual representations learned by these networks during large-scale pretraining. However, preliminary experiments revealed that different architectures exhibited varying levels of sensitivity to optimization parameters. Consequently, additional effort was dedicated to hyperparameter tuning in order to maximize classification performance.

One of the most important optimization parameters investigated during experimentation was the learning rate. The learning rate directly controls the magnitude of weight updates during optimization and can substantially influence convergence behavior. A learning rate that is too large may cause unstable training, whereas an excessively small learning rate may prevent efficient convergence. Since each pretrained architecture possesses unique characteristics and parameter distributions, a single learning rate was not expected to perform optimally across all models.

To address this issue, architecture-specific learning rates were introduced. ResNet50 and DenseNet121 were trained using a learning rate of 10^{-2} , while EfficientNetB3 was trained using a learning rate of 10^{-4} . These values were selected based on empirical experimentation and repeated evaluation of training behavior. The custom CNN utilized the default Adam optimization configuration due to its comparatively smaller parameter space. This targeted hyperparameter tuning strategy improved convergence stability and ensured that each architecture was evaluated under favorable training conditions.

All routing models were trained using the Adam optimization algorithm and sparse categorical cross-entropy loss. The use of sparse categorical cross-entropy was appropriate because the routing task involves mutually exclusive classes, where each image belongs to exactly one category. Model performance was monitored using classification accuracy, which provided a direct measure of the model's ability to correctly identify image modalities.

To ensure a fair and consistent evaluation process, all experiments were conducted using identical dataset partitions. The aggregated dataset was divided into training,

validation, and testing subsets according to a ratio of 70%, 15%, and 15%, respectively. Training was performed using a batch size of 8 and a total of 10 epochs. Maintaining a consistent experimental protocol allowed the performance differences observed between models to be attributed primarily to architectural characteristics rather than variations in the training procedure.

Following extensive experimentation, DenseNet121 was selected as the final routing model incorporated into the proposed framework. The architecture demonstrated the most favorable balance between classification performance, convergence behavior, and generalization capability. DenseNet121's dense connectivity pattern enables efficient feature reuse and improved gradient propagation, characteristics that proved particularly beneficial for distinguishing between imaging modalities exhibiting substantial structural differences. Furthermore, the architecture achieved strong performance while maintaining a relatively efficient parameter utilization strategy compared to alternative deep learning models.

The final routing model therefore consists of a DenseNet121 feature extraction backbone followed by a Global Average Pooling layer, a fully connected layer with 128 neurons, and a three-class softmax output layer. Once deployed within the complete framework, the model receives an input image, determines whether the image corresponds to a chest X-ray, a CT scan, or an out-of-distribution sample, and subsequently forwards the image to the appropriate downstream component. This routing decision constitutes the first stage of the proposed hierarchical classification pipeline and forms the foundation upon which all subsequent diagnostic analyses are performed.

4.2 Implementation of the Lung Cancer Detection Model

The second major component implemented within the proposed framework is the lung lesion detection model. Unlike conventional image classification systems that only determine whether an abnormality is present, the proposed approach performs object detection, enabling simultaneous identification and localization of suspicious pulmonary lesions within CT images. The model predicts both the spatial coordinates of lesion regions and the confidence associated with each prediction, thereby providing clinically meaningful visual outputs that can assist radiologists during image interpretation.

The detection pipeline was developed using the YOLOv8 architecture, which belongs to the family of single-stage object detectors. Single-stage detectors perform localization and classification simultaneously within a unified network, eliminating the need for separate region proposal and classification stages. This design significantly reduces computational complexity and inference time while maintaining competitive detection performance.

The complete implementation workflow consists of five primary stages:

1. Dataset preprocessing
2. Annotation generation

3. Dataset partitioning
4. YOLOv8 model training
5. Performance evaluation and visualization

4.2.1 Dataset Preparation and Annotation Generation

The original dataset contains CT images together with binary segmentation masks that identify lesion regions. Although segmentation masks provide pixel-level information regarding lesion boundaries, the YOLO detection framework requires object-level annotations represented by bounding boxes. Consequently, a dedicated annotation generation pipeline was implemented to automatically convert segmentation masks into YOLO-compatible labels.

For every image-mask pair, the following operations were performed:

Step 1: Mask Loading Each binary mask was loaded as a grayscale image where:

- Pixel value 0 represented background tissue.
- Pixel value 255 represented lesion pixels.

The masks were subsequently thresholded to ensure binary separation between foreground and background regions.

Step 2: Contour Extraction The OpenCV contour detection algorithm was applied to identify connected lesion regions:

```
contours, _ = cv2.findContours(  
    mask,  
    cv2.RETR_EXTERNAL,  
    cv2.CHAIN_APPROX_SIMPLE  
)
```

Only external contours were retained in order to avoid duplicate detections resulting from nested structures. The contour extraction stage transformed pixel-level lesion masks into geometrically defined lesion candidates.

Step 3: Bounding Box Generation For each detected contour, the minimum enclosing rectangle was computed:

```
x, y, w, h = cv2.boundingRect(contour)
```

where x, y represent the upper-left corner, w represents width, and h represents height. These coordinates define the lesion localization region used by the detector during training.

Step 4: YOLO Coordinate Conversion YOLO requires normalized coordinates expressed relative to image dimensions. The generated bounding boxes were therefore converted using:

$$x_{\text{center}} = \frac{x + w/2}{W} \quad (4.1)$$

$$y_{\text{center}} = \frac{y + h/2}{H} \quad (4.2)$$

$$w_{\text{norm}} = \frac{w}{W}, \quad h_{\text{norm}} = \frac{h}{H} \quad (4.3)$$

where W = image width and H = image height. The resulting annotation was stored using YOLO format:

```
class_id x_center y_center width height
```

Since only one lesion category was considered, all annotations were assigned `class_id = 0`.

Step 5: Label File Generation For each image (e.g., `image_001.png`), a corresponding annotation file was created (`image_001.txt`) containing all lesion bounding boxes present in the image. This automated procedure eliminated the need for manual annotation and ensured consistency throughout the dataset.

4.2.2 Dataset Organization

Following annotation generation, the dataset was reorganized according to the standard YOLO directory structure:

```
dataset/
|-- train/
|   |-- images/
|   '-- labels/
|-- valid/
|   |-- images/
|   '-- labels/
'-- test/
    |-- images/
    '-- labels/
```

Maintaining this structure allows direct integration with the Ultralytics YOLO training framework. A YAML configuration file was subsequently generated containing:

```

path: dataset
train: train/images
val: valid/images
test: test/images
nc: 1
names:
0: tumor

```

The YAML file serves as the interface between the dataset and the YOLO training engine.

4.2.3 YOLOv8 Detection Architecture

The proposed detection model employs YOLOv8 as the underlying deep learning architecture. The network consists of three major components:

- **Backbone:** The backbone is responsible for hierarchical feature extraction. Low-level layers capture edges, textures, and intensity gradients, while deeper layers learn lesion morphology, shape characteristics, and contextual lung structures.
- **Neck:** The neck aggregates information from multiple feature scales using Feature Pyramid Network (FPN) and Path Aggregation Network (PAN) mechanisms. This multi-scale representation is particularly important in lung lesion detection because lesions may vary considerably in size. Small nodules and large masses must both be represented effectively within the feature hierarchy.
- **Detection Head:** The detection head predicts bounding box coordinates, object confidence scores, and class probabilities for each candidate lesion. Unlike anchor-based detectors, YOLOv8 utilizes an anchor-free detection mechanism that simplifies prediction and improves localization accuracy.

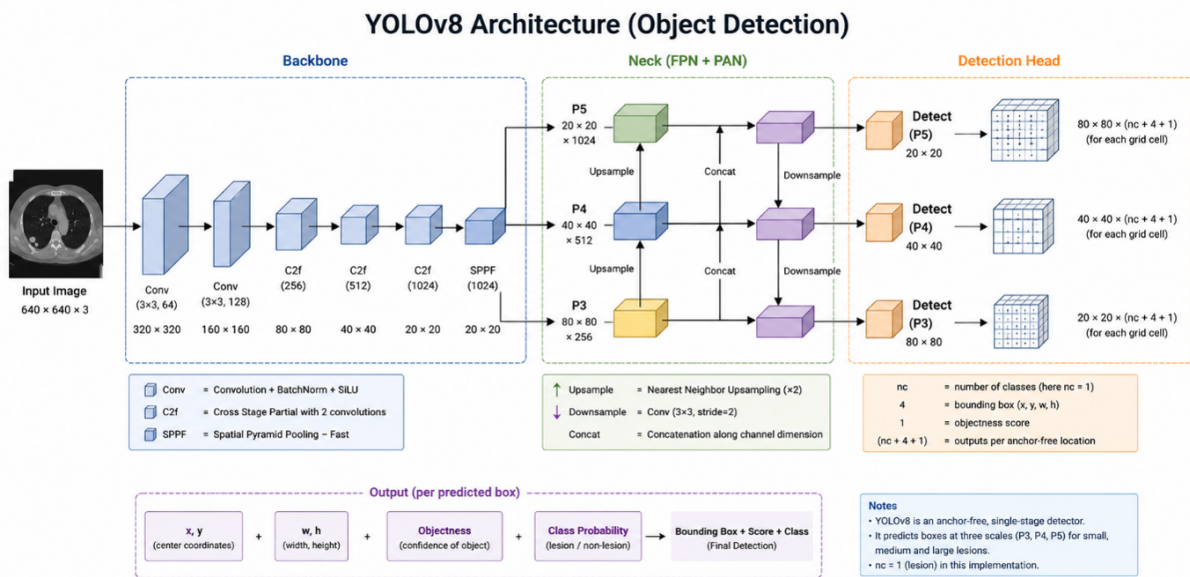


Figure 4.2: Detailed architectural diagram of the YOLOv8 object detection framework, illustrating the Backbone, Neck (FPN + PAN), and Detection Head stages for multi-scale prediction.

4.2.4 Transfer Learning Strategy

Training a deep object detector entirely from scratch typically requires extremely large medical imaging datasets. To overcome this limitation, transfer learning was employed. The detector was initialized using pretrained YOLOv8 weights trained on large-scale image datasets. The pretrained model already possesses strong low-level feature extraction capabilities, including detection of edges, contours, textures, and spatial patterns. Fine-tuning these representations on lung CT images significantly accelerates convergence while reducing the risk of overfitting.

4.2.5 Training Configuration

Model training was performed using the Ultralytics implementation of YOLOv8. The training pipeline automatically performs image loading, resizing, batching, label matching, loss computation, and weight optimization. Input images were resized to 640×640 pixels to ensure uniform tensor dimensions throughout training.

The optimization process simultaneously minimizes:

- **Localization Loss:** Measures bounding-box accuracy.
- **Classification Loss:** Measures lesion classification accuracy.
- **Distribution Focal Loss (DFL):** Improves bounding-box regression precision.

The total loss is defined as:

$$L_{\text{total}} = L_{\text{box}} + L_{\text{cls}} + L_{\text{DFL}} \quad (4.4)$$

During training, stochastic gradient optimization updates network weights through back-propagation until convergence.

4.2.6 Evaluation Methodology

After training, the best-performing model checkpoint was selected automatically according to validation performance. Performance was evaluated using standard object detection metrics:

Precision Measures the proportion of predicted lesions that are truly lesions.

$$\text{Precision} = \frac{\text{TP}}{\text{TP} + \text{FP}} \quad (4.5)$$

Recall Measures the proportion of true lesions successfully detected.

$$\text{Recall} = \frac{\text{TP}}{\text{TP} + \text{FN}} \quad (4.6)$$

Mean Average Precision (mAP) The primary detection metric used for evaluation. Two variants were reported:

- mAP@50
- mAP@50–95

mAP@50 evaluates detections using a single IoU threshold of 0.50, whereas mAP@50–95 provides a more stringent evaluation across multiple thresholds.

4.2.7 Prediction Visualization and Model Interpretability

A key advantage of the proposed detection framework is its inherent interpretability. Each prediction generated by YOLOv8 contains bounding box coordinates, a confidence score, and a class label. The confidence score reflects the model’s certainty regarding the presence of a lesion within the predicted region.

Unlike classification-only approaches that require external explainability methods such as Grad-CAM or saliency maps, the detector directly highlights suspicious anatomical regions through bounding boxes. This spatial localization capability allows clinicians to visually inspect the exact image regions influencing the model’s decision, thereby improving transparency and facilitating clinical validation of automated predictions.

CHAPTER 5 RESULTS & EVALUATION

Chapter 5 Results and Evaluation

This chapter presents preliminary results and an initial evaluation of the proposed models. The current version of the chapter is intended as a draft to demonstrate that the models have been successfully developed, trained, and integrated within the proposed system, and that a subset of the required experimental results has already been obtained. While the presented findings provide early evidence of the feasibility and effectiveness of the proposed approach, the evaluation will be expanded and refined in the final version of the manuscript to include additional experiments, metrics, and comparative analyses.

5.1 Evaluation Metrics

To quantitatively evaluate the performance of the proposed models, standard classification metrics are employed, namely precision, recall, accuracy, and F1-score. These metrics are computed based on the confusion matrix, which consists of true positives (TP), true negatives (TN), false positives (FP), and false negatives (FN).

Precision measures the proportion of correctly predicted positive samples among all samples predicted as positive and is defined as:

$$\text{Precision} = \frac{TP}{TP + FP}$$

Recall, also referred to as sensitivity, measures the proportion of correctly predicted positive samples among all actual positive samples and is given by:

$$\text{Recall} = \frac{TP}{TP + FN}$$

Accuracy represents the overall correctness of the model by measuring the proportion of correctly classified samples among all samples and is defined as:

$$\text{Accuracy} = \frac{TP + TN}{TP + TN + FP + FN}$$

The F1-score provides a single metric that balances precision and recall by computing their harmonic mean. It is particularly useful in scenarios involving class imbalance, which are common in medical imaging applications. The F1-score is defined as:

$$\text{F1-score} = 2 \times \frac{\text{Precision} \times \text{Recall}}{\text{Precision} + \text{Recall}}$$

Together, these metrics offer a comprehensive evaluation of model performance. Precision reflects the model's ability to minimize false positive predictions, recall evaluates

its effectiveness in identifying positive cases, accuracy provides an overall measure of correctness, and the F1-score captures the trade-off between precision and recall. This combination of metrics ensures a balanced and reliable assessment of the proposed models across different diagnostic tasks.

5.2 Results

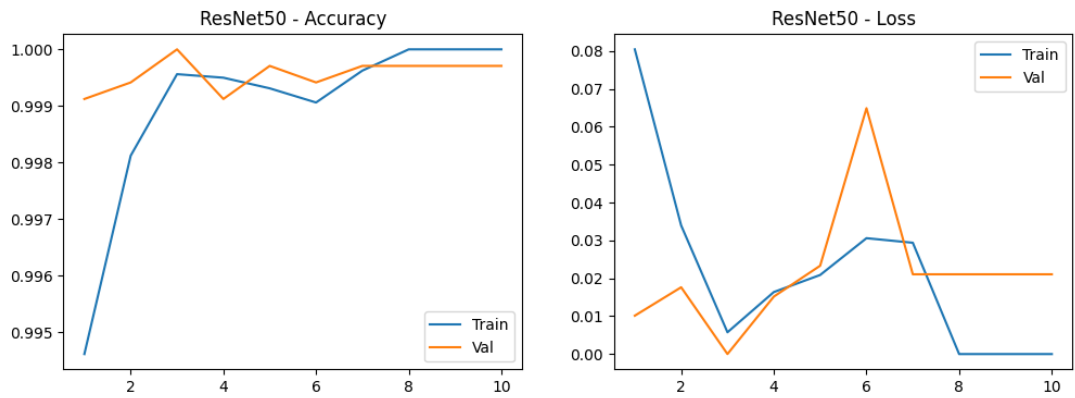


Figure 5.1: Training dynamics of the medical image routing model.

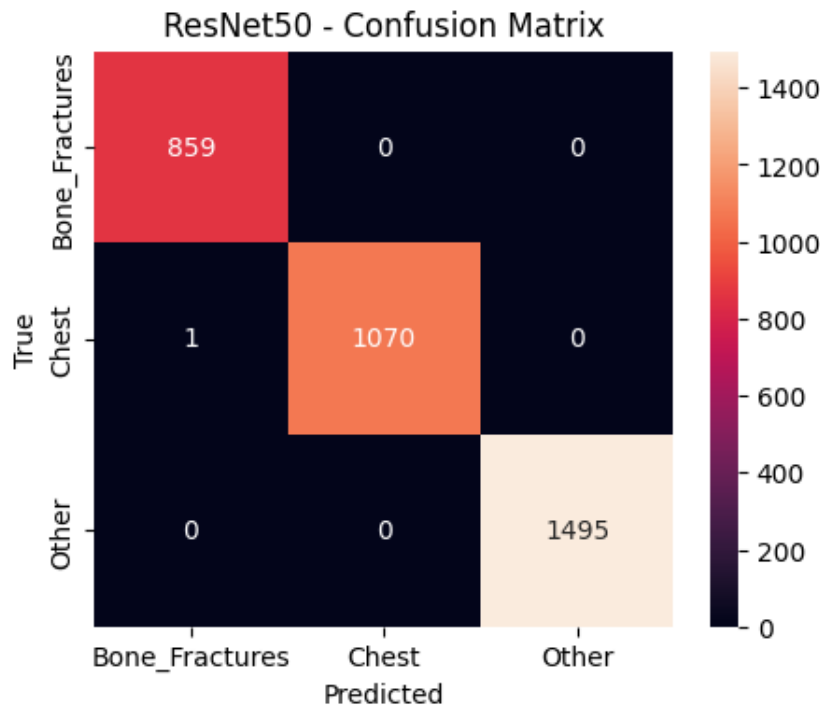


Figure 5.2: Confusion matrix of the medical image routing model.

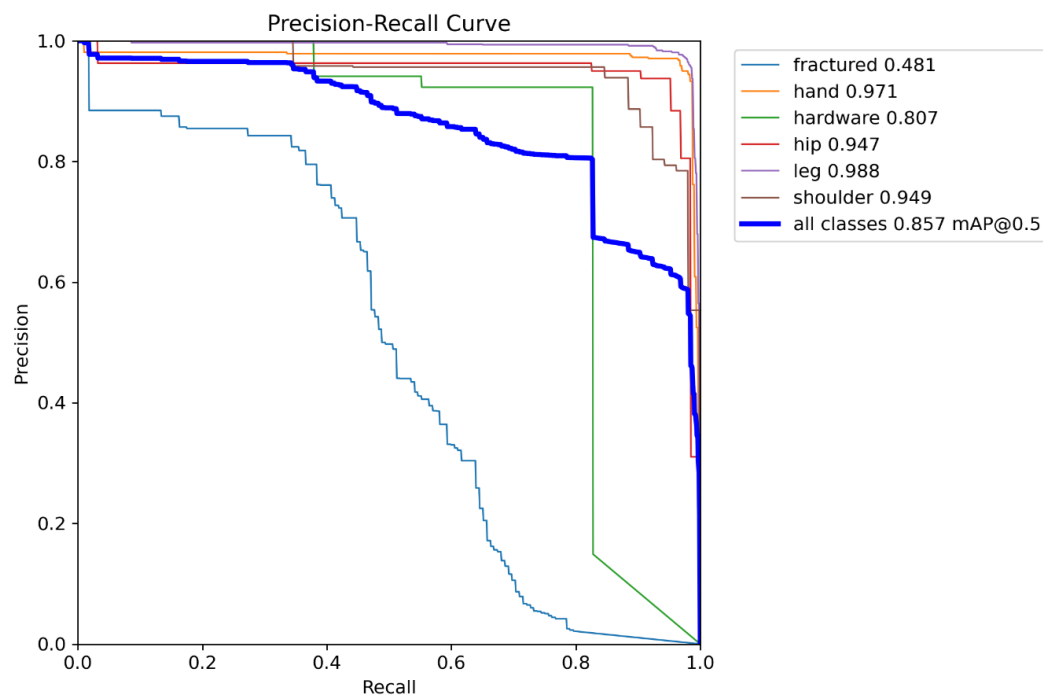


Figure 5.3: Bone fracture detection model Recall-Precision curve.

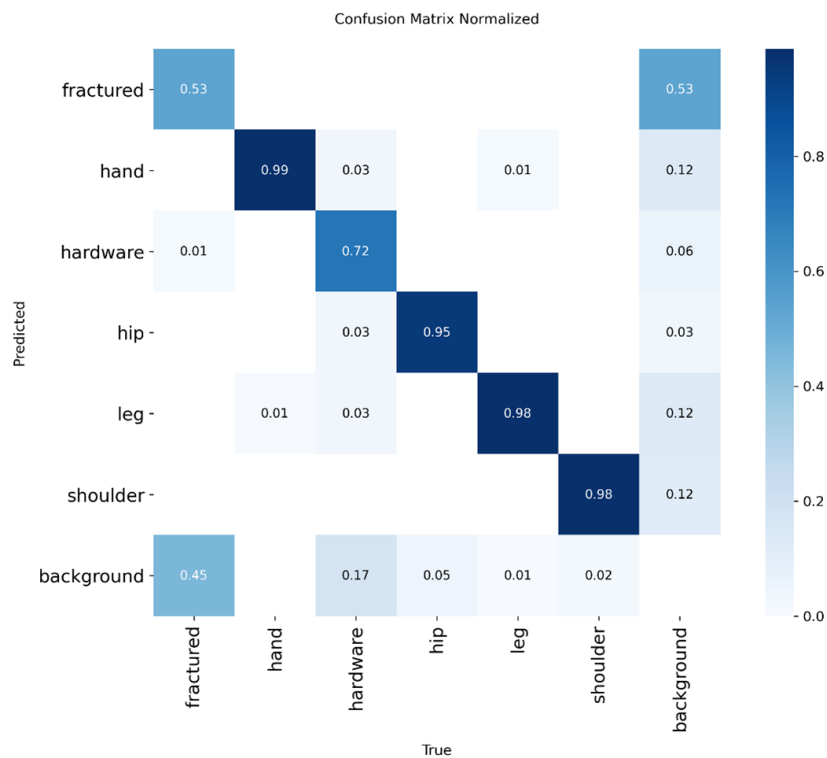


Figure 5.4: Bone fracture detection model confusion matrix.

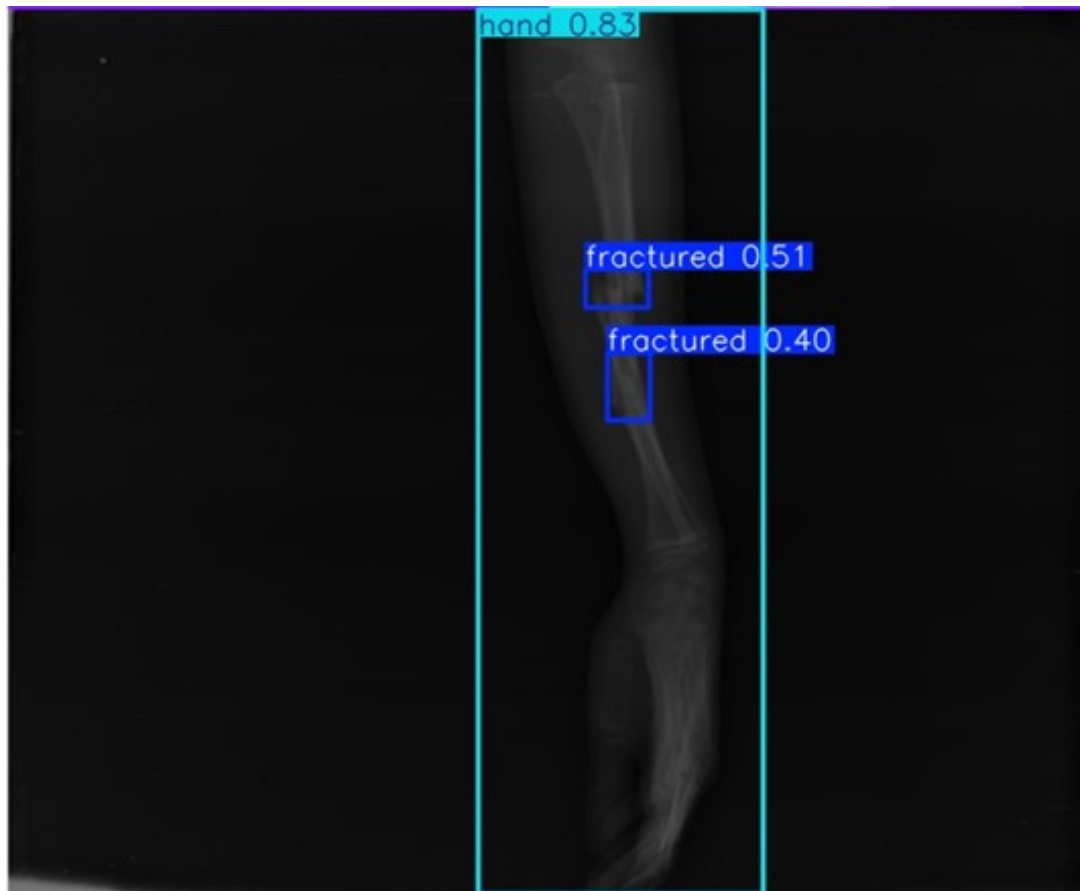


Figure 5.5: Sample output from the bone fracture detection model.

Class	Precision	Recall	F1-Score	Support
COVID19	1.00	1.00	1.00	214
NORMAL	0.94	0.97	0.95	557
PNEUMONIA	0.99	0.97	0.98	427
TUBERCULOSIS	0.96	0.93	0.95	351
Accuracy			0.97	
Macro Avg	0.97	0.97	0.97	1549
Weighted Avg	0.97	0.97	0.97	1549

Figure 5.6: Classification results of the chest disease detection model.

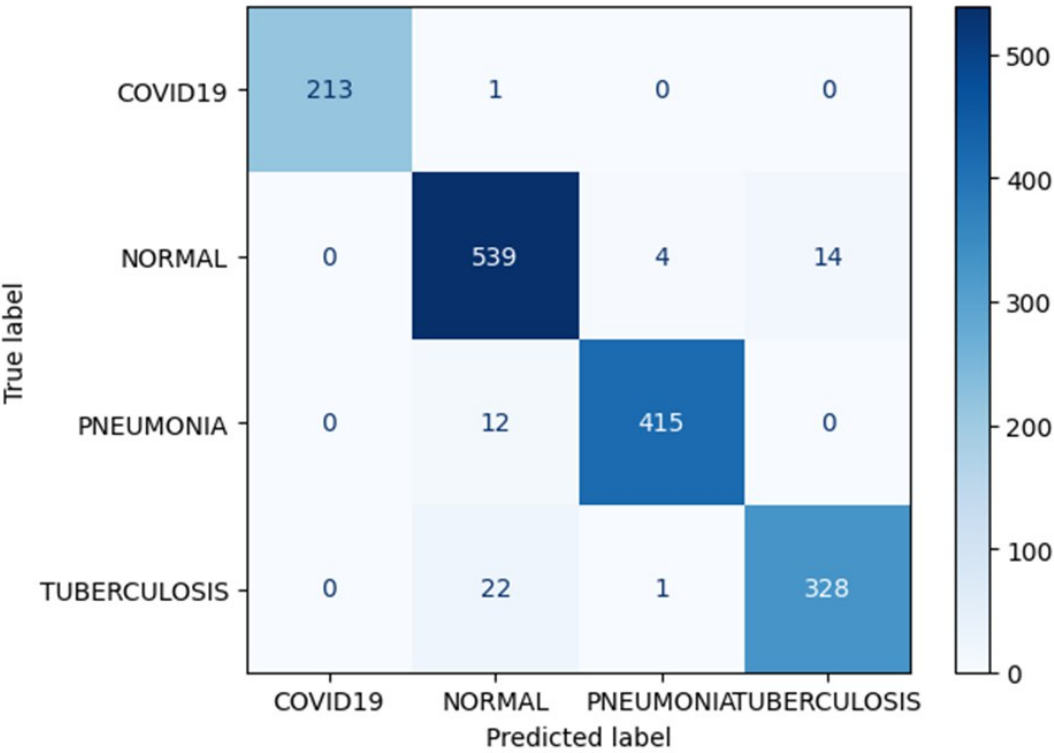


Figure 5.7: Confusion matrix of the chest disease detection model.

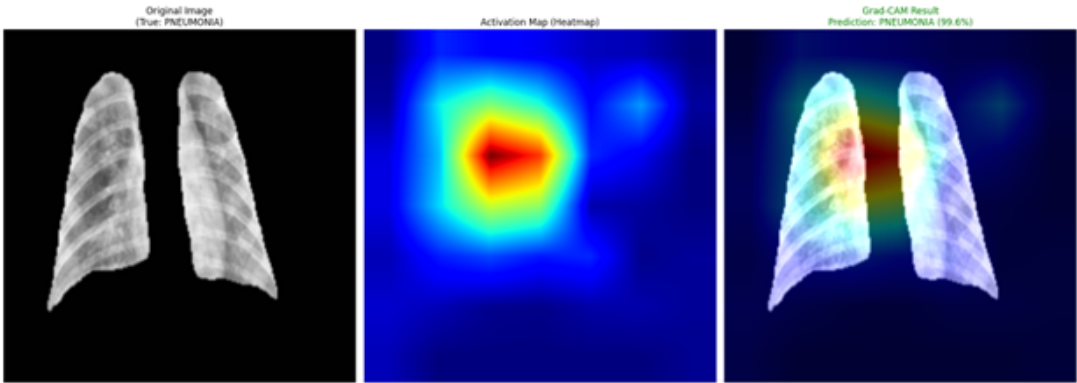


Figure 5.8: Sample output from the chest disease detection model.

The YOLOv8s had an input image size of 640×640 as raw data. It proved capable of finding normal bones, but had poor fracture recall.

In the next attempt, the resolution was increased to 1024×1024 . The performance had a noticeable drop and the higher resolution introduced noise in the input data.

The following attempt, the high resolution was kept but Contrast Limited Adaptive Histogram Equalization (CLAHE) was utilized to mitigate the problem of noise amplification. The performance recovered and the precision was high, but the recall remained a challenge.

Then, the newer YOLO11s model was tested on the MURA dataset to observe if it performed better than the previous attempts with the YOLOv8s model. It was observed that the performance of the YOLO11s model was worse than the YOLOv8 for this specific task.

In the final attempt, the batch size was configured to 8 to see if it had any significant effects on the model's performance. It was observed that reducing batch size allowed for more frequent weight updates, stabilizing gradients for the subtle fracture features. Thus, the final pipeline tested on the MURA dataset proved to be the best, and it employed the YOLOv8 model with CLAHE and a batch size of 8.

Figure 5.9 shows the results of the experiments done on the MURA dataset with the different YOLO models and configurations.

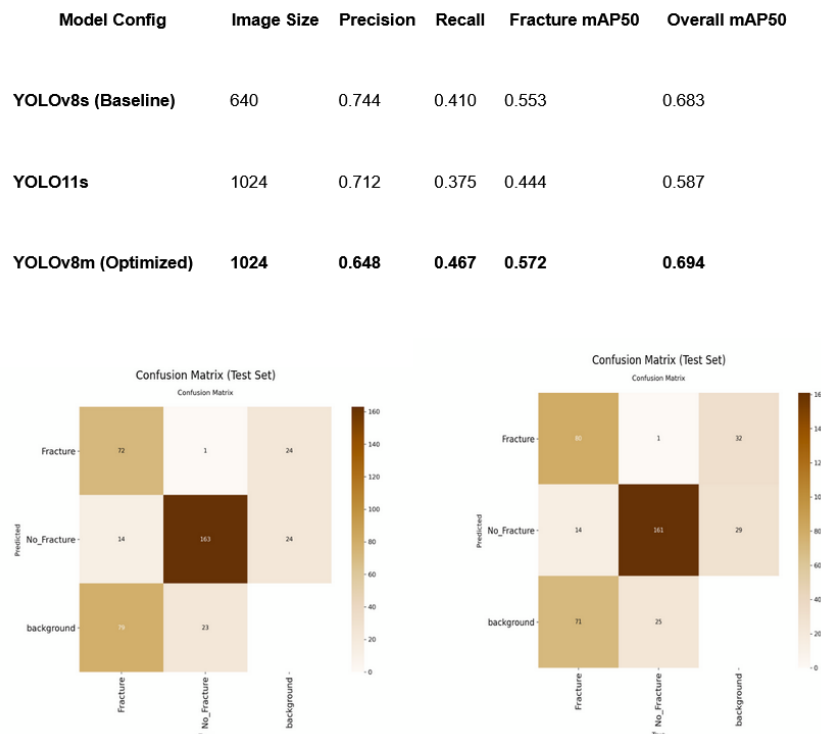


Figure 5.9: Results of successive experiments done on the MURA dataset to reach the best YOLO detection model possible.

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